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Rotating Plasmas

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Review paper

ROTATING PLASMAS

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ABSTRACT. A review is given of theoretical and experimental investigations on rotating plasmas. The basic equations are discussed from the microscopic and macroscopic points of view, including the balance of matter, momentum, and heat, as well as associated stability problems and limiting effects. Various types of devices are described, and obtained experimental results are compared with theoretical predictions. The review also summarizes the applications of rotating plasmas to fusion research, cosmical physics, and special technical arrangements such as plasma centrifuges, adjustable plasma condensers, and plasma gun and propulsion systems.

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1. INTRODUCTION

The electrodynamic force $\vec{j} \times \vec{B}$ produced by an electric current of density $\vec{j}$ flowing across a magnetic field $\vec{B}$ in an electrically conducting fluid can usually not be balanced by pressure gradients only, but gives rise to macroscopic fluid motions. Whenever the current density forms a diverging or converging pattern in the planes perpendicular to the field $\vec{B}$, a torque arises which tends to put the fluid into rotation around the magnetic field lines. Rotating motions should therefore appear rather frequently in plasmas and other electrically conducting fluids, both under laboratory conditions and in cosmical physics. Sometimes such motions may also arise from the action of time-dependent electric and magnetic fields.

When the magnetic field lines behave as equipotentials in an immersed electrically conducting fluid, the electric currents and the force $\vec{j} \times \vec{B}$ will produce a strong transmission of momentum in the magnetic field direction. In particular, for an axially symmetric, poloidal field, where the lines of force are situated in planes through the axis of symmetry, this interaction leads to the isorotation law discovered by Ferraro in 1937 [1].

According to this law a rotating star forces surrounding ionized matter to move at a uniform angular velocity all along a field line. A similar mechanism was proposed by Alfvén in a theory on the origin of the solar system, where an electrically conducting central body transmits angular momentum to a surrounding cloud of ionized gas [2-5].

In general the magnetic field becomes modified by electric currents induced in the fluid. A simplified picture of the coupling between the electric and mechanical forces leads to the concept of magnetic field lines acting as elastic strings being in a way attached to the fluid elements, and sometimes giving rise to magnetohydrodynamic waves propagated along or across the magnetic field [6]. In this review on rotating plasmas we shall mainly devote ourselves to fluid motions and macroscopic time changes which are slow compared to the propagation velocity of such waves. This corresponds to the limiting case of long wavelengths and low frequencies, where the magnetohydrodynamic restoring forces act almost instantaneously along and across the magnetic lines of force throughout the electrically conducting parts of the fluid body.

In 1921 Hull showed that charged particles can move in stable orbits in a region bounded by two coaxial cylinders acting as electrodes and containing an axial magnetic field [7]. Similar arrangements also provide a convenient method of producing rotating plasmas in the laboratory, where an applied electric field generates a radial electric current across the magnetic field. Electromagnetic 'wind generators' of this kind were constructed and investigated by Early et al. [8, 9], and a number of ion sources were studied by Smith and Luce [10]. Baker and Anderson [11] suggested the use of the same arrangement for the containment and heating of a thermonuclear plasma in the so-called 'Homopolar' device, and Boyer et al. [12] designed a modified device called 'Ixiom', having a central plasma electrode and magnetic mirrors at its ends. Further contributions to the exploration of similar systems were given at an early stage by a number of authors, both in studies of plasmas [13-40] and of electrically conducting liquids [41-47].

Up to the present date numerous investigations have been made on rotating electrically conducting magnetized fluids and their application to various branches of plasma physics, and a great number of experiments with different types of devices have been performed in the laboratory. The purpose of the present review is to sum up these results and to render earlier reviews on the subject [48-52] more complete.

Before starting a detailed analysis it may be appropriate to summarize some of the reasons for research to be conducted on rotating plasmas:

(a) Basic research

By means of the crossed field technique basic research can be carried out on fully ionized dense plasmas, even with limited resources and moderate amounts of technical complications. For the confinement and plasma balance in poloidal magnetic fields, a rotating plasma represents the most general case which also includes the momentum balance of a well-defined macroscopic fluid motion. The latter can be investigated by means of simple diagnostic methods, and from its behaviour important conclusions can be drawn on the total plasma balance. This also applies to the stability problem, where the strong centrifugal force produced by the rotation is easily varied by external means and can be used to simulate the effect of a large thermal pressure. It further seems possible to construct special rotating plasma devices in which accurate experimental tests can be made of the theoretically predicted values of various transport coefficients in a fully ionized plasma in a strong magnetic field.

(b) Thermonuclear fusion and associated problems

At the present stage it is not quite impossible but still unlikely that rotating plasmas alone would lead to the realization of a self-sustained fusion reactor. This is mainly due to the fact that such

plasmas suffer from a number of limiting effects, to be discussed later in this review. Nevertheless rotating plasmas are important to fusion research, both on account of their general plasma physical interest and for several other reasons. In principle, centrifugal confinement in a magnetic field of suitable shape should strongly reduce the end losses from magnetic mirror devices, the energy balance of which becomes quite critical on account of such losses. Further, the rotating plasma technique provides a starting and injection mechanism for magnetic bottles by which a fully ionized plasma is easily created, without difficulties in achieving high ion densities. The centrifugal force also provides an effective method for removing impurities from the plasma body. Finally, there are numerous experiments on high-current discharges where it is necessary to analyse rotating plasmas arising as a secondary effect on account of ambipolar electric fields, electric currents produced by various diffusion processes, imperfections in the design, and of unforeseen technical complications, thus affecting the stability and other properties of the confinement.

(c) Cosmical physics

It is likely that rotating plasmas play a greater role in cosmical physics than what has been realized so far, since mass motions hardly can be avoided when electric currents are flowing across a magnetic field. In addition to the problems of rotating magnetized stars [1, 6] and the origin of the solar system [5], rotating plasma phenomena should become important to the study of the atmospheres and bodies of the sun and other stars, as well as other regions of the cosmos. Particular examples are given by the sudden appearance and disappearance of solar prominences and flares [53-56], and by element and isotope separation effects in magnetized plasma bodies of limited size being surrounded by neutral gas [56-59].

(d) Special technical applications

The extremely large centrifugal force produced in rotating plasma devices makes them suitable for operation as plasma centrifuges for element and isotope separation, both in a fully [59, 60] and lowly [61, 62] ionized state. Other possible applications of rotating plasmas are provided by their ability to act as large adjustable electric condensers for electric power engineering purposes [19], and by rotating plasma guns for acceleration and propulsion systems [63, 64].

2. THEORY

We first summarize the basic theoretical principles of rotating plasmas, to give some starting points of the discussion on the experiments treated later in this review.

2.1. GENERAL FEATURES OF ROTATING PLASMA DEVICES

Rotating plasmas are usually created in the laboratory by applying an electric discharge across a magnetic field immersed in a mass of neutral gas. The discharge will then both ionize and heat the gas and put it into rotation. In some cases the plasma is created by other means than the crossed field technique, and is later put into rotation by a

transverse electric field. This review will be concentrated on plasmas the rotation of which is mainly produced by a strong transverse electric field, and where the pressure gradients only yield a minor contribution to the total fluid velocity.

Among the experimental devices which have been used so far, five types can be distinguished, as represented by the examples of Table I. Their general features are as follows:

TABLE I. MAIN DATA OF SOME ROTATING PLASMA DEVICES

Name Place References	Polarity	R_r ρ_r R_m	r_{02} r_{01} $L_{ }$	$\bar{B}_0$	n_0	n_{n0}	Gas	Special features
Type 'A'								
(a) High plasma density								
Homopolar I Berkeley [13, 19]	neg.	1 0 1	0.124 0.038 0.016	1.0 to 1.8	-	2×10^{21} to 2×10^{22}	D, He, Ne Ar, J	High ionization degree at large currents
Homopolar III Berkeley [106]	neg.	1.16 0.26 1.2	0.60 0.55 0.2	0.5	-	10^{21}	-	High ionization degree at large currents
Homopolar Stockholm [36, 38]	pos.	1.1 0.17 1.2	0.195 0.035 0.18	0.1 to 1.0	-	10^{20} to 6×10^{21}	H, D, He N, Ne Ar, O	High ionization degree at large currents
Homopolar Leatherhead [179]	neg.	1 0 1	0.07 0.025 0.025	1.0 to 1.8	-	10^{21} to 10^{23}	N	High ionization degree at large currents
Homopolar Kharkov [174]	-	- - (1)	0.065 0.008 1.8	0.2	-	2×10^{20}	H	Partially ionized during parts of holding mode
Ixion III Los Alamos [12, 158, 161]	neg.	1.48 0.55 2.2	0.12 0.015 2.6	0.4 to 1.0	10^{20}	6×10^{19} to 2×10^{20}	D	Fully ionized. Central plasma electrode. Gas injected
Kruisvuur I Amsterdam [52, 89, 143]	neg.	1(1.4) 0(0.5) 1(2)	0.17 0.02 1.5	0.1 to 0.5	3×10^{20}	3×10^{20}	H, N He, Ne Ar	Fully ionized at high currents. Brackets refer to insulators placed at mirrors
F I Stockholm [102, 160]	pos.	3 0.9 3	0.260 0.185 0.3	0.2 to 1.1	10^{21} to 3×10^{21}	10^{21} to 10^{22}	H N	Fully ionized. Neutral blanket. Large radial ratio
F II Stockholm [156, 169]	pos.	3.3 0.91 3.6	0.305 0.235 0.3	0.2 to 0.7	2×10^{21}	10^{21}	H, D	Fully ionized. Partially closed bottle. Variable radial ratio
F II A Stockholm [93]	pos.	1.7 0.66 0.9	0.308 0.243 0.3	0.25 to 0.60	10^{21} to 2×10^{21}	10^{21} to 6×10^{21}	H, D, He	Fully ionized. Partially closed bottle
F III Stockholm [186]	pos.	9.3 0.99 11.4	0.275 0.235 0.4	0.2 to 0.8	2×10^{21}	10^{21}	H	Fully ionized. Extremely large radial ratio

TABLE I. (cont.)

Name Place References	Polarity	R_r ρ_r R_m	r_{02} r_{01} L_{II}	$\bar{B}_0$	n_0	n_{n0}	Gas	Special features
<u>Type 'A'</u>								
(b) <u>Low plasma density</u>								
Ion magnetron Berkeley [17]	pos.	1.23 0.33 1.5	0.10 0.008 1.7	0.8	10^{18}	3×10^{19}	D	Partially ionized. Anode sheath formed
PIG discharge Schenectady [185]	neg.	1.41 0.5 2	0.075 0.05 0.7	0.2 to 1.8	-	3×10^{19}	H	Partially ionized
Reflex discharge Saclay [176]	neg.	- - -	0.06 - 0.9	0 to 0.2	5×10^{17} to 5×10^{18}	2×10^{19} to 3×10^{20}	H, N	Partially ionized
Penning discharge Stockholm [182]	neg.	6 0.97 20 to 100	0.1 to 0.3 0.03 to 0.17 0.8	0.02 to 0.13	10^{16} to 10^{17}	3×10^{18} to 3×10^{19}	H	Partially ionized. Variable mirror ratio
Reflex discharge Kharkov [152]	neg.	- - -	0.045 - (0.1)	0.01 to 0.04	10^{11} to 10^{12}	3×10^{16} to 3×10^{17}	N, N and O	Partially ionized
<u>Type 'B'</u>								
Homopolar IV Berkeley [31]	-	(1) (0) (1)	0.076 0.025 (1)	2	-	-	D	Fully ionized. End insulators outside central section
Homopolar V Berkeley [163]	-	(1) (0) (1)	0.076 0.025 (1.5)	0.5 to 2.5	10^{20} to 10^{21}	-	D	Fully ionized. Neutrons observed
Puffatron Risø [177, 180]	pos. and neg.	(0.7) (-1) (0.5)	0.075 0.0125 to 0.025 (0.6)	4	10^{20} to 10^{21}	-	H	Fully ionized. Anode sheath at pos. polarity. No end insulators in most experiments
Homopolar Gun I Berkeley [181]	-	- - -	0.076 0.025 -	0.5 to 2.5	10^{20}	-	D	Similar to Homopolar V, but used as gun
Alice Puffatron Livermore [175]	pos.	- - -	0.15 0.013 -	0.8	1.5×10^{19}	10^{20}	H, He	Partially ionized
Plasma source Berkeley [197]	pos.	(1.2) (0.33) 1.5	- 0.025 -	1.8 and 6	2×10^{20}	-	D	Two-stage magnetic mirror
Plasma source Madison [199]	pos.	(1.9) (0.7) 3.5	0.075 0.02 (2)	0.45	-	10^{21}	H	Electrodes con- toured to flux surfaces

TABLE I. (cont.)

Name Place References	Polarity	R_r ρ_r R_m	r_{02} r_{01} L	$\bar{B}_0$	n_0	n_{n0}	Gas	Special features
<u>Type 'C'</u>								
Hothouse I Berkeley [166]	-	(1) (0) (1)	0.073 0.025 0.86	1.5	2×10^{21} to 10^{22}	6×10^{21}	H	Ionizing waves. Fully ionized
Supper II Sydney [100, 170, 172]	pos. and neg.	(1) (0) (1)	0.105 0.076 1.7	0.5 to 2	10^{19} to 10^{22}	3×10^{19} to 3×10^{22}	D, H, He	Ionizing waves. Sometimes partially ionized. Variable polarity
<u>Type 'D'</u>								
Rotating field Culham [67]	-	1 0 1	0.015 - 0.2	0.55 to 0.73	3×10^{20}	3×10^{20}	Xe, Ar	Rotating field at 1 to 1.5 MHz
Rotating pinch Lausanne [168]	-	1 0 1	0.024 - 0.49	0.22	-	6×10^{20} to 3×10^{21}	H, He	Rotating field at 3.1 MHz
<u>Type 'E'</u>								
Stellarator Princeton [159]	pos.	(1) (0) (1)	0.025 0.01 -	2.5 to 4	2×10^{18} to 2×10^{19}	-	H D	Fully ionized. Rotational transform of magnetic field

Note: The table gives approximate values of the average radial ratio $R_r = r_0/r_w$, the centrifugal confinement factor $\rho_r = 1 - R_r^{-2}$, the average mirror ratio $R_m = B_w/B_0$, the radial extensions r_{02} and r_{01} in the midplane, the effective longitudinal length $L_{||} = V/\pi(r_{02}^2 - r_{01}^2)$, the average magnetic field strength $\bar{B}_0$ in the midplane, the average ion density n_0 in the same plane, and the neutral filling gas density n_{n0} . Corresponding references are given within square brackets. Figures within ordinary brackets are either uncertain or not well defined. Type 'A' denotes quasi-stationary systems in which a fully ionized plasma usually obeys the isorotation law at high densities and where deviations from this law arise at low densities. Type 'B' denotes non-stationary devices with pulsed neutral gas injection, Type 'C' ionizing front systems, Type 'D' rotating field systems, and Type 'E' rotating plasma devices without axial symmetry. MKSA units are used.

(A) Quasi-stationary systems

A quasi-stationary plasma can be produced and sustained by a discharge in a system containing both plasma and neutral gas. Under certain conditions the general features are as outlined in Fig. 1. The plasma is put into rotation in the $\vec{E} \times \vec{B}$ direction by an electric field $\vec{E}$ due to an electric potential difference applied on a system of electrodes.

At plasma densities n being sufficiently high to establish nearly isotropic and equal ion and electron temperatures, the component $E_{||}$ of the electric field along $\vec{B}$ will become small compared to its transverse component $E_{\perp}$. The plasma then obeys the isorotation law according to which the angular velocity becomes uniform along a magnetic field line, and the electrode system is represented by two toroidal surfaces as given by the dashed lines in the figure. Part of these surfaces may consist of real metal electrodes, but the rest may also be formed by magnetic field lines which act as equipotentials. At high plasma densities it is further possible for partially ionized boundary

regions of a limited thickness x_b to separate a central body of fully ionized plasma from a surrounding neutral gas blanket. Plasma which escapes along the magnetic field lines to the walls of the insulators recombines and forms neutral gas wall layers of thickness d_w from which neutral particles are scattered back into the plasma and become reionized. Under these conditions there are two important concepts which determine the plasma losses to the end insulators in the magnetic field direction of Fig. 1. The first is the mirror ratio B_w/B_0 between the magnetic field strengths B_w and B_0 at the insulator surfaces and in the midplane, respectively. This ratio is generally involved both in the magnetic mirror reflection process of charged particles and in the loss area at the end walls which is proportional to B_0/B_w at a given cross-sectional area of the confinement region in its midplane. The second is the radial ratio r_0/r_w between the axial distance r_0 of a field line in the midplane and the corresponding distance r_w where it cuts the surface of an end wall. The radial ratio determines the efficiency by which the

centrifugal force tends to confine the plasma in the magnetic field direction, by pushing it towards the midplane when $r_0 > r > r_w$.

At low plasma densities there also exist quasi-stationary systems containing anisotropic rotating plasmas in which the longitudinal electric field $E_{||}$ cannot be neglected, and where neutral particles penetrate into the central parts of the plasma body. Under such circumstances there do not exist any well-defined electrode surfaces or boundary regions of the type outlined in Fig.1, and the isorotation law becomes invalid. These systems include the Penning, PIG, and reflex discharges as well as the ion magnetron.

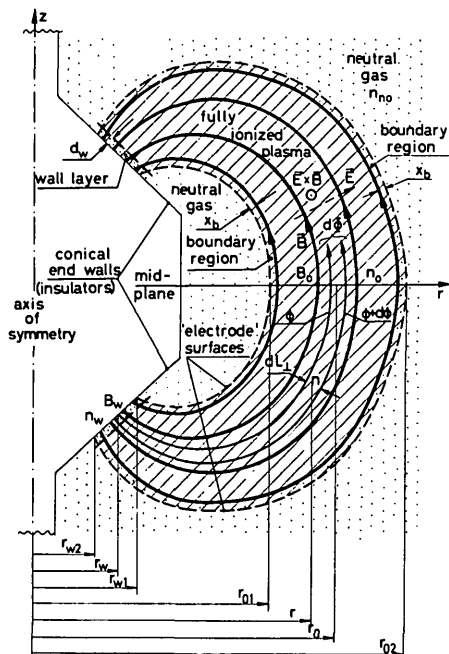


Fig. 1. Type 'A' system in which a quasi-stationary fully ionized rotating plasma is sustained.

The plasma is generated by a discharge in the crossed electric and magnetic fields E and B . Partially ionized boundary regions and wall layers separate the fully ionized plasma body from a surrounding neutral gas blanket and from the end walls. Only part of the electrode surfaces has to be made of metal walls whereas the rest may consist of magnetic field lines which act as equipotential surfaces under certain conditions.

(B) Puffatron systems

In transient operation it is possible to avoid the interaction between the plasma and the end walls by means of the 'puffatron' arrangement outlined in Fig.2(a). Just before the start of the discharge the intended plasma confinement volume is under high vacuum. An electric voltage is applied between the electrodes and a neutral gas blob is introduced into the centre of the device by means of a fast valve. Soon afterwards an electric discharge transforms the blob into a fully ionized plasma of limited axial extension. During a transient state the plasma is then bounded by high

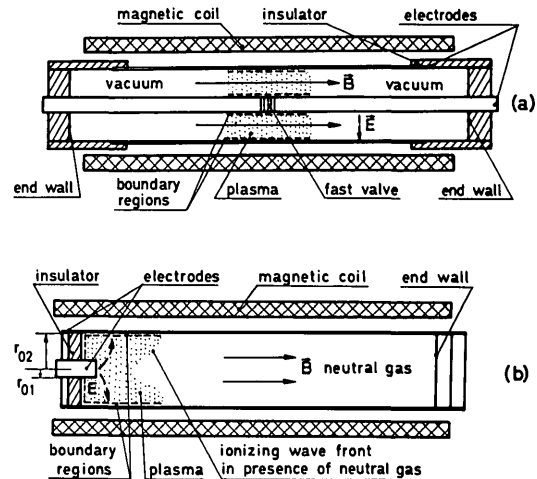


FIG. 2. Rotating plasma systems for non-stationary operation.

(a) Type 'B' system in which interaction between the plasma and the end walls is avoided by creating a plasma blob of finite axial extensions. Neutral gas is introduced by a fast valve and is ionized at the centre of the vacuum chamber [163].

(b) Type 'C' system in which ionizing fronts, plasma blobs, or plasma waves are generated by concentric electrodes at one end of the device.

vacuum regions in the magnetic field direction. After some time when the plasma has expanded far enough along the magnetic field lines, an interaction will take place with the end walls. The axial expansion can be delayed by using a magnetic mirror field.

The puffatron arrangement can also be placed at one end of the device, and by a suitable arrangement of the magnetic field it can then be converted into a type of gun system in which the plasma is made to expand axially into a vacuum region.

(C) Ionizing front systems

Transient operation of rotating plasmas can further be performed as demonstrated by Fig.2(b), where the plasma is generated by a discharge between two concentric electrodes at one end of the device. Different kinds of experiments can be carried out with this arrangement, involving studies of the motion of ionizing fronts which propagate along the field lines in a neutral gas, of coaxial rotating plasma guns in which the discharge is used in combination with a magnetic mirror field to make the plasma expand axially into the neutral gas, and of waves being generated by the electrodes and propagating in the plasma along the magnetic field.

(D) Rotating field systems

There are various possibilities of setting a plasma column into rotation by means of time-varying rotating electric and magnetic fields. One method makes use of a rotating magnetic field component which is superimposed on an axially symmetric magnetically confined plasma, and

being similar to the fields used to drive the rotor in an electric induction motor. An example of this method is given by Fig. 3.

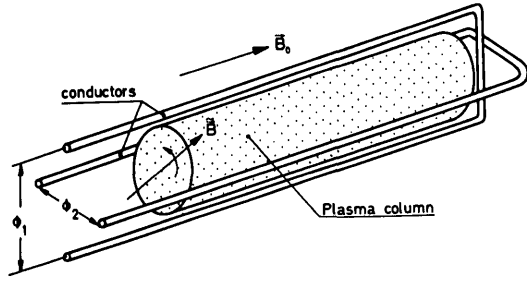


FIG. 3. Rotating plasma system of type 'D'.

A cylindrical plasma column is confined in an axial static magnetic field $\vec{B}_0$ and is set into rotation around the axis of symmetry by a superimposed rotating field $\vec{B}$. The latter is created by currents in two conductor systems, the applied voltages ϕ_1 and ϕ_2 of which are 90 degrees out of phase [67].

(E) Asymmetric field systems

There are finally systems with more complicated and asymmetric structures in which a $\vec{j} \times \vec{B}$ force puts the fluid into rotation. An example of this is given by the stellarator in which a plasma rotation sometimes arises from a radial electric field.

2.2. THE MOTION OF INDIVIDUAL PARTICLES

We begin the theoretical treatment with the motion of individual charged particles in electric and magnetic fields which are considered as given constraints. The equation of motion of a particle of mass m and charge q , having the position vector $\vec{r}(t)$ and velocity $\vec{w} = d\vec{r}/dt$, is

$$m \frac{d\vec{w}}{dt} = q\vec{E} + q\vec{w} \times \vec{B} \quad (1)$$

where

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \text{curl } \vec{A} \quad (2)$$

with ϕ and $\vec{A}$ being the electric potential and the magnetic vector potential, respectively. The present treatment is mainly restricted to systems being symmetric with respect to the z axis of a cylindrical coordinate system (r, ϕ, z) . Particle orbits in rotating fields will be shortly described in Section 2.2.4.

2.2.1. Forbidden regions in axially symmetric case

We first consider a non-stationary axially symmetric case with arbitrary fields $\vec{E}$ and $\vec{B}$. From Eqs (1) and (2), or from the corresponding Hamiltonian formalism, the generalized angular

momentum p_ϕ in the azimuthal direction is easily seen to become a constant of the motion, i.e.

$$\begin{aligned} p_\phi &= p_{\phi_0} = mrw_\phi + qrA_\phi \\ &= mr_0 w_{\phi_0} + qr_0 A_{\phi_0} = \text{const.} \end{aligned} \quad (3)$$

where subscript (0) denotes values at a particular starting point (r_0, ϕ_0, z_0) . For the velocity components in the rz plane combination of Eq. (3) with the Hamiltonian

$$\begin{aligned} H &= \frac{1}{2} m(w_r^2 + w_z^2) + (1/2 mr_\phi^2) (p_\phi - qrA_\phi)^2 + q\phi \\ &= \frac{1}{2} mw^2 + q\phi \end{aligned} \quad (4)$$

yields

$$\begin{aligned} \phi_{eq} &\equiv \frac{1}{2} m(w_r^2 + w_z^2) = H - H_0 + \frac{1}{2} mw_0^2 + q(\phi_0 - \phi) \\ &- \frac{1}{2 mr^2} \left[mr_0 w_{\phi_0} + q(r_0 A_{\phi_0} - rA_\phi) \right]^2 \geq 0 \end{aligned} \quad (5)$$

Expressions (4) and (5) do not contain A_r and A_z explicitly and depend only on the poloidal part of the magnetic field $\vec{B}$ which is generated by the component A_ϕ according to Eq. (2). The quantity ϕ_{eq} can be regarded as an equivalent potential for the particle motion in the rz plane. Negative values of the third member of Eq. (5) represent forbidden regions. For a strong poloidal magnetic field it is easily seen that each of the terms $r_0 A_{\phi_0}$ and rA_ϕ within the square bracket of Eq. (5) will give rise to contributions which are much larger than those from other terms in the equation. This implies that the particle will be trapped within a narrow strip centred around the field line $rA_\phi = r_0 A_{\phi_0}$ in the rz plane, the breadth of the strip being of the order of one Larmor radius. This result holds also for time-dependent fields, where the magnetic lines of force can be considered to move in space. The particle should then closely follow the surface, where rA_ϕ at time t is equal to the value $r_0 A_{\phi_0}$ at the starting point at time t_0 . This simply expresses the fact that an electric field is induced by $\partial \vec{A}/\partial t$, giving the particle a transverse drift in such a way that it becomes nearly 'frozen' to the field lines.

2.2.2. Centrifugal confinement and the isorotation law

From the previous section it is seen that a charged particle under quite general conditions becomes confined to a narrow potential trough centred around the lines of a poloidal magnetic field. However, Eq. (5) does not give any information of the effects on the longitudinal confinement in a rotating plasma. To study these we now restrict ourselves to a static poloidal magnetic field and consider a local frame of reference rotating at the arbitrary angular velocity $\vec{\Omega} = \vec{\Omega}(r, z, t)$ with respect to the laboratory frame. In the rotating frame the particle is observed to have the velocity $\vec{w}^*$, where

$$\vec{w} = \vec{w}^* + \vec{\Omega} \times \vec{\rho}^* \quad (6)$$

and $\vec{\rho}^*$ is the corresponding position vector. The equation of motion (1) can then be rewritten in a form associated with the rotating frame [51],

$$m \frac{d\vec{w}^*}{dt} = q\vec{E}^* + q\vec{w}^* \times \vec{B}^* \quad (7)$$

where now

$$\begin{aligned} \vec{E}^* = \vec{E} + \frac{m}{q} \Omega^2 \vec{r} + \Omega \nabla(rA_\varphi) \\ + \left(\frac{m}{q} \Omega r^2 + rA_\varphi - p_{\varphi_0}/q \right) \nabla \Omega + \frac{m}{q} \vec{r} \times \frac{\partial \vec{\Omega}}{\partial t} \end{aligned} \quad (8)$$

$$\vec{B}^* = \vec{B} + 2 \frac{m}{q} \vec{\Omega} + \frac{m}{q} (\vec{r} \times \dot{\vec{\Omega}}) \times \nabla \Omega \quad (9)$$

with $\vec{r} = (r, 0, 0)$, $\vec{\Omega} = \hat{z}\Omega$, and $\hat{z}$ representing the unit vector in the z direction. The second terms of the right-hand members of Eqs (8) and (9) are due to the centrifugal and Coriolis forces, respectively. The third term in Eq. (8) is due to the electric field arising from the transformation to the rotating frame, i.e. $\Omega \nabla(rA_\varphi) = (\vec{\Omega} \times \vec{\rho}^*) \times \vec{B}$. Especially when the angular velocity $\vec{\Omega}$ becomes uniform and time-independent, the rest of the terms of Eqs (8) and (9) vanish. By means of Eqs (6) - (9) the earlier perturbation theory of the guiding centre motion of a charged particle [6] can easily be re-formulated for an observer following the rotating frame [49, 51].

In analogy with Eqs (4) and (5) a modified potential is now obtained for the motion in the rz plane [25, 51], i.e.

$$\begin{aligned} \phi_{eq}^* \equiv \frac{1}{2} m \left(w_r^{*2} + w_z^{*2} \right) &= \frac{1}{2} m w_0^{*2} + q(\phi_0 - \phi) \\ &- q \left(\Omega_0 r_0 A_{\varphi_0} - \Omega r A_{\varphi} \right) + (\Omega_0 - \Omega) p_{\varphi_0} \\ &- \frac{1}{2} m (\Omega_0^2 r_0^2 - \Omega^2 r^2) \\ &- \frac{1}{2 m r^2} \left[m r_0 w_{\varphi_0}^* + q(r_0 A_{\varphi_0} - r A_{\varphi}) \right. \\ &\left. + m \Omega_0^2 r_0^2 - m \Omega r^2 \right]^2 \geq 0 \end{aligned} \quad (10)$$

This result can also be deduced directly by inserting expression (6) into the equivalent potential (5). Since the angular velocity $\vec{\Omega}$ has not yet been specified, no definite conclusion can be drawn about the confinement from expression (10). However, if we consider the situation outlined in Fig. 1, where the magnetic field lines become equipotentials in the first approximation, expression (10) can be

reduced to a form which gives more information about the forbidden regions than that obtained from direct inspection of the inequality (5). Thus, for a narrow cylindrical shell of local thickness $dL_\perp$, situated between two field lines which have the potentials $\phi + d\phi$ and ϕ , the electric field strength becomes $E = -d\phi/dL_\perp$. In the first approximation all charged particles move at the corresponding drift velocity E/B . We now separate this drift from the rest of the particle motion by defining the angular velocity as

$$\vec{\Omega} = \vec{E} \times \vec{B} / rB^2 \quad (11)$$

The magnetic flux enclosed by the shell in Fig. 1 is $d\Phi = 2\pi r B dL_\perp$ and is related to $d\phi$ and Ω by

$$\Omega = -E/Br = 2\pi d\phi/d\Phi = d\phi/d(rA_\varphi) \quad (12)$$

Since $d\phi$ and $d\Phi$ are both constant along the shell, we have $\Omega = \Omega_0 = \text{const.}$ for all points belonging to the shell. This is the isorotation law earlier deduced by Ferraro in a different way [1]. In the limit of strong magnetic fields and for Ω being much smaller than the gyrofrequency $\omega_g = qB/m$ we can then insert the result (12) into expression (10) and obtain [26, 51]

$$\begin{aligned} \phi_{eq}^* &= \frac{1}{2} m w_0^{*2} - \frac{1}{2} m \Omega_0^2 (r_0^2 - r^2) \\ &- \frac{1}{2 m r^2} \left[m r_0 w_{\varphi_0}^* + q(r_0 A_{\varphi_0} - r A_{\varphi}) \right. \\ &\left. + m \Omega_0^2 (r_0^2 - r^2) \right]^2 \geq 0 \end{aligned} \quad (13)$$

Here the second term of the right-hand member arises from the centrifugal force and the last within the square bracket from the Coriolis force.

Relation (13) shows that for $r > r_w$ there is absolute confinement for a particle inside the shell defined by the field lines in Fig. 1, provided that the kinetic energy $\frac{1}{2} m w_0^{*2}$ at the starting point r_0 is less than the centrifugal work $\frac{1}{2} m \Omega_0^2 (r_0^2 - r_w^2)$. The latter approaches its maximum value $\frac{1}{2} m \Omega_0^2 r_0^2$ at large radial ratios r_0/r_w .

Further, due to the equivalence between Eqs (1) and (7) we can immediately apply the results from the ordinary first-order orbit theory to the rotating frame by introducing the equivalent fields (8) and (9). The modified equivalent magnetic moment $M^* = mW^{*2}/2B^*$ produced by the velocity W^* of the gyromotion then becomes an adiabatic invariant for a particle moving at the total velocity $\vec{w}^*$ in the modified magnetic field $\vec{B}^*$. After some deductions involving the longitudinal component of the corresponding equation of motion of the guiding centre and the longitudinal velocity $u_\parallel^*$ we obtain [49, 51]

$$\begin{aligned} u_\parallel^{*2} &= u_{\parallel 0}^{*2} - W_0^{*2} (B^*/B_0^* - 1) \\ &- \Omega_0^2 r_0^2 (1 - r^2/r_0^2) \geq 0 \end{aligned} \quad (14)$$

where the equality refers to the turning points of the particle. The second term in the intermediate member of expression (14) is due to the mirror confinement which is controlled by the mirror ratio B^*/B_0^* , whereas the last term is due to the centrifugal confinement which is controlled by the radial ratio r/r_0 . Boyer et al. [12] have earlier deduced a similar relation, but where the centrifugal confinement term contains the mirror ratio instead of the radial ratio. However, this is a special case of expression (14) which only applies to mirror ratios close to unity where $B = \text{const.}/r^2$ and where the centrifugal force will only have a weak effect on the confinement.

Consequently, we see that full use can be made of the centrifugal confinement effect in rotating plasma devices having large radial ratios r_0/r_w , provided that the thermal velocity w_0^* of the plasma particles is small compared to the velocity $\Omega_0 r_0$ of rotation.

In concluding this discussion it should be pointed out that there are important situations in which the magnetic field lines do not become equipotentials. This is the case in some low-density plasmas in strongly inhomogeneous magnetic fields, where large magnetic mirror forces balance the forces from a strong field $E_{||}$ which is produced by ion and electron velocity distributions having unequal pitch-angles and turning points [6, 65, 66]. In such systems the isorotation law (12) does not apply and the angular velocity Ω may vary appreciably along a field line. Such a differential rotation also arises from incomplete line-tying due to

finite resistivity and to the specific boundary conditions which prevail in the direction of the magnetic field lines.

2.2.3. Orbits in the planes perpendicular to the magnetic field

In Sections 2.2.1. and 2.2.2. it has been shown that a charged particle is confined within a trough centred around a field line and having the breadth of about one Larmor radius. We shall now examine the orbits of the transverse motion a little more in detail and start with the simple plane case of crossed homogeneous fields as outlined in Fig.4(a). With the coordinates defined in the figure, Eq.(1) yields the following expressions for the transverse velocity components [51]

$$w_y = w_{y0} \cos \omega_g t - w_{x0} \sin \omega_g t + (E_0/B) \cos \omega_g t - E/B + (1/B) \cos \omega_g t \int_0^t \frac{dE}{dt} \cos \omega_g t dt \quad (15)$$

$$+ (1/B) \sin \omega_g t \int_0^t \frac{dE}{dt} \sin \omega_g t dt$$

$$w_x = w_{x0} \cos \omega_g t + w_{y0} \sin \omega_g t + (E_0/B) \sin \omega_g t + (1/\omega_g B) \frac{dE}{dt} - (1/\omega_g B) \left(\frac{dE}{dt} \right)_0 \cos \omega_g t \quad (16)$$

$$- (1/\omega_g B) \cos \omega_g t \int_0^t \frac{d^2 E}{dt^2} \cos \omega_g t dt - (1/\omega_g B) \sin \omega_g t \int_0^t \frac{d^2 E}{dt^2} \sin \omega_g t dt$$

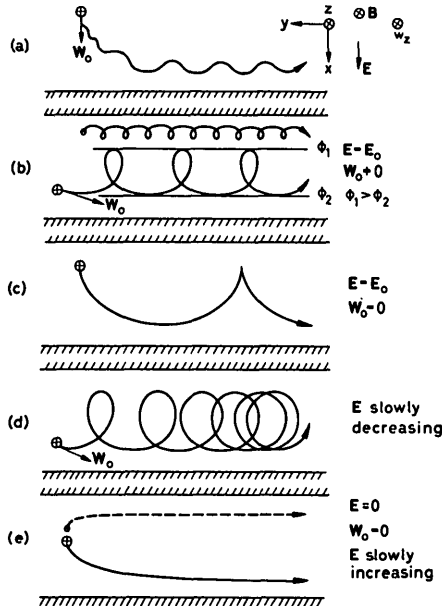


FIG.4. Orbits of a charged particle in a homogeneous magnetostatic field $\vec{B}$ and a homogeneous, time-dependent electric field $\vec{E}$.

- (a) Definitions of coordinates.
- (b) Constant electric field $\vec{E}_0$. The electric drift is the same for particles of both polarities.
- (c) Constant electric field and zero initial velocity $\vec{w}_0$.
- (d) Electric field slowly decreasing. The rate of change of the electric field drift is shown on a strongly exaggerated scale.
- (e) Electric field slowly increasing from zero. No initial velocity. Displacements in x direction strongly exaggerated.

in a homogeneous magnetostatic field $\vec{B}$ and a homogeneous time-dependent electric field $\vec{E}(t)$. Subscript (0) refers to initial values at time $t = 0$. By means of Eqs (15) and (16) we now discuss the following special cases:

(a) In a constant electric field $\vec{E} = \vec{E}_0$ and with a finite initial velocity $\vec{w}_0 = (w_{x0}, w_{y0}, 0)$ of gyration, an ion will drift in trochoidal orbits at the velocity $\vec{u}_E = \vec{E} \times \vec{B}/B^2$ in the negative y direction, in the space between the equipotential surfaces $\phi = \phi_1$ and $\phi = \phi_2$ as shown by Fig.4(b). An electron drifts at the same velocity $\vec{u}_E$ but the direction of its Larmor motion and the size of the corresponding Larmor radius differs from that of the ion. Thus, in a plasma the electric field drift defined by $\vec{u}_E$ does not produce any charge separation or electric current.

(b) In the special case of a vanishing initial velocity $\vec{w}_0$, the orbit of an ion becomes a cycloid as demonstrated by Fig.4(c). This situation arises e.g. when the ion is suddenly formed by an ionizing or charge exchange collision from a neutral particle being nearly at rest. It also arises if the electric field is switched on and reaches its final value $\vec{E}_0$

during a time interval which is short compared to the gyroperiod $2\pi/|\omega_g|$.

(c) When the electric field changes slowly compared to the gyroperiod, the integrals of Eqs (15) and (16) can be neglected and the motion is approximated by a gyration at the velocity $\vec{W}_0$ superimposed on a slowly changing electric field drift $\vec{u}_E = \vec{E}(t) \times \vec{B}/B^2$. A sketch of the corresponding orbit is shown in Fig. 4(d) on a strongly exaggerated scale and with the small displacements in the x direction being left out. Consequently, slow changes in the electric field will be followed by corresponding changes in the guiding centre drifts of ions and electrons in a rotating plasma.

(d) Especially if the particles start from rest with $\vec{W}_0 = 0$ in an electric field which slowly and monotonically starts from the initial value $E_0 = 0$, the ion and electron orbits become as shown by Fig. 4(e). In the figure the displacements in the x direction have been strongly exaggerated. We observe that during this transient process ions and electrons move oppositely in the x direction. In a rotating plasma this gives rise to an electric polarization during the acceleration period, being associated with an equivalent electric capacity.

(e) The drift $\vec{u}_E$ described under (a) can be considered as the result of an acceleration of the particle in the electric field direction. Thus, the instantaneous radius of curvature of the orbit increases when an ion 'falls' from the surface $\phi = \phi_1$ to $\phi = \phi_2$. A similar drift will be produced in the $\vec{F} \times \vec{B}$ direction by any transverse force field $\vec{F}$ which acts on the particle. This also applies to the case where $\vec{F}$ appears during short time intervals such as by collisions. In this latter case the corresponding average drift velocity can also be deduced from the displacement of the guiding centre produced during a collision [6]. We observe that the force $\vec{F}$ in all these cases has to be a real force acting on the individual particle, and should therefore not be confused with the pressure gradient which represents an equivalent force acting on the volume elements of a macroscopic fluid model.

In a rotating plasma the effects of the curvature of the cylindrical geometry will modify the results just obtained. This can be studied by returning to the results of Section 2.2.2. and choosing a rotating frame as specified by Eq. (11). In this frame $\vec{E} = -(\vec{\Omega} \times \vec{r}) \times \vec{B}$ and the first and third terms of the right-hand member of Eq. (8) cancel. In the case of a uniform stationary angular velocity $\vec{\Omega}$ the equivalent fields (8) and (9) in the rotating frame then become

$$\vec{E}^* = m\Omega^2 \vec{r}/q; \quad \vec{B}^* = \vec{B} + 2m\vec{\Omega}/q \quad (17)$$

where $\vec{E}^*$ and $2m\vec{\Omega}/q$ originate from the centrifugal and Coriolis forces. When $\omega_g \gg |\Omega| = u_E/r$ there is an additional azimuthal drift being mainly due to the centrifugal force. It becomes

$$\vec{u}_E^* = \vec{E}^* \times \vec{B}^*/B^{*2} \approx \Omega^2 \vec{r} \times \vec{B}/\omega_g B; \quad |\vec{u}_E^*| = u_E^2/\omega_g r \quad (18)$$

where $\vec{u}_E = \vec{E} \times \vec{B}/B^2$ is the lowest order electric guiding centre drift in the laboratory frame. Consequently, heavy particles will have a larger differential drift $\vec{u}_E^*$ than light particles. This effect is of importance to rotating plasma centrifuges, as discussed later in Sections 2.8. and 3.9.

2.2.4. Orbits in rotating fields

A plasma column can be put into rotation not only by stationary crossed $\vec{E}$ and $\vec{B}$ fields, but also by applying rotating time-dependent fields as outlined in Fig. 3 [67]. The particle orbits in such rotating field systems have been analysed by Insinger et al. [68] who have shown that resonance phenomena arise between the particles and the fields under certain conditions. This combination of fields can be made to act either as a velocity discriminator, accelerator, reflector, or injector. Reviews on rotating field systems are given by Johnston [69] and Insinger [52] among others.

2.3. BASIC MACROSCOPIC EQUATIONS

We now turn to the macroscopic plasma balance equations which can be obtained from the first three moments of the Boltzmann equation by integration over velocity space (see e. g. Refs [70-72]). In this paper we use subscript (ν) to denote any of the different species which consist of ions (i), electrons (e), and neutral particles (n). Further, we introduce the notations n_ν for the particle density in coordinate space, $\vec{v}_\nu$ for the macroscopic fluid velocity, $\vec{w}_\nu$ for the velocity vector of an individual particle, q_ν and m_ν for the particle charge and mass, $\vec{\pi}_\nu$ for the pressure tensor,

$U_{\nu k} = \frac{1}{2} m_\nu (w_{\nu k} - v_{\nu k})^2$ for the thermal particle energy associated with the k direction as obtained from a mean value formation in velocity space, and use $\vec{q}_\nu$ to denote the heat flow vector.

2.3.1. The moment equations for each species

The first moment of the Boltzmann equation determines the particle balance which is expressed by

$$\frac{\partial n_\nu}{\partial t} + \text{div} (n_\nu \vec{v}_\nu) = P_\nu \quad (19)$$

where

$$P_\nu = \int \int \int \left(\frac{\partial f}{\partial t} \right)_\nu dw_x dw_y dw_z \quad (20)$$

is the net rate at which particles of kind (ν) are produced by collisions and $(\partial f/\partial t)_\nu$ is the corresponding change in the distribution function f per unit time. By combining Eq. (19) with the second moment, the momentum balance equation is obtained having the form

$$n_\nu m_\nu \left(\frac{\partial}{\partial t} + \vec{v}_\nu \cdot \nabla \right) \vec{v}_\nu = q_\nu n_\nu (\vec{E} + \vec{v}_\nu \times \vec{B}) - \text{div} \vec{\pi}_\nu + \vec{M}_\nu - m_\nu \vec{v}_\nu P_\nu \quad (21)$$

where

$$\vec{M}_\nu = \int \int_{-\infty}^{+\infty} m_\nu \vec{w}_\nu \left(\frac{\partial f}{\partial t} \right)_\nu dw_x dw_y dw_z \quad (22)$$

is the net gain of momentum by collisions. Finally, by combining Eqs (19) and (21) with that obtained from the third moment, the heat balance equation results which has the form

$$\begin{aligned} \frac{\partial}{\partial t} (n_\nu U_\nu) + \text{div} (n_\nu U_\nu \vec{v}_\nu) + \text{div} \vec{q}_\nu + \vec{\pi}_\nu : \nabla \vec{v}_\nu \\ = E_\nu - \vec{v}_\nu \cdot \vec{M}_\nu + \frac{1}{2} m_\nu v_\nu^2 P_\nu = H_\nu \end{aligned} \quad (23)$$

where $U_\nu = \sum_k U_{\nu k}$ is the total thermal energy and

$$E_\nu = \int \int_{-\infty}^{+\infty} \int \frac{1}{2} m_\nu w_\nu^2 \left(\frac{\partial f}{\partial t} \right)_\nu dw_x dw_y dw_z \quad (24)$$

$$H_\nu = \int \int_{-\infty}^{+\infty} \int \frac{1}{2} m_\nu (\vec{w}_\nu - \vec{v}_\nu)^2 \left(\frac{\partial f}{\partial t} \right)_\nu dw_x dw_y dw_z \quad (25)$$

are the net gain of total energy and heat by collisions, respectively. Collisions leading to energy losses by radiation are also included in H_ν .

The form of the pressure tensor $\vec{\pi}_\nu$ depends on the particular problem to be discussed. In collisionless anisotropic plasmas $\vec{\pi}_\nu$ can in the first approximation be expressed in terms of the scalar pressures $p_{\parallel\nu}$ and $p_{\perp\nu}$ as shown by Chew et al. [73] and Chandrasekhar et al. [74]. More accurate forms of this tensor can be obtained by including finite Larmor radius effects as given by Marshall [75], Thompson [76], Roberts and Taylor [77], and especially for cylindrical systems by Haines [78] and Berge [79, 80]. Further, the effects of viscosity have been included in the pressure tensor by Chapman and Cowling [81], Marshall [75], Kaufman [82], Spitzer [83], and Thompson [84]. In particular, when the pressure tensor is mainly due to a scalar pressure p_ν , it is convenient to split it into two parts defined by

$$\text{div} \vec{\pi}_\nu = \nabla p_\nu + \text{div} \vec{\lambda}_\nu \quad (26)$$

which also implies that

$$\vec{\pi}_\nu : \nabla \vec{v}_\nu = p_\nu \text{div} \vec{v}_\nu + \vec{\lambda}_\nu : \nabla \vec{v}_\nu \quad (27)$$

We finally notice that also the heat flow vector $\vec{q}_\nu$ will have a form depending on the magnetic field and the collision processes involved in a magnetized plasma. Expressions for $\vec{q}_\nu$ have been given by Spitzer [83], Thompson [84], and Braginskij [85].

2.3.2. The plasma fluid equations

The treatment is now restricted to a dense quasi-neutral plasma in a strong magnetic field, where $n_i \approx n_e \equiv n$ and the space charge density σ is small compared to $q_\nu n_\nu$, the Larmor radii a_i and a_e of ions and electrons and the Debye distance are much smaller than the macroscopic dimensions, the gyrofrequencies $\omega_i = eB/m_i$ and $\omega_e = eB/m_e$ of ions and electrons are much larger than the characteristic frequencies of the macroscopic time variations, and where the velocity $\vec{v}_n$ and temperature T_n of the neutral gas are negligible compared to the corresponding velocities and temperatures of ions and electrons. We assume only one kind of positively and negatively charged particles and neutrals to be present, and that the mass m_n of a neutral particle is comparable to m_i . For the various collision processes which are discussed in more detail elsewhere [72] we introduce ξ for the net frequency at which charged particles are produced by ionization and which sometimes has to be modified by recombination processes at low plasma temperatures. We further use ν_{ei} , ν_{en} and ν_{in} for the frequencies of momentum transfer by electron-ion, electron-neutral and ion-neutral collisions, ϵ_{in} for the fraction of the heating power from the ion-neutral drag force which is shared by the ions, R for the radiation losses except those by excitation, $f_{in} \nu_{in}$ for the effective frequency of heat transfer between ions and neutrals by elastic and charge exchange collisions, $f_{en} \nu_{en}$ for the corresponding transfer between electrons and neutrals including both elastic collisions and those leading to excitation and other inelastic processes except dissociation and ionization, and W for the total dissociation and ionization work.

We now define the centre-of-mass velocity $\vec{v} = (m_i \vec{v}_i + m_e \vec{v}_e)/m$ and current density $\vec{j}$ of the plasma and introduce the notations $m = m_i + m_e$, $\vec{\pi} = \vec{\pi}_i + \vec{\pi}_e$, $U = U_i + U_e$, $\vec{q} = \vec{q}_i + \vec{q}_e$, $d/dt = \partial/\partial t + \vec{v} \cdot \nabla$, $\nu_i = \nu_{in} + \xi$, $\nu_e = \nu_{en} + \xi$, $\alpha = (m_i \nu_i + m_e \nu_e)/m$, $\gamma = m_i m_e (\nu_e - \nu_i)/em$, and $\eta = m_e (\nu_{ei} + \nu_e + m_e \nu_i/m)/e^2 n$. From Eqs (19), (21), and (23) we then obtain the corresponding balance equations for a one-fluid model of the plasma immersed in a background of neutral gas. The balance of matter is given by

$$\frac{\partial n}{\partial t} + \text{div} (n \vec{v}) = n \xi \equiv n n_n \xi \quad (28)$$

$$\text{div} \vec{j} = - \frac{\partial \sigma}{\partial t} \quad (29)$$

the balance of momentum at negligible electrostatic forces $\sigma \vec{E}$ by

$$\begin{aligned} nm \frac{d\vec{v}}{dt} + \frac{m_i m_e}{e^2 m} (\vec{j} \cdot \nabla) (\vec{j}/n) \\ = \vec{j} \times \vec{B} - \text{div} \vec{\pi} - nm \alpha \vec{v} + \gamma \vec{v} \end{aligned} \quad (30)$$

$$\begin{aligned}
& \frac{m_i m_e}{e^2 m} \frac{d}{dt} (\vec{j}/n) \\
& + \frac{m_i m_e}{e^2 n m} (\vec{j} \cdot \nabla) \vec{v} - \frac{m_i m_e (m_i - m_e)}{e^3 n m^2} (\vec{j} \cdot \nabla) (\vec{j}/n) \\
& = \vec{E} + \vec{v} \times \vec{B} - \frac{m_i - m_e}{e n m} \vec{j} \times \vec{B} \\
& + (m_i \operatorname{div} \vec{\pi}_e - m_e \operatorname{div} \vec{\pi}_i)/e n m + \gamma \vec{v} - \eta \vec{j}
\end{aligned} \quad (31)$$

and the total heat balance of the plasma considered as one entity by

$$\begin{aligned}
& \frac{\partial}{\partial t} (nU) + \operatorname{div} (nU \vec{v}) - \operatorname{div} [(m_i U_e - m_e U_i) \vec{j}/e n] \\
& + \vec{\pi} : \nabla \vec{v} - (m_i \vec{\pi}_e - m_e \vec{\pi}_i) : \nabla (\vec{j}/e n m) \\
& + \operatorname{div} \vec{q} + R = m_e \left(\nu_{ei} + \nu_{en} \right) \\
& + \epsilon_{in} m_e \nu_{in}/m + \frac{1}{2} \xi \vec{j}^2/e n \\
& + n m_i \left(\epsilon_{in} \nu_{in} + m_e \nu_{en}/m_i + \frac{1}{2} \xi \right) \vec{v}^2 \\
& + 2 m_e (\epsilon_{in} \nu_{in} - \nu_{en}) \vec{v} \cdot \vec{j}/e \\
& - n W \xi - f_{in} \nu_{in} n U_i - f_{en} \nu_{en} n U_e
\end{aligned} \quad (32)$$

Especially for a rotating plasma of high density in which the macroscopic relaxation times are large compared to the collision times, the pressure tensor becomes nearly isotropic and is dominated by the scalar pressures p_i and p_e . We can then introduce the isotropic ion and electron temperatures T_i and T_e and write $U_i = \frac{3}{2} k T_i$, $U_e = \frac{3}{2} k T_e$ and $p = p_i + p_e = nk(T_i + T_e) = \frac{2}{3} nU$ with good approximation. It is then also convenient to use expressions (26) and (27) in the balance equations (30) - (32), and to write $\operatorname{div} \vec{\pi} = \nabla p + \operatorname{div} \vec{\lambda}$ and $\vec{\pi} : \nabla \vec{v} = p \operatorname{div} \vec{v} + \vec{\lambda} : \nabla \vec{v}$.

2.3.3. Estimations and approximations

For the coefficients involved in the basic Eqs (28) - (32) we introduce the expressions [83, 84, 86, 87]

$$\nu_{ei} = 0.91 \times 10^{-6} n Z^2 (\ln \Lambda) / T_e^{3/2} \quad (33)$$

$$\nu_{ii} = 1.20 \times 10^{-8} n Z^4 (\ln \Lambda) / A^{1/2} T_i^{3/2} \quad (34)$$

$$\begin{aligned}
t_{eq} &= 1/\nu_{eq} \\
&= 2.54 \times 10^8 T_e^{3/2} A (1 + 5.5 \times 10^{-4} T_i / A T_e)^{3/2} \\
&/ n Z^2 (\ln \Lambda) \approx 1/f_{ei} \nu_{ei}
\end{aligned} \quad (35)$$

$$\eta_{||} = 65.3 (\ln \Lambda) / T_e^{3/2} \quad \eta_{\perp} = 129 (\ln \Lambda) / T_e^{3/2} \quad (36)$$

$$\begin{aligned}
\mu_{||} &= nk T_i / 3 \nu_{ii} \\
\mu^* &= 3 nk T_i \nu_{ii} / 16 \omega_i^2 = k_1 n^2 A^{3/2} (\ln \Lambda) / B^2 T_i^{1/2}
\end{aligned} \quad (37)$$

$$\begin{aligned}
\lambda_{||} &= nk^2 T / m_e \nu_{ei} \\
\lambda^* &= 1.25 nk^2 T_i \nu_{ii} / m_i \omega_i^2 = k_2 n^2 A^{1/2} (\ln \Lambda) / B^2 T_i^{1/2}
\end{aligned} \quad (38)$$

$$R_b = 1.7 \times 10^{-40} n^2 Z^2 T_e^{1/2} \quad (39)$$

$$\xi = 820 a_z k (2e/\pi m_e)^{1/2} (W/e)^{3/2} \chi^{3/2} (1 + 1/2\chi) \exp(-1/\chi) \quad (40)$$

where MKSA units are used, $k_1 = 3.39 \times 10^{-48}$, $k_2 = 1.87 \times 10^{-43}$ for the values given by Thompson [84], and $k_1 = 2.71 \times 10^{-47}$, $k_2 = 1.50 \times 10^{-42}$ for those by Spitzer [83], A and Z are the mass and charge numbers, and Λ is the ratio between the Debye distance and the impact parameter. Further ν_{ii} denotes the ion-ion collision frequency, t_{eq} is the equipartition time for thermal equilibrium between ions and electrons, $\eta_{||}$ and $\eta_{\perp}$ are the resistivity, $\mu_{||}$ and μ^* the viscosity, $\lambda_{||}$ and λ^* the heat conductivity parallel and perpendicular to a strong magnetic field, R_b is the bremsstrahlung radiation loss, and ξ is the rate of thermal ionization by electrons, with $\chi = k T_e / W$ and a_z being a constant with a numerical value equal to that given by von Engel and Steenbeck [86]. The rate expressed by Eq. (40) has under certain conditions to be replaced by a more relevant formula which also includes ionization by ions and neutrals [88, 89] as well as recombination processes. With the exception of k_1 and k_2 there are variations by a factor of about 2 in the values of the coefficients of Eqs (33) - (40) as given by different authors.

When Eqs (28) - (32) and expressions (33) - (40) are applied to rotating plasmas at densities and magnetic field strengths being roughly of the order of $n = 10^{21} \text{ m}^{-3}$ and $B = 1 \text{ V} \cdot \text{s} \cdot \text{m}^{-2}$, the following general statements can be made [72]:

(a) If the ion temperature is anisotropic and has the values $T_{i||}$ and $T_{i\perp}$ parallel and perpendicular to the magnetic field, ion-ion collisions will tend to equalize $T_{i||}$ and $T_{i\perp}$. When the plasma is heated by viscous shear forces due to a velocity gradient $|\nabla_{\perp} v_{\perp}| \equiv v_{\perp} / L_v$ perpendicular to the magnetic field, the corresponding power will mainly tend to raise the transverse temperature $T_{i\perp}$. The degree of anisotropy then becomes $|T_{i\perp} - T_{i||}| / (T_{i\perp} + T_{i||}) \approx 3 v_{\perp}^2 / 32 \omega_i^2 L_v^2$ which is usually a small quantity in a rotating plasma. Similar conclusions can be drawn for the electrons, where electron-ion and electron-electron collisions tend to produce isotropy, whereas a power loss due to cyclotron radiation tends to decrease the transverse temperature $T_{e\perp}$ as compared to the longitudinal $T_{e||}$. As long

as this power loss is less than the bremsstrahlung power R_b of Eq. (39), the anisotropy of the electron gas should become quite small. However, there are also systems with small relaxation times where the deviation from isotropy may not become negligible. This is the case when there are large particle losses along the magnetic field and a strong back-scatter of neutral particles from the end walls. Then, charge-exchange collisions produce a large group of ions moving in cycloidal orbits and $T_{i\perp}$ becomes greater than $T_{i\parallel}$ [88, 90].

(b) In a clean fully ionized hydrogen plasma the main radiation losses are by electron bremsstrahlung and cyclotron radiation. In cases where the former losses dominate, the temperature difference $T_i - T_e$ between the ion and electron gases will be determined by the balance between radiation and transfer of heat from ions to electrons by collisions, as given by $|T_i - T_e|/T_e = 8 \times 10^{-9} A T_e / (\ln \Lambda)$. For $T_e < 10^6$ K this ratio is less than 10^{-3} . The electron temperature will only decrease noticeably below the ion temperature when the radiation from impurities, recombination and cyclotron motions strongly exceeds that from hydrogen bremsstrahlung.

(c) The ratio between the longitudinal and transverse heat conductivities in a strong magnetic field is $\lambda_{\parallel}/\lambda^* = \omega_i \omega_e / \nu_{ei} \nu_{ii}$. This ratio usually becomes a large quantity which implies that in the hot parts of a fully ionized plasma the magnetic field lines will nearly coincide with the isothermal surfaces. The conditions for such a situation to prevail are also connected with the conditions for the magnetic field lines to become equipotentials as discussed earlier in Sections 2.1 and 2.2.2.

(d) The heat loss by thermal conduction across a strong magnetic field is $\lambda^* |\nabla_{\perp} T_i| \equiv \lambda^* T_i / L_{\perp}$. The ratio between the heat losses from bremsstrahlung and thermal conduction is small in most rotating plasma experiments where $(BL_{\perp})^2 \ll 1$, but not necessarily in a full-scale fusion device.

(e) In cases where volume recombination is important, the right-hand member of Eq. (28) has to be modified to include a term $-n^2 \rho$, where ρ is the corresponding recombination coefficient, and $t_{\rho} = 1/\rho$ the time constant. At electron temperatures above some 10^4 K the rate of volume recombination becomes quite small in a fully ionized plasma [87]. In a partially ionized plasma this rate is speeded up by electron attachment which produces negative ions [86, 91].

(f) Under most conditions of practical interest the differential velocity $|\vec{v}_i - \vec{v}_e|$ does not exceed the centre-of-mass velocity $|\vec{v}|$ by orders of magnitude. When $\omega_i t_c \gg 1$ and $\omega_e t_c \gg 1$, with t_c being a characteristic value of the macroscopic time variations, it is then seen that the terms containing $\vec{j} \cdot \nabla$ and $\vec{j}/n$ in Eqs (30) - (32) can be neglected with good approximation. This also applies to the terms containing γ in Eqs (30) and (31), as well as to the term containing $\vec{v} \cdot \vec{j}$ in Eq. (32) and to those parts of the $\vec{j}^2$ term of the same equation not containing the contribution from ν_{ei} .

(g) In the case of negligible dissipation and neutral gas interaction, of moderately large acce-

leration forces $n m d\vec{v}/dt$, and of a nearly isotropic plasma with $\vec{\pi}_i \approx \vec{\pi}_e$, scalar multiplication of Eqs (30) and (31) by $\vec{B}$ yields the result $E_{\parallel} \ll E_{\perp}$.

Consequently, when statements (a) - (g) are valid, Eqs (28) - (32) can be simplified and restricted to a nearly isotropic plasma with scalar ion and electron temperatures T_i and T_e . In the fully ionized parts of the plasma we have $T_i \approx T_e \equiv T$, volume recombination is neglected, and the magnetic field lines are nearly equipotentials and isothermal surfaces at the same time. In the partially ionized regions deviations may arise from isotropy and between T_i and T_e , radiation losses become important, and volume recombination has sometimes to be taken into account. The relevant terms of the momentum and heat balance which we are going to consider here have been underlined in Eqs (30) - (32).

2.4. PARTIALLY IONIZED REGIONS AND BOUNDARY LAYERS

When a fully ionized plasma is created within a limited part of a neutral gas cloud, when ions and electrons escape from the plasma and recombine at surrounding vessel walls, or when neutral gas is released from such walls by plasma interaction, there arise partially ionized boundary regions which separate the plasma from the neutral gas and the vessel walls. These regions play an important role in the boundary conditions of the system, and we shall discuss them in this section before turning to the balance of the main plasma body.

Under the present conditions a certain minimum power input is required for a fully ionized plasma to be created and sustained, as will be further explained in Sections 2.7.3. and 2.7.5. Here we assume such a power to be available and first consider a simplified one-dimensional stationary model in which a fully ionized static plasma of density n_b occupies the volume $x \geq 0$ in a rectangular frame xyz of reference, and where there is neutral gas of density $n_n = n_{n0}$ in the region $x < 0$. The neutral gas then diffuses into the plasma according to Eqs (21) and (28). Its density decays exponentially in the direction perpendicular to the plasma boundary with the e-folding penetration length [72, 92]

$$L_n = (1/n_b)(kT/m\xi_n\xi)^{1/2} \quad (41)$$

Here the plasma and the neutral gas are assumed to have the same temperature T in the boundary region and $\xi_n = \nu_{in}/n_n$. Usually $\xi \ll \xi_n$ in the cold outer regions of the plasma, and the mean free path λ_{ni} of neutrals then becomes much smaller than L_n . Since L_n is a slow function of T , ξ_n , and ξ , uncertainties in the latter quantities do not affect the result of Eq. (41) to any large extent. At high plasma densities and sufficiently large ionization rates the penetration length L_n becomes quite small. An example of this is given by thermal ionization by electrons. With the rate (40) and at a temperature $T > 10^5$ K we then obtain

$n_b L_n < 4 \times 10^{18} \text{ m}^{-2}$ for hydrogen, which implies that $L_n < 4 \times 10^{-3} \text{ m}$ for $n_b > 10^{21} \text{ m}^{-3}$.

It should also be noticed that neutral particles reaching far enough into a rotating plasma, will be accelerated to the full rotational velocity by collisions with the ions. A strong centrifugal force then pushes them in the radial direction, resulting in a modified and reduced penetration length L_n in the outer boundary region [27].

For a fully ionized plasma communicating with regions occupied by neutral gas, the plasma balance will depend critically on the relative magnitudes of the penetration length L_n and the characteristic macroscopic dimensions L_c of the system. There are in principle two existing possibilities:

(a) When $L_n \ll L_c$ the plasma becomes 'impermeable' in the sense that its body screens itself from penetrating neutral particles. Well-defined boundary regions arise in such a case which separate the plasma from the neutral gas. In the type 'A' systems defined in Section 2.1. and in the wall-near parts of type 'B', 'C', 'D' and 'E' systems, a quasi-stationary balance between the plasma and the neutral gas is established in the boundary regions. In type 'C' systems a non-stationary boundary region arises in the form of an ionizing front and its associated shock layer. In this latter case L_c should be defined as the width of the layer, and $L_n \ll L_c$ is then the condition for making the ionizing front and shock piston impermeable to the neutral gas.

(b) When L_n becomes comparable to or greater than L_c , the plasma may be denoted as 'permeable'. The neutral gas will then be able to penetrate into the interior parts of the plasma body and will strongly affect its balance. The boundary regions become diffuse in such a case.

2.4.1. Wall layers produced by longitudinal particle flow

When a plasma with isotropic temperatures is confined in an open magnetic bottle like that shown in Fig. 1, ions and electrons will be able to escape along the magnetic field lines and recombine at the end walls. Provided that the resulting neutral particles are not absorbed by the walls or lost elsewhere, they are scattered back into the plasma and become reionized within partially ionized layers situated close to the walls and having the thickness d_w . When the times of flight of the particles across the distance d_w are short compared to the macroscopic changes of the system, a quasi-stationary balance establishes itself within these wall layers.

For ions the path of energy exchange with the neutral gas becomes

$$\lambda_{in} = 1/f_{in} \sigma_{in} n_n \quad (42)$$

and for electrons the path λ_e of energy exchange with neutrals and ions is given by

$$\begin{aligned} 1/\lambda_e &= 1/\lambda_{en} + 1/\lambda_{ei}; \\ \lambda_{en} &= 1/f_{en} \sigma_{en} n_n; \quad \lambda_{ei} = \bar{w}_e t_{eq}. \end{aligned} \quad (43)$$

where σ_{in} and σ_{en} are the corresponding cross-sections for collisions with the neutral gas, $\bar{w}_e$ is the average thermal velocity of an electron, and t_{eq} is given by Eq. (35). From a comparison between the paths λ_{in} , λ_e and the thickness d_w we now obtain three possible types of wall layers [72, 92, 93]:

(a) If at high plasma temperatures $d_w \lesssim \lambda_{in} < \lambda_e$ a 'thin' wall layer is formed in which both ions and electrons impinge on the end insulators, nearly at the thermal velocity of the hot fully ionized plasma. There is free streaming for most of the ions, electrons and neutrals through the layer which has the thickness

$$\begin{aligned} d_w &\approx (k_w/n_{nw} \xi_w) (kT/m_i)^{1/2} \\ &\approx (f_n k_t/n_w \xi_w) (kT_w/m_n)^{1/2} \end{aligned} \quad (44)$$

Here ξ_w is the average ionization rate within the layer, k_w and k_t are dimensionless constants of order unity, n_{nw} is the mean density of neutrals in the layer, n_w is the plasma density at the edge of the Debye sheath, T and T_w are the temperatures of the plasma and the end wall, and it has been assumed that f_n incoming ions give rise to one back-scattered neutral gas molecule. For a non-conducting wall we have $k_w \approx 1 + 1/\sqrt{2\pi}$, where the unity term is due to electrons which charge the wall negatively and accelerate the ions through the Debye sheath, and the last term stands for the thermal ion flux [94]. Similarly $k_t \approx 1/\sqrt{2\pi}$ stands for the thermal counter-flux of neutrals. The thickness d_w decreases very steeply with increasing plasma temperature T and easily becomes much smaller than the characteristic dimensions L_c of a fully ionized high-density plasma. An example of this is given by Eq. (44) for thermal ionization by electrons in hydrogen at plasma temperatures $T \geq 10^5 \text{ °K}$ and a wall temperature $T_w = 300 \text{ °K}$. This yields $n_w d_w \leq 8 \times 10^{18} \text{ m}^{-2}$ and $d_w \leq 8 \times 10^{-3} \text{ m}$ for $n_w = 10^{21} \text{ m}^{-3}$. Even in cases where the mean free path λ_{in} of Eq. (42) somewhat exceeds the layer thickness d_w of Eq. (44) and there are one or two mean free paths within the distance d_w , the thin wall layer model should become a rather good approximation of the real physical situation.

(b) At decreasing temperatures a 'thick' wall layer soon develops for which $\lambda_{in} < d_w$ and ions diffuse towards the insulator wall, thereby losing part of their energy to the neutral gas by collisions. As long as $d_w < \lambda_e$, the electrons still remain hot and are able to ionize the neutral gas within the entire layer. Detailed deductions of the layer thickness in this case are given elsewhere [72, 93].

(c) When the plasma temperature decreases far enough for λ_e to become smaller than d_w , most of the electrons within the layer will be cooled to a

low temperature by collisions with the neutral gas. Then, they will no longer contribute to the ion production, and the ionization balance in the layer breaks down.

2.4.2. Boundary regions produced by the transverse particle flow

Under experimental conditions to be further discussed in Section 2.5.1. the balance of matter in the direction across the magnetic field often becomes decoupled from the longitudinal balance described in the previous section. In a stable quasi-stationary state, the outflux of plasma by transverse diffusion across the magnetic field is then balanced by an influx of neutral gas from a surrounding neutral gas blanket. The corresponding partially ionized boundary regions are shown in Fig. 1, having the thickness x_b . These regions consist of an outer part of thickness $x_b - L_n$ in which the ion production can be neglected and only diffusion takes place, and an inner part of thickness L_n , where the incoming flux of neutral particles is absorbed by ionization. The density $n_0 \equiv n_B N_0$ of the fully ionized plasma and the density of the surrounding neutral gas blanket $n_{n0} = n_B N_{n0}$ are connected by a general relationship between the dimensionless functions N_0 and N_{n0} and by the density

$$n_B = B / (\eta_{\perp} m_i \sigma_{in} \bar{w}_i)^{1/2} \quad (45)$$

associated with the magnetic field [92, 95]. In Eq. (45) $\bar{w}_i$ is the average thermal velocity of an ion. A similar result has earlier been obtained by Wienecke [96] by means of a computer. The present model does not change its general features in presence of photoionization in the neutral gas blanket, as long as the latter remains partially ionized.

A boundary region of the described type can only be formed when the penetration length L_n of Eq. (41) is smaller than the dimensions of the fully ionized plasma body. Further, for the values of x_b which correspond to a stable heat balance there exist certain conditions on the flow of heat from the plasma into the boundary region, as well as on the heat losses due to conduction and radiation [92].

The time required to establish a quasi-stationary state in the boundary region is of the order of $100 \mu s$ under the conditions of a number of rotating plasma experiments [92]. This implies that the boundary region can only be considered as stationary in laboratory experiments for certain systems of types 'A' and 'E', and hardly for any of the systems of types 'B', 'C', and 'D' described in Section 2.1. On the other hand, the time-scale for the development of the wall layers described in the previous section is usually shorter than $1 \mu s$, and such layers should therefore become quasi-stationary in most systems.

In this connection should finally be mentioned that the partially ionized transition region between a plasma body of limited size and a surrounding neutral gas blanket may act as an efficient separation

system for isotopes and elements. Since x_b is likely to be much smaller than the characteristic dimensions L_c of cosmical systems, this mechanism may become more pronounced in cosmos than the separation effect by the centrifugal force discussed later in Section 2.8. The present diffusion process provides a possible explanation of the observed abundance anomalies of some rare elements in magnetized stars and of helium in the solar wind during active events [56-58, 97].

2.4.3. Ionizing fronts

In the previous sections we have treated two examples of quasi-stationary boundary regions separating a plasma from a neutral gas. We shall now proceed to the non-stationary case represented by ionizing fronts. These can be considered as a kind of hydromagnetic shock waves, where neutral gas becomes ionized by passing through a shock layer.

A theory on hydromagnetic ionizing waves has earlier been developed by Kunkel [98], and Kunkel and Gross [99] under the following assumptions:

(a) A plane geometry is considered in which uniform crossed electric and magnetic fields $\vec{E}$ and $\vec{B}_0$ are applied.

(b) The gas in front of the shock is neutral, at rest, non-conducting, and has the mass density $\rho_{n0} = m_n n_{n0}$.

(c) The electric conductivity and viscous dissipation can be neglected at the boundaries of the flow.

(d) The rarefaction wave following the shock remains attached to the front.

(e) A transverse magnetic field component b_1 becomes associated with the wave and should be much smaller than the applied steady magnetic field B_0 .

(f) The gas behind the shock is essentially a fully ionized plasma having the mass density $\rho = mn$. The total energy of dissociation and ionization per unit mass of gas is $e_0 = W/m$.

With these assumptions Eqs (28) - (32) and Maxwell's equations can be combined to the corresponding equations for the conservation of matter, momentum and heat for the gas passing through the shock front. This yields a front velocity U_f determined by

$$U_f^2 = \frac{16}{9} e_0 \left[-1 + \left(1 + 9 B_0^2 b_1^2 / 16 \mu_0 \rho_{n0}^2 e_0^2 \right)^{1/2} \right] \quad (46)$$

and a plasma density behind the shock

$$\rho = \frac{8}{5} \rho_{n0} \left(1 - b_1^2 / 2 \mu_0 \rho_{n0} U_f^2 \right)^{-1} \quad (47)$$

where $\mu_0 = 4\pi \times 10^{-7}$ is the magnetic permeability in vacuo.

The corresponding cylindrical case of ionizing fronts in rotating plasma devices of type 'C' has been treated by Millar and Watson-Munro [100].

These authors have adopted conditions (b) - (e) assumed by Kunkel and Gross, but allow for partial ionization behind the shock front. They further assume that the neutrals which have escaped ionization are tightly coupled to the ions, because of the high ion-neutral collision frequency in actual experiments, and should then acquire the same momentum and energy as the ions. When the azimuthal field component b_ϕ of the wave is small compared to B_0 and the plasma pressure can be neglected compared to the energy densities associated with the macroscopic fluid motion and with the ionization work, the front velocity becomes

$$U_f \approx B_0 b_\phi / \mu_0 (2e_0 \rho \rho_{n0})^{1/2} \quad (48)$$

where ρ stands for the ion mass density behind the front. In the limit of small fields b_1 Eq.(46) yields a similar expression for U_f , but ρ_{n0}^2 is replaced by $\rho \rho_{n0}$ in Eq.(48). Further, conservation of the azimuthal momentum and energy yields

$$\begin{aligned} (\rho / \rho_{n0})^{1/2} &\approx v_\phi / (2e_0)^{1/2} \\ &\approx \phi_{rz} / B_0 (b-a) (2e_0)^{1/2} \end{aligned} \quad (49)$$

where v_ϕ is the azimuthal fluid velocity of the rotating plasma, r_{01} and r_{02} are the radii of the inner and outer concentric electrodes at the end of a device of the type shown in Fig. 2(b), and ϕ_{rz} is the corresponding applied voltage. Finally, the voltage ϕ_{rz} and the total driving current J_{rz} become related by

$$U_f \approx B_0^2 J_{rz} (\ln r_{02} / r_{01}) / 2\pi \rho_{n0} \phi_{rz} \quad (50)$$

In concluding this section it should be observed that the described results on ionizing fronts may have to be modified in certain situations where the penetration length of the neutral gas does not fulfil the conditions $L_n \ll L_c$ and the plasma becomes more or less permeable to the neutral gas. Under such circumstances there will arise additional momentum and energy losses due to a recirculation of neutrals which easily move from the neutral gas regions and the vessel walls into the central parts of the plasma, and which are rapidly forced out again from the plasma by collisions with the ions and by the action of the associated centrifugal force.

2.5. THE PLASMA BALANCE EQUATIONS IN THE AXISYMMETRIC CASE

We now turn to the balance of a fully ionized high-density plasma confined in a poloidal field, being subject both to losses along and across the magnetic field lines, and interacting with the partially ionized boundary regions treated in Section 2.4. In this section mean values will be

taken over the plasma body to give rough estimates of the total plasma balance. This should lead to a correct dependence on the parameters involved, such as the magnetic field strength, density, temperature, and linear dimensions, at the same time as it only yields the order of magnitude of the associated numerical coefficients. For this purpose we shall use $c_0, c_1, \dots$ to indicate dimensionless constants of order unity, including the uncertainties in the numerical coefficients of the present mean value formations. We further adopt the notations of Fig. 1, introduce subscripts (0), (w), and (b) to indicate quantities inside the plasma in the equatorial plane, at the end walls, and in the boundary regions, respectively. A bar denotes mean value formation over the plasma body, and the effective longitudinal length is $L_{||} = V / \pi (r_{02}^2 - r_{01}^2)$, where V is the volume of the plasma body. We finally restrict ourselves to configurations with conical end walls as given by Fig. 1. A discussion of devices with spoke-shaped rods being situated inside the confinement volume in planes through the axis of symmetry is given elsewhere [72, 93]. Only the thin wall layer case of Section 2.4.1.(a) will be treated explicitly here, and for other types of layers reference is made to earlier investigations [72, 93].

The deductions of the present section mainly apply to dense fully ionized plasmas in type 'A' and 'B' systems where the magnetic field lines nearly coincide with the equipotential and isothermal surfaces, and where $L_n \ll L_c$ such as to screen the main plasma body from neutral gas interaction. Further, on account of the viscosity (37) and finite heat conductivity (38) involved in the transverse components of the pressure tensor in Eq. (30) and the heat flow vector of Eq. (32), the azimuthal velocity of rotation $v_{\phi 0}$ and the temperature T in the midplane of Fig. 1 will have maximum values v_0 and T_0 somewhere inside the plasma body and will decrease to low values close to the boundary regions. This is the case when the plasma is coupled with the electrodes by an efficient line-tying mechanism. We then write

$$v_{\phi 0} / v_0 \equiv G \leq 1 \quad T / T_0 \equiv F^2 \leq 1 \quad (51)$$

2.5.1. Particle balance

The particle balance is determined by the outflux of plasma along and across the magnetic field, and by the influx of reionized neutrals from the layers at the end walls and from the exterior of the confinement region through the boundary regions of Fig. 1. When the macroscopic changes of the main plasma body are slow compared to the build-up times of the layers and boundary regions, the latter will develop as described in Sections 2.4.1. and 2.4.2. In a general case a complicated recycling process of neutral gas may take place. However, under experimental conditions the thickness d_w of the wall layers usually becomes very small. On account of this, recycling of neutrals

cannot take place in a way to connect the wall layers with the external regions, and the longitudinal and transverse particle balance of the plasma are then effectively separated [72, 93].

In the first approximation combination of Eqs (30) and (31) yields a diffusion velocity of the plasma across the field lines

$$\vec{v}_{pm} = \vec{v}_p + \vec{v}_m = -(\eta/B^2) \nabla_{\perp} p - (nm\eta/B^2)(d\vec{v}/dt)_{\perp} \quad (52)$$

where the first term is due to the transverse pressure gradient and the second to the centrifugal force. We write v_{pm} and v_p for the moduli of $\vec{v}_{pm}$ and $\vec{v}_p$, $v_{pm} = K_p v_p$ and

$$K_p = \delta_K [1 + \delta_p m v_0^2 (r_{02} - r_{02}) / 2kT_0(r_{02} + r_{02})] \quad (53)$$

with δ_K and δ_p being dimensionless coefficients in the range between 0 and 1.

Equation (28) is now integrated over the entire plasma volume V , including the narrow partially ionized wall layers and boundary regions. After division by $n_0 V$, where n_0 is the average plasma density in the equatorial plane, the result becomes

$$\frac{1}{n_0} \cdot \frac{\partial n_0}{\partial t} = -c_1/t - c_2/t_{\perp} + c_3/t_w + c_4/t_b \equiv -1/t_n \quad (54)$$

where

$$1/t_n = 2k_w(\overline{B_0/B_w}) \overline{C_w}(kT_0/m_i)^{1/2}/L_{\parallel}; \quad C_w = n_w/n_0 \quad (55)$$

$$1/t_{\perp} = 8K_p \eta_{10} n_0 kT_0 / B_0^2 (r_{02} - r_{02})^2 \quad (56)$$

$$1/t_w = 2(\overline{B_0/B_w}) \overline{C_w} n_{nw} \xi_w d_w / L_{\parallel} \quad (57)$$

$$1/t_b = 2n_{n0} \xi_b x_b / (r_{02} - r_{02}) \quad (58)$$

In the second member of Eq. (54) the first two terms represent the longitudinal and transverse particle losses, and the last two terms the reionization of neutrals in the wall layers and boundary regions, respectively. The coefficient $C_w \leq 1$, where the inequality sign represents the effect of centrifugal confinement in a rotating plasma.

When the recycling of neutrals is restricted as described at the beginning of this section and a neutral gas density n_{n0} is maintained in the surrounding blanket, the particle balance will essentially be divided into a longitudinal part given by the first and third terms of the second member of Eq. (54) and a transverse part given by the second and fourth terms of the same member:

(a) Both ions and neutrals mainly perform free streaming through the wall layers. Since d_w is very small compared to the macroscopic dimensions, an individual ion will not get lost from the

system after a single passage through the wall layer. Instead it carries out a kind of random walk in the transverse magnetic field direction, each time when it passes the layer, thereby spending a short part of its time in the layer as a back-scattered neutral.

(b) At high magnetic field strengths the transverse plasma loss by diffusion becomes strongly reduced according to Eq. (56). It is easily compensated by a counter-stream of reionized neutrals which penetrate into the boundary regions as described in Section 2.4.2. and represented by expression (58).

(c) Under these conditions the terms in Eq. (54) which contain $t_{\parallel}$ and t_w will almost cancel each other separately, as well as the terms containing $t_{\perp}$ and t_b . The characteristic time-scale t_n of the density changes of the plasma body then becomes much longer than each of the relaxation times determined by Eqs (55) - (58).

2.5.2. Momentum balance

We now turn to the local momentum balance expressed by the underlined terms of Eqs (30) and (31), under the assumption of nearly isotropic conditions and equal ion and electron temperatures. We further assume that in a cylindrical frame of reference $r\phi z$ the azimuthal fluid velocity component v_{ϕ} is much larger than the components v_r and v_z associated with the macroscopic fluid motions in the planes through the axis of symmetry. With $\hat{r} = (1, 0, 0)$ and $\hat{\phi} = (0, 1, 0)$, Eqs (30) and (31) then reduce to

$$\frac{\partial}{\partial t} (nm\vec{v}) = \vec{j} \times \vec{B} - \nabla p - nm\alpha\vec{v} - nm(\nu_{\mu} + \nu_{wb})v_{\phi}\hat{\phi} \quad (59)$$

$$+ g_c \hat{r} - nm[(\vec{v} \cdot \nabla)\vec{v}]_{rz} + m\vec{v} \frac{\partial n}{\partial t}$$

$$\eta \vec{j} = \vec{E} + \vec{v} \times \vec{B} - \frac{m_i - m_e}{enm} \vec{j} \times \vec{B} + \frac{m_i - m_e}{2enm} \nabla p \quad (60)$$

where α is the effective frequency of collisions between the plasma and neutral particles being located inside the main plasma body,

$$\nu_{\mu} = (1/nm v_{\phi}) |\nabla_{\perp} (\mu^* \nabla_{\perp} v_{\phi})| \quad (61)$$

represents an equivalent collision frequently arising from transverse gradients in the velocity field v_{ϕ} and from the plasma viscosity (37),

$$\nu_{wb} = (1/v_{\phi}) \left| \left(v_r \frac{\partial}{\partial r} + v_z \frac{\partial}{\partial z} \right) v_{\phi} \right| \quad (62)$$

can be considered as an equivalent collision frequency produced by convective momentum losses of plasma in the rz planes,

$$g_c = nm v_{\phi}^2 / r \quad (63)$$

is the centrifugal force, and $[(\vec{v} \cdot \nabla) \vec{v}]_{rz}$ stands for the part of $(\vec{v} \cdot \nabla) \vec{v}$ situated in the rz planes. Further

$$\text{curl} \vec{B}/\mu_0 = \vec{j} + \epsilon_0 \partial \vec{E}/\partial t \quad (64)$$

where ϵ_0 is the dielectric constant in vacuo.

We now study the motion of the plasma perpendicular to the magnetic field lines in the special case of a time-independent density n , and return later at the end of this section to the motions along $\vec{B}$. After vector multiplication of Eqs (59) and (60) by $\vec{B}$, the resulting system of equations can be divided into two sets. The first set corresponds to the electric field $\vec{E}_{rz}$ and current density $\vec{j}_{rz}$ in the rz planes and becomes with $\nu = \alpha + \nu_\mu + \nu_{wb}$

$$\vec{j}_{rz} = nm\vec{B} \times \left(\frac{\partial \vec{v}_{rz}}{\partial t} \right) / B^2 \quad (65)$$

$$+ nm\nu\vec{B} \times \vec{v}_{rz} / B^2 \equiv \vec{j}_{\epsilon rz} + \vec{j}_{\nu rz}$$

$$\eta \vec{j}_{rz} = \vec{E}_{rz} - \vec{E}_{orz}$$

$$+ \vec{v}_\varphi \times \vec{B} - \frac{m_i - m_e}{enm} \vec{j}_\varphi \times \vec{B} + \frac{m_i - m_e}{2enm} \nabla_{rz} p$$

$$\equiv \vec{E}_{rz} - \vec{E}_{orz} - \vec{E}_{\epsilon rz} - \vec{E}_{H\varphi} - \vec{E}_{prz} ;$$

$$\vec{E}_{H\varphi} = \frac{m_i - m_e}{enm\eta} (\vec{E}_\varphi - \vec{E}_{\epsilon\varphi}) \times \vec{B} + \left(\frac{m_i - m_e}{enm} \right)^2 \frac{B^2}{\eta} \vec{j}_{rz} \quad (66)$$

$$(\text{curl} \vec{B}/\mu_0)_{rz} = \vec{j}_{rz} + \epsilon_0 \partial \vec{E}_{rz}/\partial t \quad (67)$$

The second set corresponds to the azimuthal electric field $\vec{E}_\varphi$ and current density $\vec{j}_\varphi$ and becomes

$$\vec{j}_\varphi = nm\vec{B} \times \left(\frac{\partial \vec{v}_{rz}}{\partial t} \right) / B^2 + nm\alpha\vec{B} \times \vec{v}_{rz} / B^2 - \vec{B} \times \left\{ \nabla_{rz} p - g_c \hat{r} + nm [(\vec{v} \cdot \nabla) \vec{v}]_{rz} \right\} / B^2 \quad (68)$$

$$\equiv \vec{j}_{\epsilon\varphi} + \vec{j}_{\alpha\varphi} + \vec{j}_{pc\varphi}$$

$$\eta \vec{j}_\varphi = \vec{E}_\varphi + \vec{v}_{rz} \times \vec{B} - \frac{m_i - m_e}{enm} \vec{j}_{rz} \times \vec{B} \quad (69)$$

$$\equiv \vec{E}_\varphi - \vec{E}_{\epsilon\varphi} - \vec{E}_{Hrz}$$

$$(\text{curl} \vec{B}/\mu_0)_\varphi = \vec{j}_\varphi + \epsilon_0 \partial \vec{E}_\varphi / \partial t \quad (70)$$

These equations correspond to the equivalent circuit of Fig. 5, where $\vec{E}_{rz}$ is the electric field representing the applied voltage in the rz direction and $\vec{E}_\varphi$ is the corresponding induced voltage in the φ direction. The field $\vec{E}_{orz}$ is due to the effects of space charges and electric sheath formation. Further, $\epsilon = nm/B^2$ is the dielectric constant of the plasma due to polarization effects by ion inertia, and $\vec{E}_{\epsilon rz}$ and $\vec{E}_{\epsilon\varphi}$ are the corresponding electric fields which represent the charging of the equivalent capacities associated with ϵ and the currents $\vec{j}_{\epsilon rz}$ and $\vec{j}_{\epsilon\varphi}$. The momentum losses due to 'frictional' forces in the φ direction correspond to the leakage currents $\vec{j}_{\alpha rz}$, $\vec{j}_{\mu rz}$ and $\vec{j}_{wb rz}$, whereas

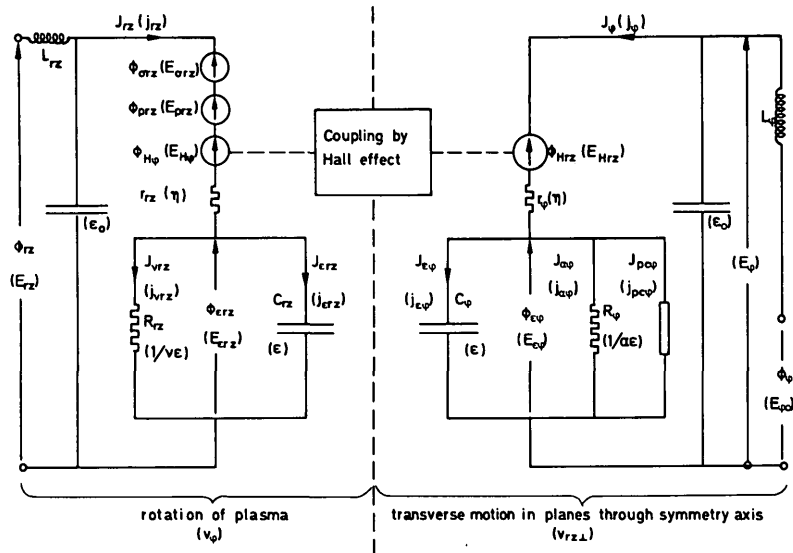


FIG. 5. Equivalent electric circuit of the rotating plasma for the transverse directions of the magnetic field.

Quantities within brackets correspond to the local properties of small volume elements, and quantities without brackets to the average properties obtained from a rough mean value formation over the entire plasma body. The circuit elements are given by Eqs (75) - (78).

the momentum losses associated with the same forces in the rz planes correspond to the current $\vec{j}_{\alpha\varphi}$. Further, $\vec{E}_{H\varphi}$ and $\vec{E}_{Hz}$ are the fields representing the Hall couplings between the two sets, and $\vec{E}_{prz}$ is the electric field set up by the pressure gradient in the rz planes. Whenever necessary, the Hall field $\vec{E}_{H\varphi}$ in the circuit in Fig. 5 can be transformed to the branch of the radial current $\vec{j}_{rz}$, as shown by the last expression in Eq. (66). Finally, $\vec{j}_{pc\varphi}$ represents the azimuthal currents which balance the pressure gradient, the centrifugal force, and other inertia forces in the rz planes. In cases where $2\mu_0 n m v_\varphi^2 (r_{02} - r_{01}) / (r_{02} + r_{01}) B^2$ becomes comparable to unity, the centrifugal force will induce appreciable azimuthal currents and the energy of the rotational motion $\vec{v}_\varphi$ is not only stored in the field $\vec{E}_{\epsilon rz}$ of the equivalent capacity, but also as energy in an induced magnetic field. This effect can be included in a modified expression for the equivalent capacity as shown later in this section.

When the macroscopic time variations are slow compared to the propagation time of a sound wave, when the isorotation law $v_\varphi/r = \Omega = \text{const.}$ is valid, and $\partial n/\partial t = 0$, the scalar product of Eq. (59) with $\vec{B}$ yields

$$n(r) = n_0 \exp \left[- (m_i \Omega^2 / 4kT) (r_0^2 - r^2) \right]; \quad (71)$$

$$C_w = n(r_w) / n_0$$

for a particular field line which cuts the midplane of Fig. 1 at the radial distance r_0 . Equation (71) expresses the longitudinal momentum balance between the centrifugal force and the pressure gradient [23, 25, 26].

The local balance Eqs (59) - (70) are less suitable for a comparison with experiments than the corresponding relations which describe the average behaviour of the entire plasma body. We therefore multiply the azimuthal component of Eq. (59) by r and integrate it over the plasma volume including the wall layers and boundary regions. After division by $n_0 m \bar{r}_0 v_0 V$ the result becomes

$$(1/n_0 v_0) \frac{\partial}{\partial t} (n_0 v_0) = c_1 (\vec{j} \times \vec{B})_\varphi / n_0 m v_0$$

$$- c_2 \alpha - c_3 / \tau_{||} - c_4 / \tau_{\perp} - c_5 (1/n_0) \frac{\partial n_0}{\partial t} \quad (72)$$

where

$$1/\tau_{||} = 2k_w (B_0 r_w^2 / B_w r_0^2) C_w (kT_0 / m_i)^{1/2}$$

$$(1 + \alpha_w / \zeta_w) / L_{||} \quad (73)$$

$$1/\tau_{\perp} = 4 k_1 (1 + 0.44 K_p / A^{1/2}) n_0 A^{1/2}$$

$$(\ln \Lambda) / m_p B_0^2 T_0^{1/2} (r_{02} - r_{01})^2 \quad (74)$$

In Eq. (72) the first term of the right-hand member represents the driving force of the rotation. The second term arises from the angular momentum losses produced by collisions with neutral particles located inside the main plasma body. The longitudinal time constant $\tau_{||}$ is due to one contribution from the angular momentum being carried away by plasma streaming along the field lines, as given by the unity term within the bracket of expression (73). An additional contribution arises from the second term within the same bracket which represents the plasma-neutral gas friction within the wall layers at the end insulators. In systems with large longitudinal losses and considerable amounts of back-scattered neutral particles, charge-exchange collisions may produce anisotropic ion temperatures. In such cases T_0 in Eq. (73) will have to be replaced by the parallel temperature $T_{||}$. Further, the transverse time constant $\tau_{\perp}$ arises from the superimposed effects of plasma viscosity as given by the unity term within the bracket of expression (74), and of transverse diffusion by the pressure gradient and the centrifugal force being represented by the second term within the same bracket and by Eq. (53). The losses due to viscosity, free streaming and diffusion are derived from the terms in Eq. (59) specified by expressions (61) and (62).

A similar mean value formation can also be made for the alternative forms (65) - (70) of the momentum balance equations. We then define $\phi_{rz} = (r_{02} - r_{01}) E_{rz}$, $\phi_{\sigma rz} = (r_{02} - r_{01}) E_{\sigma rz}$,

$$\phi_\varphi = \pi (r_{02} + r_{01}) E_{\varphi_0}, \quad \phi_{\epsilon rz} = (r_{02} - r_{01}) E_{\epsilon rz} = (r_{02} - r_{01}) \overline{v_\varphi B_0},$$

$$\phi_{\epsilon\varphi} = \pi (r_{02} + r_{01}) E_{\epsilon\varphi} = \pi (r_{02} + r_{01}) |\overline{\vec{v}_{rz} \times \vec{B}}|,$$

$$\phi_{H\varphi} = (m_i - m_e) (r_{02} - r_{01}) |\overline{\vec{j}_\varphi \times \vec{B}}| / enm,$$

$$\phi_{Hz} = \pi (m_i - m_e) (r_{02} + r_{01}) |\overline{\vec{j}_{rz} \times \vec{B}}| / enm, \text{ and}$$

$\phi_{prz} = (m_i - m_e) (r_{02} - r_{01}) |\overline{\nabla_{rz} p}| / 2enm$ as the total voltages corresponding to the electric fields in Eqs (65) - (70) and in Fig. 5. Here E_{φ_0} is the azimuthal field induced by external sources. Further the total currents are

$$J_{rz} = \pi (r_{02} + r_{01}) L_{||} |\overline{\vec{j}_{rz}}|, \quad J_\varphi = (r_{02} - r_{01}) L_{||} |\overline{\vec{j}_\varphi}|,$$

$$J_{\epsilon rz} = \pi (r_{02} + r_{01}) L_{||} |\overline{\vec{j}_{\epsilon rz}}|, \quad J_{\epsilon\varphi} = (r_{02} - r_{01}) L_{||} |\overline{\vec{j}_{\epsilon\varphi}}|,$$

$$J_{\nu rz} = \pi (r_{02} + r_{01}) L_{||} |\overline{\vec{j}_{\nu rz}}|, \quad J_{\alpha\varphi} = (r_{02} - r_{01}) L_{||} |\overline{\vec{j}_{\alpha\varphi}}|, \text{ and}$$

$$J_{pc\varphi} = (r_{02} - r_{01}) L_{||} |\overline{\vec{B} \times (\nabla_{rz} p - g_c \hat{r})}| / B_0^2$$

The equivalent electric circuit then becomes as outlined in Fig. 5, where quantities within brackets refer to the local properties of small volume elements and quantities without brackets represent the average properties obtained from integration

over the plasma body. For the latter the circuit elements consist of the equivalent capacities

$$C_{rz} = \frac{\pi(r_{02} + r_{01}) L_{\parallel} n_0 m}{(r_{02} - r_{01}) \bar{B}_0^2} \quad (75)$$

$$C_{\varphi} = \frac{(r_{02} - r_{01}) L_{\parallel} n_0 m}{\pi(r_{02} + r_{01}) \bar{B}_0^2}$$

their leakage resistances

$$R_{rz} = 1/\nu C_{rz} \quad R_{\varphi} = 1/\alpha C_{\varphi} \quad (76)$$

the series resistances

$$r_{rz} = \frac{(r_{02} - r_{01}) \eta_{\perp 0}}{\pi(r_{02} + r_{01}) L_{\parallel}} \quad (77)$$

$$r_{\varphi} = \frac{\pi(r_{02} + r_{01}) \eta_{\perp 0}}{(r_{02} - r_{01}) L_{\parallel}}$$

and the azimuthal inductance

$$L_{\varphi} = \pi \mu_0 (r_{02} + r_{01})^2 / 4 L_{\parallel} \quad (78)$$

A corresponding inductance L_{rz} can also be included in the branch in Fig. 5 which represents the current in the rz plane. In this connection it should finally be mentioned that Chang and Lundgren [44] have earlier deduced an equivalent circuit for the special case of an incompressible rotating fluid.

A refined description of the electric behaviour of the plasma can be obtained from the integration over thin shells of the plasma volume having the thickness $dL_{\perp}$ as shown in Fig. 1. This yields a set of infinitesimal circuits similar to that of Fig. 5, each having potential drops $d\phi_{rz}$ and $d\phi_{\varphi}$ and being connected in series to represent the entire plasma body with its electrode voltage ϕ_{rz} .

Among the deduced circuit elements the equivalent capacity C_{rz} of Eq. (75) is of special interest. The associated electric charge

$$Q_{\epsilon}(t) = C_{rz} \phi_{\epsilon rz}(t) = \int_0^t J_{\epsilon rz} dt \quad (79)$$

$$\approx \pi(r_{02} + r_{01}) L_{\parallel} n_0 m |\vec{v}_{\varphi}(t)| / \bar{B}_0$$

is a direct measure of the angular momentum stored in the plasma. In absence of dissipation this momentum is reversible in the sense that the charge Q_{ϵ} fed into the plasma can be entirely recovered by means of a short-circuit between the electrodes. This corresponds to a certain reversible velocity distribution $v_{\varphi}(r, z)$. Especially in

the long cylinder approximation, where the plasma consists of a shell of constant density n_0 uniformly extended in the z direction and immersed in a homogeneous axial magnetic field, the reversible velocity distribution is obtained from Eq. (79) by replacing $\frac{1}{2}(r_{02} + r_{01})$ by the radial distance r . In this case rv_{φ} becomes constant. This implies that a rapid acceleration of the plasma by means of a strong current $J_{\epsilon rz}$ will lead to a nearly reversible velocity distribution $v_{\varphi}(r, z)$. On the other hand, if the plasma is stopped by a short-circuit being applied late enough for dissipative effects to alter the velocity distribution, the latter no longer remains reversible and only parts of its associated momentum can then be recovered. This is also immediately seen from the refined equivalent circuit mentioned in the last paragraph. Thus, in the reversible case all the condensers of the set of infinitesimal circuits can be charged and emptied by the same total current J_{rz} , but in the non-reversible case some of the condensers may have remaining charges and $d\phi_{\epsilon rz} \neq 0$ even if the total voltage $\phi_{rz} = 0$ after an applied short-circuit.

In a dissipation-free state a more accurate expression can further be deduced for C_{rz} in the long cylinder approximation, also including the potential energy of the magnetic field induced by the centrifugal force. From Maxwell's equations and from Eqs (28) - (31) we obtain [40, 101]

$$\left(\frac{\partial}{\partial t} + \vec{v} \cdot \nabla \right) (\vec{B}/nm) = [(\vec{B}/nm) \cdot \nabla] \vec{v} \quad (80)$$

when pressure and ionization effects are neglected. We now consider two special cases [40, 102]:

(a) The external wall can be placed at the radius r_2 and act as outer electrode, thereby having a sufficiently high electric resistance not to affect the magnetic field by induced currents. As long as the centrifugal force remains small, the inner electrode will at the same time be connected with field lines situated at the radial distance r_1 from the axis in the midplane of the device. When a strong centrifugal force is acting, the field lines are pushed out radially. The wall at r_2 still defines the outer boundary, but part of the plasma will now be 'scraped off' here, due to the radial expansion. The inner plasma boundary will be displaced from r_1 to $r_1^* > r_1$. We further assume a wall of infinite conductivity to be wrapped around the device at $r = R_0 > r_2$. With $x_{\epsilon}^2 = \mu_0 Q_{\epsilon}^2 / 8 \pi^2 n_0 m r_2^2$, $x_{12} = r_1 / r_2$, and $x = r_1^* / r_2$ the displacement of the inner plasma boundary in the midplane is then determined by

$$x^2 = (x_{12}^2 + x_{\epsilon}^2) / [1 + x_{\epsilon}^2 + (r_2 / R_0)^2 x_{\epsilon}^2 \ln(1/x^2)] \quad (81)$$

and the equivalent capacity of a length $L_{\parallel}$ of the plasma column becomes

$$C_{rz} = C_0 / \psi_{12} \quad C_0 = 2 \pi n_0 m L_{\parallel} / B_0^2 \ln(r_2 / r_1) \quad (82)$$

where

$$2\psi_{12}\ln(r_2/r_1)=[1+x_\epsilon^2+x_\epsilon^2\ln(1/x_\epsilon^2)]\ln(1/x_\epsilon^2)+x_\epsilon^3(1-1/x_\epsilon^2) \quad (83)$$

For small stored charges Q_ϵ and small centrifugal forces Eqs (82) and (83) yield $\psi_{12}=1$, and C_{rz} tends to the value C_0 earlier deduced by Anderson et al. [19]. In the limit where $r_2 - r_1 \ll r_2 + r_1$ the capacity C_0 becomes equal to the expression for C_{rz} given by Eq.(75). At large stored charges ψ_{12} will deviate from unity on account of the centrifugal expansion and of the fact that plasma is being scraped off by the wall at $r=r_2$. In the special case $r_2=R_0$, the capacity C_{rz} increases beyond C_0 in a way earlier discussed by Anderson et al. [19].

(b) When the outer electrode is placed at the ends of the configuration in a way to allow the outer plasma boundary to expand radially without being scraped off at a wall, the result (81) has to be modified. With r_2^* and r_2^* now denoting the undeformed and deformed outer boundary in the mid-plane, we obtain

$$(r_2^*/r_2)^2 = 1 + x_\epsilon^2 \ln \left[\frac{(r_2^*/r_2^2 + x_\epsilon^2)}{(x_{12}^2 + x_\epsilon^2)} \right] \quad (84)$$

$$(r_1^*/r_1)^2 = (1 + x_\epsilon^2/x_{12}^2)/(1 + x_\epsilon^2 r_2^2/r_2^{*2}) \quad (85)$$

in the absence of an infinitely conducting wall. Eqs (84) and (85), as well as Eq.(81), can be used to estimate the magnetic field which is necessary for radial containment of a stable plasma having a certain initial density $n_0 m$. For further discussions on this problem reference is made to earlier investigations [40, 102].

So far we have only treated the acceleration of the plasma in the directions across the magnetic field. However, in devices such as rotating plasma guns there is also an acceleration in the direction along the field lines. The magnetic field is then usually made to diverge in the z direction and an axial force $-j_\phi B_r$ produces the desired acceleration. This is easily seen from Eqs (59) and (60) in a simplified case where the dissipation and pressure terms are neglected, the magnetic field diverges only slightly such that $B_r \ll B_z$ and $v_r \ll v_z$, and where only the parts of the plasma close to the symmetry axis are being studied. The acceleration of the plasma can then be treated by means of Eqs (59) and (60) for a stationary state, including the $(\vec{v} \cdot \nabla) \vec{v}$ term of Eq.(59) in the direction of $\vec{B}$. After integration and by applying the condition $\text{div } \vec{B} = 0$, the axial velocity v_z of the plasma becomes $(E/B) \cdot [1 - B(z)/B_0]^{1/2}$, where E is the applied electric field, space charges have been neglected within the plasma, and the axial velocity is assumed to be zero at $z=0$, where the magnetic field $B(z) \leq B_0 \equiv B(0)$ starts to diverge. A similar result has been obtained by Minardi [64] from a self-consistent analysis of the electric field. Further discussions on the longitudinal acceleration processes of rotating plasmas have been summarized by Kash [63] among others.

2.5.3. Heat balance

To deduce the heat balance equation of the entire plasma body we now integrate Eq.(32) over the plasma volume V and apply the approximations of Section 2.3.3. The heat balance depends on the type of wall layer approximation to be applied. This review is restricted to thin layers represented by Eq.(44) for which recycling of neutrals is limited as described at the beginning of Section 2.5.1. Excitation radiation losses are included in the last term of Eq.(32). The heat balance is then expressed by

$$\begin{aligned} 3kV \frac{\partial}{\partial t} (n_0 T_0) &= P - \Lambda \\ &\equiv c_1 P_s + c_2 P_\mu + c_3 P_{rz} + c_4 P_{\eta rz} + c_5 P_{\eta \phi} \\ &\quad - c_6 \Lambda_s - c_7 \Lambda_w - c_8 \Lambda_\lambda - c_9 \Lambda_{rz} - c_{10} \Lambda_R \end{aligned} \quad (86)$$

Here P and Λ are the total heating power and heat losses, and

$$\begin{aligned} P_s &= \pi \bar{C}_w k_w (kT_0/m_i)^{1/2} n_0 m (1 + 2\epsilon_{in} \nu_{in}/\xi) v_0^2 \\ &\quad \times \overline{(B_0 r_w^2/B_w r^2)} (r_{02}^2 - r_{01}^2) \end{aligned} \quad (87)$$

$$\begin{aligned} P_\mu &= 4\pi k_1 n_0^2 A^{3/2} (\ln \Lambda) v_0^2 (r_{02} + r_{01}) L_\parallel / (r_{02} - r_{01}) \\ &\quad B_0^2 T_0^{1/2} \end{aligned} \quad (88)$$

$$P_{rz} = \delta_{rz} (K_p/A^{1/2}) (r_w/r_0)^2 P_\mu \quad (89)$$

$$P_{\eta rz} = (r_{02} - r_{01}) \eta_{\perp 0} J_{rz}^2 / \pi (r_{02} + r_{01}) L_\parallel \quad (90)$$

$$P_{\eta \phi} = \pi (r_{02} + r_{01}) \eta_{\perp 0} J_\phi^2 / (r_{02} - r_{01}) L_\parallel \quad (91)$$

$$\Lambda_s = 10\pi \bar{C}_w k_w (kT_0/m_i)^{1/2} n_0 kT_0 (\overline{B_0/B_w}) (r_{02}^2 - r_{01}^2) \quad (92)$$

$$\begin{aligned} \Lambda_w &= 2\pi \bar{C}_w k_w (kT_0/m_i)^{1/2} n_0 W [1 + 3kT_0 (\epsilon_w f_{in} \nu_{in} \\ &\quad + f_{en} \nu_{en}) / 2W\xi] \overline{(B_0/B_w)} (r_{02}^2 - r_{01}^2) \end{aligned} \quad (93)$$

$$\Lambda_\lambda = 4\pi k_2 n_0^2 A^{1/2} (\ln \Lambda) T_0^{1/2} (r_{02} + r_{01}) L_\parallel / (r_{02} - r_{01}) B_0^2 \quad (94)$$

$$\Lambda_{rz} = \delta_{rz} (K_p/A^{1/2}) \Lambda_\lambda \quad (95)$$

$$\Lambda_R = \pi (r_{02}^2 - r_{01}^2) L_\parallel R \quad (96)$$

where use has been made of Eqs (37), (38), (44) and (53), and δ_{rz} and ϵ_w are dimensionless constants with values between 0 and 1. The physical explanation of these results is as follows:

(a) P_s is the 'frictional' heating due to interaction between the rotating plasma and the end walls and their neutral gas layers. Part of the heating is produced by ions which are reionized at the rate ξ and start their motion in cycloidal orbits, where half of the energy is translatory and the other half is 'thermal'. Another part is due to charge exchange collisions which have a similar effect, and to elastic collisions between ions and neutrals in the wall layer. This part gives rise to the term $2\epsilon_{in}\nu_{in}/\xi$ in Eq. (87).

(b) P_μ is the viscous heating by ion-ion collisions in the sheared velocity field given by Eq. (51). It arises from the term $|\vec{\lambda} : \nabla \vec{v}| \approx |\mu^*(\nabla_\perp v_\phi)^2|$ given by Eqs (27) and (32).

(c) P_{rz} is a frictional heating similar to the first part of P_s described in (a), but being produced by reionization of those plasma particles which escape by transverse diffusion at the velocity of Eq. (52). The coefficient $\delta_{rz}(r_w/r_0)^2$ in Eq. (89) is due to the fact that reionization of the neutrals may take place all along the field lines in the boundary regions of Fig. 1.

(d) $P_{\eta rz}$ is the ohmic heating resulting from the electric current J_{rz} between the electrodes.

(e) $P_{\eta\phi}$ is the ohmic heating being partly due to the current $J_{pc\phi}$ of Fig. 5 which balances the pressure gradient and the centrifugal force, partly to other currents flowing in the azimuthal direction as shown in the same figure.

(f) Λ_s is due to the total thermal power loss of the plasma particles escaping to the end insulators. We observe that the second term in Eq. (32) is $(3/2) \text{div}(\vec{p}\vec{v})$ which gives rise to an outflux of thermal energy, and that the fourth term of the same equation gives rise to the contribution $p \text{div} \vec{v}$ which is an expansion work being lost when the plasma particles hit the end insulators. Further, since the Debye sheaths at the end insulators will return the main part of the electrons back into the plasma when there is ambipolar free streaming, the longitudinal heat losses will be determined by the net flow of ions to the end walls. In this connection we also observe that the end walls do not emit electrons and that an electron heat transport by conduction along the field lines to these walls therefore becomes impossible, i.e. the heat transport is produced by impinging particles only.

(g) Λ_w is partly due to the ionization and dissociation energy W being lost by plasma particles escaping to the end insulators, partly to energy losses by collisions with the neutral gas within the wall layers. In Eq. (93) the term $f_{en}\nu_{en}$ is defined to include all losses from collisions between electrons and neutrals, except those leading to dissociation and ionization, and $\epsilon_w f_{in}\nu_{in}$ has an analogous meaning for ions. The factor ϵ_w has been introduced because part of the energy given off by the ions to the neutrals is re-gained in cases where the latter re-enter the plasma.

(h) Λ_λ is due to heat conduction across the magnetic field on account of the temperature distribution given by Eq. (51). It arises from the term $|\text{div} \vec{q}| \approx |\text{div}(\lambda^* \nabla T)|$ in Eq. (32). We observe that the main power loss in the boundary regions, which arises from reionization of neutrals and other inelastic electron-neutral collisions, is covered by the transverse heat flow from the fully ionized plasma [92]. We therefore consider this loss to be included in Λ_λ .

(i) Λ_{rz} is the heat loss due to diffusion across the magnetic field at the velocity given by Eq. (52). This loss consists of one part originating from convection of matter, and one part from the corresponding expansion work. We further observe that the cooling produced by this expansion work is compensated by the part of the ohmic heating power $P_{\eta\phi}$ which arises from the current balancing the thermal pressure [93].

(j) Λ_R is the radiation loss by other sources than the excitation radiation from electron-neutral collisions. It includes bremsstrahlung as well as possible radiation losses from impurities inside the fully ionized plasma.

The balance Eq. (86) can be divided by $3kVn_0T_0$ to give a corresponding expression for the time constants of the thermal energy and the heating and loss mechanisms involved. In a stationary state Eq. (86) yields a relation between the velocity v_0 and temperature T_0 which also can be expressed in terms of the ratio $\Theta = m_i v_0^2 / 3kT_0$ between the energies of the directed and thermal motions of the ions. This parameter is important to the centrifugal confinement expressed by Eq. (68). In the limiting case of a plane slab approximation where all longitudinal effects are neglected and only the contributions from P_μ and Λ_λ remain, Eq. (86) has the solution $\Theta = \Theta_1 \approx 6.9$, when the values of k_1 and k_2 given in Section 2.3.3. are inserted [103].

The radiation losses from impurities may strongly affect the heat balance. They will, however, only have a moderate and indirect influence on the particle and momentum losses described in Sections 2.5.1. and 2.5.2., i.e. by their effects on the plasma temperature as determined by Eqs (36), (55), (56), (73) and (74).

2.6. SPACE CHARGE AND SHEATH FORMATION

So far no attention has been paid to the part $\vec{E}_{orz}$ of the electric field which arises from space charges in addition to those produced by the diamagnetism of the plasma and being included in the dielectric constant ϵ and the equivalent capacity C_{rz} of Eq. (75). The additional space charge effects represented by E_{orz} can be localized either inside the fully ionized plasma or in the transition regions between the plasma and the electrode surfaces. In this connection it should be kept in mind that the space-charges in an open-ended rotating plasma device may be partly or fully removed along the magnetic field lines in a way to reduce the field E_{orz} , but this depends on prevailing plasma conditions and associated line-tying effects [104].

2.6.1. Space charges in the main plasma volume

The electric space charge σ is determined by

$$\text{div } \vec{E} = \sigma/\epsilon_0 \quad (97)$$

By combination of expressions (29) - (31) and (97) the space charge in the quasi-neutral plasma region can be written as

$$\sigma/\epsilon_0 = -\eta \frac{\partial \sigma}{\partial t} + \vec{j} \cdot \nabla \eta - \text{div}(\vec{v} \times \vec{B}) \quad (98)$$

$$+ \frac{m_i - m_e}{e} \text{div} \left[\frac{1}{2nm} (\nabla p - \text{div} \vec{\lambda}) + \alpha \vec{v} + \frac{d\vec{v}}{dt} \right]$$

It is quite obvious that both differential motions between ions and electrons and compressible motions which 'pile up' one of each species more than the other will lead to space charges in the plasma as described by this equation. In general the electric field from such charges cannot be compensated simply by imposing an extra electric vacuum field between the electrodes. Further physical explanations of this result are easily given in the case of axially symmetric configurations, where the terms of Eq. (98) can be explained as follows:

(a) The contributions involving η are due to the friction between the electron gas and ions and neutrals which piles up space charges inside the plasma.

(b) The field originating from $\vec{v} \times \vec{B}$ is not necessarily divergence free. This implies that space charges have sometimes to be distributed within the plasma volume to satisfy the conditions imposed on the electric field by the plasma motions through the balance equations and Maxwell's equations. Especially in the long cylinder approximation discussed in connection with Eq. (79), only a reversible velocity field $v_\phi = \text{const.}/r$ would make the term $\text{div}(\vec{v} \times \vec{B})$ vanish in a homogeneous magnetic field $\vec{B}_0$.

(c) The contributions from ∇p and $d\vec{v}/dt$ are due to the fact that there is a difference in the forces by which ions and electrons are displaced and tend to escape from the plasma volume along and across the magnetic field. As a result, a charge separation arises which sets up an ambipolar electric field.

(d) The term $\alpha \vec{v}$ yields a contribution mainly from the radial diffusion of ions. This occurs in cases where $\alpha r v_r$ does not become constant and ion-space charges are piled up with respect to the electron gas, in a way similar to the resistive effect on electrons described under (a) above.

(e) When the term $\text{div} \vec{\lambda}$ is due to viscosity as expressed by Eqs (59) and (61) and becomes purely azimuthal, it does not lead to charge separation. On the other hand, if a radial component of $\text{div} \vec{\lambda}$ exists, there is a contribution to Eq. (98) from this term.

From the present discussion it is seen that there is some arbitrariness in the distribution of the electric fields $\vec{E}_{rz}$ and $\vec{E}_{\sigma rz}$ in every rotating plasma system as long as not all of its properties are specified in detail. The exact solution of the electric field $\vec{E}$ has to be determined in a self-consistent way which often leads to rather involved deductions. A special example of this has been demonstrated by Minardi [64, 105] in a double-adiabatic treatment of the plasma acceleration in a rotating plasma gun.

2.6.2. Electrode sheaths

In practice electrodes and other metal surfaces in a plasma do not become perfect sources and sinks for ions and electrons [104]. Complicated transition regions in the form of anode and cathode sheaths will therefore develop between the plasma and such surfaces to secure the continuity of the electric current. This involves processes such as the surface emission and absorption of ions and electrons and the formation of neutral gas and space-charge sheaths near the electrodes.

In particular, since the ion Larmor radius is much larger than that of the electrons, the radial electric current is mainly due to ions in plasmas of axial symmetry [106]. In a stable state one should therefore expect an uncompensated number of electrons to arise within a sheath in front of the anode. Such anode sheaths are discussed to some detail by Longmire [107]. They may break down when there is neutral gas near the anode which enables the electrons to diffuse across the magnetic field on account of collisions with the neutrals. Another possibility for the electrons to reach the anode is by Buneman's two-stream instability [108] which enhances the electron diffusion.

2.7. LIMITING EFFECTS

A rotating plasma is subject to a number of limiting effects which are both of physical and practical interest. They are listed in Table II and will be shortly described in this section, in the same order of succession as they appear in an experiment.

2.7.1. Minimum neutral gas density

We first consider the earlier parts of the ionization period of a crossed field discharge which starts in a mass of neutral gas having the uniform density n_n . To obtain an increasing plasma density we must have $\partial n/\partial t > 0$ in Eq. (28). Integration over the confinement volume then yields the condition [109]

$$n_n > n_{n\min.} = (n v_N)_S / n_V V \xi_{\max.} \quad (99)$$

where subscripts (V) and (S) denote mean values over the volume V and its boundary surface S, v_N is the normal component of $\vec{v}$ at the same surface,

TABLE II. LIMITING EFFECTS OF ROTATING PLASMAS

Effect	Basic mechanisms	Relevant parameters and conditions	Possible methods for avoidance or reduction of effect
<u>Minimum neutral density</u>			
Lower limit of neutral gas density during starting process	Balance between ionization rate and particle losses breaks down	Maximum value $\xi_{\max}$ of ionization rate, and plasma outflow to walls	Reduction of plasma losses and deviations from isotropy decrease density limit
<u>Velocity limitation</u>			
Upper limit of plasma fluid velocity during interaction with neutral gas	Enhanced ionization rate arises when a plasma cloud tends to penetrate a neutral gas cloud at high mutual velocities	The critical velocity v_c , the state of ionization of the plasma body, the radial ratio r_0/r_w , and the boundary conditions	By choosing large radial ratios, by avoiding wall layers, or possibly by a short-circuit of the electric field producing the enhanced interaction, or by decreasing the density
<u>Burn-out power</u>			
Lower limit of power input for burn-out of the neutral gas within the confinement region	The azimuthal drag force between ions and neutrals	Burn-out current J_{rzc} , critical velocity v_c , and gas density n_{g0} before start of discharge	Reduction of initial gas density lowers burn-out current and power
<u>Longitudinal losses</u>			
Lower limit of plasma losses determined by longitudinal effects	Ambipolar free streaming of plasma to end walls	Radial and mirror ratios r_0/r_w and B_w/B_0 , ion energy ratio, and anisotropy conditions of plasma	By choosing large radial, mirror and ion energy ratios, and by establishing anisotropy with small longitudinal pressures
<u>Minimum power effect</u>			
Lower power and temperature limit to sustain a fully ionized plasma	Sudden breakdown of particle or heat balance of the plasma body, or of heat balance of boundary regions. It takes place when power input is reduced below a certain value	Wall layer thickness d_w , ratio P_p/Λ_p between power input and loss as function of temperature T_0 , and boundary region thickness x_b	Minimum power P_m and temperature T_{0m} decrease somewhat when confinement of plasma is improved

and $\xi_{\max}$ is the maximum value of the ionization rate ξ . In principle such a maximum exists both for thermal and non-thermal ionization by electrons, ions or neutrals. When there are isotropic ion and electron temperatures T_i and T_e and the plasma escapes by ambipolar free streaming along the field lines, condition (99) reduces to

$$n_n > n_{\min.}$$

$$\approx (n_s/n_v) (S/V\xi_{\max.})^{1/2} (k/m_i)^{1/2} \left[T_e^{1/2} + (T_i/2\pi)^{1/2} \right] \quad (100)$$

A special case of this relation has earlier been treated by Poletaev [110].

From relations (99) and (100) it is seen that, for a given geometry and given plasma conditions, there exists a minimum neutral gas density $n_{\min.}$ below which a plasma cannot be produced. This limit

applies also to non-rotating plasmas. The magnetic field and the centrifugal confinement are involved in the coefficients of Eqs (99) and (100). In cases where the plasma is anisotropic and magnetic mirror effects strongly reduce the outflux $(nv_N)_s$, such as in ionization gauges and Penning-magnetron discharges, quite low values can be obtained for the minimum density (99).

2.7.2. Velocity limitation

The neutral gas in the partially ionized boundary regions of a rotating plasma is usually coupled in an efficient way with surrounding vessel walls. As compared to the high velocities of the rotating plasma it is therefore nearly at rest with respect to the walls. It is likely that the differential velocity between the plasma and the neutral gas provides an energy source for ionization in the partially ionized regions, as soon as this becomes energetically possible. Consequently, Alfvén [5] has postu-

lated that a strongly enhanced ionization process should arise when the mutual velocity $|\vec{v} - \vec{v}_n|$ reaches the critical value

$$v_c = (2e\phi_i/m_i)^{1/2} \quad (101)$$

where ϕ_i is the ionization potential. However, even if this hypothesis is plausible and there are several experiments which support it, the problem of the underlying physical mechanism has not yet been settled. A review of the theoretical and experimental investigations on this problem has recently been prepared by Block, Sherman and Danielsson [111].

In fact, it is likely that the critical velocity v_c represents a universal transition process from the third to the fourth state of matter which can be provided by several different mechanisms operating under various conditions:

(a) A direct application of the macroscopic conservation equations to partially ionized plasmas leads under certain circumstances to velocities v_ϕ being roughly equal to the critical velocity. Such results have been obtained by Lin [112] who considered the heating of the electron gas by Coulomb collisions, by Drobyshevskij [113] who stated that the principal energy source of the electrons arises directly from the electric field, by Hassan [114] who considered the conservation equations on general grounds, by Sockol [115] and Fay and Sockol [116] who have introduced boundary layers at the end walls, by Sherman [88, 90] who has included ionization by ions and by neutral particles involved in charge exchange collisions, and by Kunkel and Gross [99] and Patric and Pugh [117] who have treated the shock relations of ionizing waves.

(b) There are also more involved mechanisms which may become responsible for an enhanced ionization process. Colgate [118] and Beard [119] considered the effects of an electrostatic layer being produced in the same way as that arising in the boundary region between a plasma and a vacuum region in the presence of a magnetic field. Chodura [120] stated that the electrons are prevented by the ions from drifting freely in the presence of a density gradient, and should therefore move along an applied transverse electric field. It has further been shown by the author that space-charge effects can arise from the finite Larmor radius of non-thermal ions produced by charge-exchange and ionizing collisions in combination with a transverse density gradient [121]. As a consequence, a strongly oscillating electric potential pattern arises which produces high electric fields inside the plasma. This mechanism should operate provided that the density of the non-thermal ions is at least half as large as that of the thermal background plasma [122]. In any case, large amounts of non-thermal ions should be created by the reionization of the back-flux of neutral particles at the end insulators of certain rotating plasma devices [123]. This is due to the fact that the ratio $d_w/\lambda_{ni} \approx (f_n \sigma_{in}/\xi_w)(kT/2\pi m_i)^{1/2}$ between the wall

layer thickness (44) and the mean free path λ_{ni} for non-ionizing neutral-ion collisions becomes comparable to and sometimes even less than unity when $T \geq 10^5$ °K. Finally, Sherman [124] has considered the differential motion between ions and electrons in a rapidly accelerated or decelerated plasma and has found that such a motion can lead to a two-stream instability producing an electric field which heats the electrons to high energies.

Whenever the critical velocity phenomenon exists, it will put a strong constraint on the plasma motion and the mutual velocity $|\vec{v} - \vec{v}_n|$. In most cases discussed here $v_n \ll v$ and the plasma velocity is then limited to v_c , but it should also be kept in mind that v_n may become comparable to v , such as to make v exceed v_c under certain experimental conditions.

During the starting period of rotating plasmas in systems like that shown in Fig. 1, there is neutral gas within the entire confinement volume and the velocity of rotation is expected to be limited to v_c even in the midplane of Fig. 1. The voltage between the electrodes then becomes

$$\phi_{12c} \approx \int_{r_{01}}^{r_{02}} v_\phi B_0 dr_0 \approx v_c \bar{B}_0 (r_{02} - r_{01}) \quad (102)$$

Equation (102) holds when the voltage ϕ_{erz} in Fig. 5 represents the main part of the total voltage ϕ_{rz} , and when $r \leq r_0$ such as to make the integral of $B_0 L_\perp = d\Phi/2\pi r$ across the confinement region a minimum in the midplane.

In the fully ionized case, however, the plasma will only interact with the neutral gas at the end walls and in the boundary regions. The local velocity on a particular field line is then expected to become limited first when it reaches v_c at the end walls. In such a case the isorotation law (12) yields a maximum velocity in the midplane [51, 103]

$$v_{\max.} = (r_0/r_w) v_c \quad (103)$$

For velocity distributions of the form (51) the corresponding voltage limit becomes

$$\phi_{12c} \approx v_c \bar{B}_0 (r_0/r_w) (r_{02} - r_{01}) \quad (104)$$

Consequently we see that the velocity limit often enters in two ways into the momentum balance of a rotating plasma; first during the starting process when there is neutral gas inside the whole plasma body, and later when the neutral gas only remains in the wall layers and boundary regions.

In this connection it should be pointed out that the centrifugal expansion effect described by Eqs (81) - (85) may interfere with experimental studies of the voltages connected with the velocity limiting effect. Thus, a voltage limitation is not only produced by v_c but also by 'scaping off' parts of the outer plasma regions at a surrounding vessel wall when the centrifugal force becomes large enough to expand the plasma strongly in the radial direction [40].

In a fully ionized dense plasma the interaction process leading to the critical velocity phenomenon becomes localized to the wall layers having the limited thickness d_w . It may therefore become possible to avoid the velocity limit or suppress the underlying mechanism by means of the following special measures:

(a) The radial ratio r_0/r_w can be chosen large enough for obtaining the desired maximum velocity (103) for stationary operation in type 'A' systems.

(b) The formation of wall layers can be avoided in a transient state of type 'B' systems, where the plasma is bounded by vacuum in the longitudinal direction, or in a similar state in type 'A' systems when plasma is all the time being removed at the ends, and the back-flux of neutrals is being suppressed.

(c) If the velocity limitation is produced by non-thermal ions [121] or by a type of two-stream instability [124], the associated non-uniform electric field in the azimuthal direction may be short-circuited by means of a set of narrow, concentric metal rings placed at the end insulators of type 'A' systems [123, 125]. For the non-thermal ions the rings should have a spacing being smaller than or equal to the local ion Larmor radius in the wall layers.

(d) At low plasma densities it should be possible both to avoid isorotation of the plasma and to establish anisotropic temperature distributions, in a way to reduce the interaction with the end walls. Possibly this may also eliminate the velocity limitation effect.

2.7.3. Burn-out power

Even if the plasma density can be made to increase during the earlier phases of the starting process, it is not sure that the available power input is sufficient to complete a transition of the plasma to a fully ionized state.

In a quasi-stationary state the results of Section 2.5.2. and Fig. 5 can be used to estimate the necessary driving current which has to be applied between the electrodes for this purpose during the partially ionized starting process. It becomes

$$J_{rz} \approx J_{\nu rz} = \phi_{\epsilon rz} \nu C_{rz} \approx \pi (r_{02} + r_{01}) L_n M \bar{n} \bar{n}_n \bar{v}_\phi < \sigma_{in} w_{in} > / \bar{B}_0 \quad (105)$$

according to Eqs (61), (62), (75) and (76) when there is a strong volume interaction between the plasma and the neutral gas by collisions such as to justify the approximations $\nu \approx \alpha \approx < \sigma_{in} w_{in} >$ and $\alpha \gg \nu_\mu, \nu_{wb}$. In Eq.(105) the relative velocity w_{in} between ions and neutrals consists of one contribution from the fluid velocity which is $\vec{v}_i \approx \vec{v}_\phi$ and one contribution $\vec{w}_{in} \equiv \vec{v} |v_\phi|$ from the thermal motion of ions. If we assume the applied voltage ϕ_{rz} in Fig. 5 to be large enough for v_ϕ to reach the critical velocity (101), the velocity dependent part of Eq.(105) should be replaced by the factor $v_c^2 \sigma_{in} (1 + \vec{v})$.

We further have to discuss the product $\bar{n} \bar{n}_n$ in Eq.(105) which depends both on the burn-out process of the neutral gas taking place within the plasma confinement volume V , and on the flow of particles across the bounding surface S of the same volume. Consequently, those neutrals which collide once or several times with the ions will be rapidly scattered and centrifuged out of the confinement region. This is the case when, during the main phase of the burn-out process, the density inside the plasma body permits free neutral gas streaming. As long as the collision frequency $\alpha \approx \nu_{in}$ is much higher than the ionization frequency ξ this outflux of rapidly moving neutrals through the surface S should be nearly balanced by an equal influx of cold neutral gas at a counter-streaming velocity $w_{nw} \approx (8 k T_{nw} / \pi m_n)^{1/2}$. Here T_{nw} is the neutral gas temperature in the volume V_w of the spacing between the plasma confinement volume V and the surrounding walls. Under the condition that $(r_{02} - r_{01})(e\phi_i)^{1/2} V_w \gg \lambda_{in} (k T_{nw})^{1/2} V$ and when $V_w \neq 0$ which usually applies to experiments in systems of the type shown in Fig. 1, the driving current (105) is then found to have the maximum value [126]

$$J_{rzc} \approx n_{00} m w_{nw} v_c (1 + V/V_w) / 4 (r_{02} - r_{01}) \bar{B}_0 \quad (106)$$

Here $n_{00}(V + V_w)$ is the original number of gas molecules of density n_{00} in the device before the start of the discharge, and we observe that $m w_{nw} v_c$ is independent of the mass number A and does not contain σ_{in} . For devices with dimensions of the order of $r_{02} - r_{01} \approx 0.1$ m, $T_w \approx 300^\circ K$, and V_w comparable to V , we have $J_{rzc} \approx 10^{-18} n_{00} / \bar{B}_0$. The corresponding critical power becomes

$$P_{rzc} = \phi_{\epsilon rz} J_{rzc} \approx n_{00} m w_{nw} v_c^2 (1 + V/V_w) / 4 \quad (107)$$

The current (106) is a consequence of the momentum balance, simply due to the fact that the azimuthal drag force between ions and neutrals has a maximum at some intermediate ionization degree, as is also understood from inspection of Eq.(105). To make the transition all the way from a lowly to a fully ionized state and to burn out the neutral gas completely within the volume V , it is therefore necessary that the current and power input are chosen higher than the maximum values (106) and (107). On the other hand, this mechanism is not necessarily the only one which limits the burn-out process. There may also exist other limiting phenomena such as those associated with the heat balance and the radiation losses.

2.7.4. Longitudinal losses

When a crossed-field discharge has been converted into a fully ionized state, and the parameters can be chosen such as to avoid the velocity limit and instability phenomena, the dissipative effects treated in Section 2.5. will still put a limit on the plasma balance. Since rotating plasma devices are open

magnetic bottles, there are residual losses along the magnetic field lines under isotropic plasma conditions, even for infinitely strong magnetic fields. These losses are represented by the longitudinal time constants of the plasma density and angular momentum given by Eqs (55) and (73) and by the longitudinal heat losses of Eqs (92) and (93). They can either be reduced by establishing strongly anisotropic conditions with small longitudinal pressures, or by achieving large ion energy ratios Θ as defined at the end of Section 2.5.3., thus reducing the longitudinal loss factor C_w of Eq.(55). Almost all ions would then be able to satisfy conditions (13) and (14) for single particle confinement.

2.7.5. Minimum power effect

We have seen in Section 2.7.3. that a certain burn-out power is required to convert a neutral gas body into a fully ionized plasma. The underlying process is not reversible. Thus, as soon as a fully ionized quasi-stationary plasma has established itself, somewhat as outlined in Fig.1, such a plasma can often be sustained at a much lower power input than that given by Eq.(107). The reason for this is obviously that the neutral gas interaction is only localized to the wall layers and boundary regions in the fully ionized state.

However, there also exists a limit for the fully ionized plasma to be kept in a quasi-stationary state. Thus, if the power supply is being reduced gradually during a slowly decaying holding mode, the plasma temperature decreases as well, and finally reaches a level at which the plasma balance breaks down and there is a transition to a lowly ionized mode. There are several mechanisms which produce such a transition [56, 72]:

(a) In a quasi-stationary state the outflux of plasma along the field lines is compensated by re-ionization of back-scattered neutrals within the wall layers discussed in Section 2.4.1. At decreasing plasma temperatures the particle balance finally breaks down as described in Section 2.4.1. (c).

(b) The ratio between the total power input P_p and the total power loss Λ_p of the fully ionized plasma body is usually not a monotonic function of the characteristic temperature T_0 . In certain systems there are ranges of the parameters involved for which $\Lambda_p > P_p$ for all possible values of T_0 , as well as a marginal state with a maximum of P_p/Λ_p at which $\Lambda_p = P_p = P_{pm}$ at a certain temperature $T_0 = T_{0m}$. When the power input is decreased below the value P_{pm} , the heat balance breaks down and a sudden transition takes place to a lowly ionized mode.

(c) The mechanisms described under (a) and (b) are connected with the over-all particle and heat balance of the main plasma body. In addition, there exist critical conditions on the balance of the partially ionized boundary regions described in Section 2.4.2. If these cannot be satisfied the boundary region grows indefinitely in size and the plasma balance breaks down [56, 92].

All the mechanisms described here give rise to a sharply defined transition from a fully to a lowly

ionized state at a minimum level of the power input ('minimum-P effect'), even if this input is allowed to decrease gradually and slowly. They are not connected in a trivial manner with the ionization potential, and the minimum temperature T_{0m} is not related in a simple manner to the quantity $e\phi_i/k$ but depends on the detailed plasma balance. The minimum power and temperature can be somewhat decreased by improving the confinement of the plasma.

The present effect, as well as the burn-out power mechanism of Section 2.7.3., are not only important to laboratory plasmas but may also be connected with certain cosmical phenomena [56]. A possible application is given by the sudden appearance and disappearance of solar prominences.

2.8. MULTICOMPONENT PLASMAS

When there are several ion species present in a plasma, interactions will take place between these which, among other things, produce diffusion and dissipation processes in addition to those treated in Section 2.5. In particular, the strong centrifugal force which exists in a rotating plasma provides a mechanism for element and isotope separation which becomes important both to the removal of impurities from the plasma and to its use as a centrifuge. Reviews on these problems have been given by Bonnevier [59, 60] and Gough [127].

To discuss these phenomena we now return to the basic equations of Section 2.3.1. and restrict ourselves to stationary states where the production rate (20) of ions can be neglected. We further observe that, even if high rotational speeds can be reached, the relative velocity $|\vec{v} - \vec{v}_n|$ between the plasma and the neutral gas becomes usually limited to the critical value v_c , either in the wall layers or within the main plasma body. The net gain of momentum by collisions between non-identical particles is given by

$$\vec{M}_k = \sum_{j \neq k} \alpha_{jk} n_j n_k (\vec{v}_j - \vec{v}_k); \quad (108)$$

$$\alpha_{jk} = m_j m_k \langle \sigma_{jk} w_{jk} \rangle / (m_j + m_k)$$

where subscripts (j) and (k) denote non-identical charged particle species throughout this section and $\sigma_{jk} w_{jk}$ is the corresponding mutual collision rate.

2.8.1. Transverse momentum balance

To study the transverse motions of the multi-component plasma we limit the discussion to a long cylinder geometry with a homogeneous magnetic field $\vec{B}$ along the axis of symmetry. Such systems can be used as centrifuges both for fully and partially ionized plasmas:

(a) In the fully ionized case vector multiplication of Eq.(21) and insertion of expressions (26) and

(108) then yields the azimuthal and radial velocity components

$$v_{k\phi} = -E_r/B + (f_k - M_{kr})/q_k n_k B; \quad (109)$$

$$f_k = \frac{dp_k}{dr} - n_k m_k \left(\frac{v_{kr}^2}{r} + v_{kr} \frac{dv_{kr}}{dr} \right)$$

$$v_{kr} = M_{k\phi}/q_k n_k B - m_k v_{kr} \left(\frac{\partial v_{k\phi}}{\partial r} + \frac{v_{k\phi}}{r} \right) / q_k B$$

$$- (\text{div } \vec{\lambda}_k)_\phi / q_k n_k B \quad (110)$$

When there is a sufficiently strong magnetic field for the gyrotimes $1/\omega_k = m_k/q_k B$ to be much shorter than the characteristic macroscopic time scale t_c we should have $v_{kr} \ll v_{k\phi}$. The terms including v_{kr} , M_{kr} and $\text{div } \vec{\lambda}_k$ in Eqs (109) and (110) can then be neglected and a radial diffusion is produced by collisions between non-identical particles. The corresponding velocity becomes

$$v_{kr} \approx \sum_{j \neq k} \frac{\alpha_{jk} n_j}{q_n B^2} \left(\frac{f_j}{q_j n_j} - \frac{f_k}{q_k n_k} \right) \quad (111)$$

as earlier shown by Bonnevier [60]. When the contributions from the pressure can be neglected, we see that the radial diffusion is produced by a frictional force due to the differential motion $v_{k\phi} - v_{j\phi}$ in the azimuthal direction. The latter, in its turn, is due to the mass dependence of the centrifugal force, as already discussed in connection with Eqs (17) and (18).

We now investigate the conditions for ion-ion collisions to provide the dominating contribution to the radial velocity (110), in combination with the centrifugal force. The pressure gradients will have a small effect on the radial diffusion when $r/3 \Theta_k L_{pk} \ll 1$, where $L_{pk} = p_k/|dp_k/dr|$ and $\Theta_k = m_k v_{k0}^2/3 k T_0$. Further, the effect of $\text{div } \vec{\lambda}_k$ in Eq. (110) is mainly due to the viscous force from collisions between identical ions in a sheared velocity distribution. It gives a contribution to the radial flux of matter and produces a radial electric current. This effect can be neglected in a first approximation provided that

$$\Theta_k v_{k\phi} r Z_k (m_j + m_k) \nu_{kk} / 16 \omega_k L_{vk}^2 (Z_j m_j - Z_k m_k) \nu_{jk} \ll 1,$$

where $L_{vk} = v_{k\phi}/|dv_{k\phi}/dr|$ and ν_{jk} is the rate of collisions between ions of types subscripts (j) and (k). To investigate the effect of electron collisions we finally consider a three-component plasma, with two kinds of ions having the mass numbers A_i and A_j and the charge numbers $Z_1 = Z_2 = 1$. For the radial velocity (110) of ions of kind (i), the ratio between the contributions from electron collisions and from collisions with ions of kind (j) then becomes about

$$m_e n_e (A_i + A_j) \langle \sigma_{ie} w_{ie} \rangle / m_p n_j (A_i - A_j) A_j \langle \sigma_{ij} w_{ij} \rangle$$

$$\approx 4.1 \times 10^{-2} n_e (A_i + A_j) / n_j (A_i - A_j) A_i^{1/2}$$

$$\text{when } |A_i - A_j| \ll A_i + A_j$$

From Eq. (111), or from Eqs (109) and (110) with $\omega_k t_c$ being comparable to or less than unity, we can then derive radial diffusion equations for a quasi-neutral multicomponent plasma in which $q_k n_k = 0$ [60]. These equations result in different fluxes $n_k v_{kr}$ for different ion species, thus leading to element and isotope separation.

(b) In the partially ionized case the element and isotope separation takes place mainly by a radial magnetically unconfined diffusion of neutral gas, as in ordinary mechanical centrifuges. Favourable values can be obtained for Θ_k when $|\vec{v} - \vec{v}_n|$ is increased to the limit v_c , whereas the temperature remains low, due to heat losses from ionization, excitation, and other types of plasma-neutral gas interaction. By providing the rotating plasma device by a set of concentric electrode rings at one or both end insulators, a number of further advantages can be achieved as compared to mechanical centrifuges [61]. Firstly, as is also the case with fully ionized plasmas, the driving force is by electric means acting within the entire gas volume by means of the $\vec{j} \times \vec{B}$ force and not being limited by the tensile strength of a mechanical rotor. Secondly, a convenient distribution of this driving force can be chosen by adjusting the current flow to the rings. The velocity profile can then be optimized for the highest possible velocity at which stability is secured. These conditions are related to the coupling between the plasma and the neutral gas and to line-tying effects which are discussed elsewhere [61].

2.8.2. Longitudinal momentum balance

Also the component of the centrifugal force along $\vec{B}$ can be used for separation purposes [60]. The longitudinal momentum balance in a fully ionized static state is obtained from scalar multiplication of Eq. (21) by $\vec{B}$. With s as the corresponding co-ordinate we have

$$n_k \frac{\partial}{\partial s} \left(q_k \phi - \frac{1}{2} m_k \Omega^2 r^2 \right) + k T \frac{\partial n_k}{\partial s} = 0 \quad (112)$$

when the isorotation law (12) is applicable and $v_{j\phi}/r \approx v_{k\phi}/r \equiv \Omega$ is constant along a field line as well as the temperatures $T_j = T_k \equiv T$. We integrate Eq. (112) for two ion species having the densities n_{j0} and n_{k0} at $r = r_0$ in the midplane of the plasma. The corresponding density ratio n_j/n_k at r is then given by

$$\frac{n_j}{n_k} / \frac{n_{j0}}{n_{k0}} \approx \exp \left[- \frac{m_p \Omega^2}{2 k T} (A_j - A_k) (r_0^2 - r^2) \right] \quad (113)$$

where m_p is the proton mass. Thus, the ratio of Eq. (113) becomes less than unity when $A_j > A_k$, as

is also expected from the corresponding result (71) applying to a one-component plasma. The difference in centrifugal forces acting on various ion species will therefore lead to a separation effect which can be made use of by extracting matter at the ends of a rotating plasma device.

2.9. STABILITY

There are at least three conditions which have to be taken into account when considering the stability of a magnetically confined plasma, namely:

(a) The spatial distribution of the magnetic field within the confinement region.

(b) The distribution of the plasma in phase space, including both the velocity distributions of ions and electrons, and the distributions of the macroscopic plasma properties in coordinate space.

(c) The boundary conditions.

One or two of these three conditions alone are not always sufficient to specify the stability properties uniquely. Consequently, no general statement can be made about the stability of rotating plasmas. This problem requires detailed examination of every particular case.

Among the great number of unstable modes which arise in a magnetically confined plasma we shall in this paper pay special attention to those associated with the macroscopic rotation. These are the flute instability produced by the centrifugal force and the thermal pressure, the neutral drag instability arising from the frictional force between ions and neutrals, the Kelvin-Helmholtz instability due to the velocity shear, and micro-instabilities such as the cyclotron resonance mode associated with the angular frequency of rotation.

2.9.1. Flute instability

The instability of the interface between a heavy fluid placed on top of a lighter one is a well-known phenomenon in hydrodynamics first treated by Lord Raleigh. A corresponding mode also arises in an electrically conducting fluid supported by a magnetic field against gravity, a centrifugal force, or a thermal pressure. This mode has later been treated in connection with astrophysical problems, by Kruskal and Schwarzschild [128], Chandrasekhar [129], Pacholczyk and Stodolkiewicz [130], and Chakraborty [131] among others. The charge separation mechanism which feeds energy into the growing disturbance in a plasma was first described by Rosenbluth and Longmire [132]. The instability, which is also denoted as the 'flute' or 'interchange' mode, is of particular interest to rotating plasmas on account of the large existing centrifugal force $\vec{g}_c = \Omega^2 r \hat{r} = (v_\phi^2/r) \hat{r} = g_c \hat{r}$. Keeping the statements at the beginning of this section in mind, we now summarize the theoretical work on the flute instability as follows:

(a) The stability of a low-beta plasma has been further examined in plane geometry by the author [51, 133, 134], Rosenbluth et al. [135], Roberts

and Taylor [136], Berge [137], and Stringer and Schmidt [138], and in cylindrical geometry for special types of plasma distributions in space by Taylor [139], Lehnert [51], Haines [78], and Chen [140].

In plane geometry with a homogeneous external field $\vec{B}$ along z , the gravitation force $\vec{g}_c$ along the positive y direction, an unperturbed density distribution satisfying $y_0 = n(y)/(dn/dy) = \text{const.}$, and a spoke-shaped flute disturbance varying as $\exp(ik_x x + i\omega t)$, the dispersion relation becomes [51]

$$2\omega/k_x = \frac{g_c}{\omega_i} - \frac{\omega_i a_i^2}{2y_0} \pm \left[\left(\frac{g_c}{\omega_i} - \frac{\omega_i a_i^2}{2y_0} \right)^2 + 4g_c/y_0 k_x^2 \right]^{1/2} \quad (114)$$

when the plasma density is large enough for $\epsilon = nm/B^2$ to be much greater than ϵ_0 . The last term within the bracket of Eq. (114) tends to produce a flute instability when $y_0 < 0$ and the force $\vec{g}_c$ acts in the direction of decreasing density. This destabilizing effect is counteracted by the term containing g_c/ω_i which is due to the overlapping of flutes [133], and by the term containing the finite ion Larmor radius a_i [135]. At low plasma densities where $\epsilon \leq \epsilon_0$ Eq. (114) becomes modified and the former effect is reinforced [134, 141, 142].

In cylindrical geometry there is also a contribution from the Coriolis force. In particular, if the finite Larmor radius effects are neglected and normal modes of the form $r\gamma \exp(im_\phi \phi + i\omega t)$ with constant γ are studied for unperturbed density distributions with $r_0 = n(r)/(dn/dr)$, a localized perturbation analysis yields the dispersion relation [51]

$$\omega \left(1 - \frac{\gamma^2}{m_\phi^2} - \frac{\gamma r}{r_0 m_\phi^2} \right) / \Omega = - \frac{r}{r_0 m_\phi} \pm \left[\left(\frac{r}{r_0 m_\phi} \right)^2 + \frac{r}{r_0} \left(1 - \frac{\gamma^2}{m_\phi^2} - \frac{\gamma r}{r_0 m_\phi^2} \right) \right]^{1/2} \quad (115)$$

Equation (115) reduces to earlier results by Chen [140] and Barbican and Rasmussen [143], and by Taylor [139] when γ is put equal to 0 and m_ϕ , respectively. The last term inside the square root of Eq. (115) arises from the centrifugal force which is destabilizing when $r_0 < 0$, and the last term yields a stabilizing effect due to the Coriolis force in this particular case.

(b) The results under (a) have been obtained from simple localized theory. Results which better correspond to real physical conditions can be obtained by treating the long cylinder case with a Gaussian density profile $n(r) = n_0 \exp(-r^2/r_0^2)$ and by applying certain boundary conditions on the disturbed state [80, 135, 140, 144]. For a free outer plasma boundary and for flute disturbances propagating only perpendicularly to the magnetic field the stability condition then becomes [80, 135]

$$\frac{1}{2} (1 - \sqrt{\gamma}) < \Omega r_0^2 / a_i^2 \omega_i < \frac{1}{2} (1 + \sqrt{\gamma}) \quad (116)$$

where $\gamma > 2$ is an integer and the sign of Ω is defined as in Eq. (12) with $B_z > 0$.

The analysis for the Gaussian density profile can be modified to include the effects of shear and resistivity on flute disturbances having the wave number $k_{\parallel}$ in the magnetic field direction [140,144]. It is then found that the corresponding stability condition for finite $k_{\parallel}$ and η is more difficult to satisfy than that given by relation (116). Further considerations of the instability of the $m_{\phi} = 1$ mode of a rotating theta pinch at $k_{\parallel} > 0$ are due to Freidberg and Wesson [196] among others.

(c) It is not sure that the Gaussian density profile is always a relevant approximation of the spatial distribution in a real plasma. In certain cases such as in systems of type 'A' described in Fig. 1 it is likely that the density increases steeply to its full value in narrow boundary regions close to the electrode surfaces [92]. Whenever line-tying effects are strongly pronounced along these surfaces, a constraint will be put on the plasma which, in combination with the finite Larmor radius effects, sometimes will be able to stabilize the flute disturbances. As an example we consider a simple plane case where there is a homogeneous magnetic field along z , and an unperturbed plasma density $n = n_0 \exp(y/y_0)$, in a boundary region of small thickness d . This region is bounded at the surface $y = d$ by a plasma of constant density $n_0 \exp(d/y_0)$, and at the surface $y = 0$ by magnetic field lines which graze a metal surface and can be considered as equipotentials. The stability condition for modes of the form $f(y) \cdot \exp(ik_x x + i\omega t)$ then becomes [145]

$$-4(g_c k_x^2 / y_0 \Lambda^2) / k_x^2 [(g_c / \omega_i) - \frac{1}{2} a_i^2 \omega_i / y_0]^2 < 1 \quad (117)$$

where $y_0 < 0$ when gravity points in the direction of decreasing density and $\Lambda^2 \equiv k_x^2 + (1/2y_0)^2 + \gamma^2$ with $\gamma d / tg(\gamma d) = (1/2y_0 - k_x)d$. In relation (117) the term containing a_i^2 is due to the stabilizing effect of the finite Larmor radius and the term g_c / ω_i is arising from the overlapping of flutes. At steep density gradients the former contribution dominates. Further, the factor $(\Lambda/k_x)^2$ is due to the constraint imposed on the flutes by the line-tying effects at the boundary $y = 0$ in combination with the reduced size of the boundary region, from the characteristic width r_0 of the Gaussian profile to the width $d \ll r_0$ in the present case. This diminishes the driving force of the instability by a factor of about $|\Lambda/k_x| \approx 1/2 |y_0| k_x$ for small $|y_0|$.

In this connection it should finally be pointed out that the condition (117) has been obtained from short-circuit effects acting on the outermost plasma boundary region only. Stabilization by line-tying can also be achieved by placing conducting walls all the way across the magnetic field lines of an open magnetic bottle [104]. However, since this will short-circuit the radial electric field it cannot be applied to a rotating plasma. It is also important to note that line-tying of the plasma to metal walls depends on the efficiency by which an electric contact can be established between the plasma and the walls [104, 145].

We summarize the present stability theories by pointing out that there are several mechanisms such as those from the finite Larmor radius, line-tying, overlapping of flutes, and the Coriolis force which under certain conditions will counteract the development of flutes in a rotating plasma. Whether these mechanisms become efficient enough to stabilize the plasma completely depends on the properties of the particular system under consideration, i.e. on the shapes of the density and temperature distributions, on the question whether the outer boundary should be considered to be free as in a theta pinch or subject to constraints imposed by the electrodes, and on the state of the surrounding neutral gas.

2.9.2. Neutral drag instability

It has earlier been shown by Simon [146] and Hoh [147] that the plasma rotation due to a radial electric field and the associated drag force from ion-neutral collisions produce a charge separation in the azimuthal direction. When the applied electric field points in the outward radial direction, the azimuthal electric field produced by the drag will generate a radial motion towards the axis of rotation. A reversed polarity of the applied field also reverses the radial motion and leads sometimes to instability. This mechanism affects rotating plasmas in the following way:

(a) As long as there is neutral gas within the whole plasma body, such as during the starting process, the formation of flute-like spoke-shaped plasma disturbances moving radially outwards will be reinforced by the neutral drag mechanism when the inner electrode acts as cathode. At opposite polarity the outward motion of the spokes will be counteracted by the neutral drag.

(b) When the main plasma body is fully ionized as in Fig. 1, the neutral drag is located to the wall layers and boundary regions only. If the outer electrode acts as cathode, this will help to suppress the formation of flutes at the outer plasma boundary. Since the velocity of rotation (51) in type 'A' systems is usually quite small in the boundary region, however, the influence of the neutral drag is expected to become quite small in such systems.

2.9.3. Velocity shear instability

According to Section 2.5.2. the velocity field is expected to become sheared in many rotating plasmas. We should then expect the Kelvin-Helmholtz (or velocity shear) instability to arise under certain conditions [148, 149] which shall not be treated in detail here.

2.9.4. Gravitational electromagnetic mode

The flute instability of Section 2.9.1. is an electrostatic mode which does not produce any significant currents which perturb the magnetic field. A corresponding gravitational electromagnetic mode may sometimes arise which is associated

with perturbations of the magnetic field. This mode has earlier been described by Kulsrud [150] among others, and recently by Ariel [151] who has shown that the Hall effect and a sufficiently high conductivity $1/\eta$ will suppress the instability in a rotating plasma.

2.9.5. Cyclotron resonance instability

When the angular frequency Ω of the rotating plasma is tuned to some of the natural frequencies of oscillation, a resonance phenomenon may arise which leads to perturbations of increasing amplitude. An example of this has been given by Rozhkov et al. [152] who found a resonant coupling with ion cyclotron oscillations when $\Omega = \omega_i/2$, where ω_i is the ion cyclotron frequency.

3. EXPERIMENTS

The rotating plasma experiments performed so far will now be described and compared to theory. Since earlier investigations have already been reviewed [48-52], this section will be largely devoted to recently obtained results.

3.1. DEVICES

Rotating plasmas are usually produced by discharges in externally imposed crossed magnetic and electric fields. It should, however, be observed that a torque may sometimes arise under less evident conditions such as those prevailing in theta pinches and other systems with complicated boundary conditions and mechanisms for angular momentum transfer [78, 153, 154]. The conventional way of producing a rotation has already been demonstrated by Figs 1-3 and is used in practically all the rotating plasma projects of Table I. In the table part of the more relevant technical data have been listed for various devices. Some concrete examples of apparatus design are further given by the disc-shaped Homopolar I of Fig. 6, the Ixion of

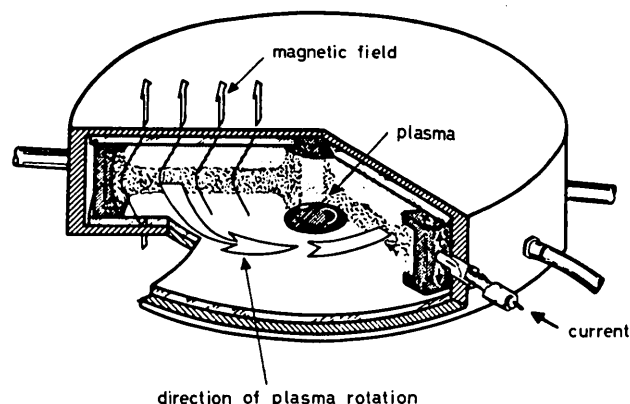


FIG. 6. Homopolar I consisting of a flat disc-shaped discharge chamber placed in a nearly homogeneous magnetic field [13].

Fig. 7, and the F I apparatus of Fig. 8, all being type 'A' systems. Among the type 'B' systems the Homopolar V puffatron device should be mentioned which is similar to that outlined in Fig. 2(a). Most of the type 'C' ionizing front systems and rotating plasma guns are somewhat like that shown in Fig. 2(b). The rotating plasma guns are usually type 'B' or 'C' systems which have a magnetic field diverging in the axial direction, away from the central electrode. Examples of type 'D' and 'E' systems are finally given by the rotating field pinch which is similar to the system of Fig. 3, and by the stellarator, respectively.

3.2. METHODS OF MEASUREMENT

The diagnostical methods which have been applied to explore rotating plasmas can be summarized as follows:

(a) Direct inspection of the damage of the insulators and electrodes, and an analysis of the gas in the discharge chamber after a shot gives certain information about the interaction between the plasma and the walls [37, 39, 50, 155, 156].

(b) Measurements of the voltage and current applied between the electrodes can be used to determine the average properties of the plasma body. The interpretation is particularly straightforward for rotating plasmas, on account of the existing mass motion. By a short-circuit between the electrodes, the angular momentum can be recovered in the form of a reversed current pulse as predicted by Eq. (79). In a free-wheeling mode where the plasma is released from its energy source, the decay time of the rotation becomes a direct measure of the internal momentum losses. A simple example of a circuit used for this purpose is given in Fig. 9.

(c) Small 'flags' can be inserted [17] or small glass spheres can be made to fall through the plasma [52, 157] to indicate the macroscopic fluid motion and stored angular momentum.

(d) A division of the electrode surfaces into separate segments or 'current buttons', or the introduction of electrostatic probes and a measurement of the corresponding voltage and current characteristics gives information about the velocity of rotation and the presence of azimuthal inhomogeneities in the discharge [89, 102, 106].

(e) Double and single electric probes can in some cases be used to measure the potential distribution and electric fields in the plasma [100, 106, 117, 158, 159]. Electric probes may also give information about the behaviour of the outer plasma boundary as shown e.g. in Fig. 8 [160]. Care is necessary in the interpretation of the probe signals when such arrangements are used inside a dense fully ionized rotating plasma which interacts heavily with the probes.

(f) Magnetic probes and loops measure the local or average magnetic fields induced by the centrifugal force and by the thermal pressure of the plasma [13, 19, 20, 52, 100, 158]. They can also

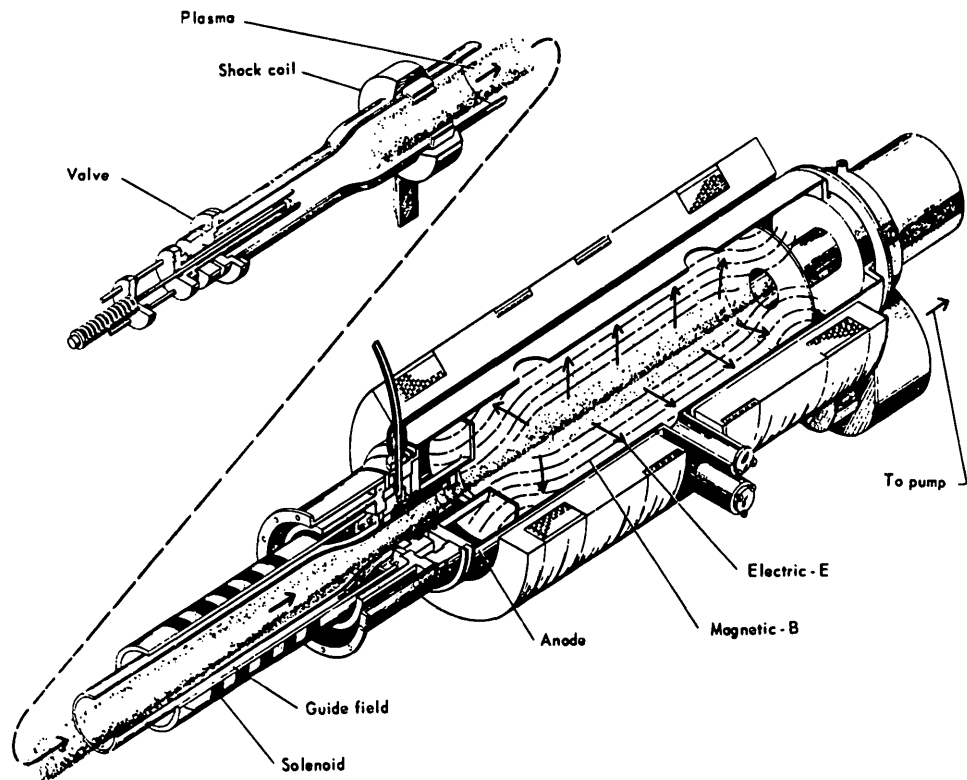


FIG. 7. Ixion device consisting of an elongated discharge chamber with a magnetic mirror field. A central plasma column is produced by a pulse of ionized gas and is used as cathode [12].

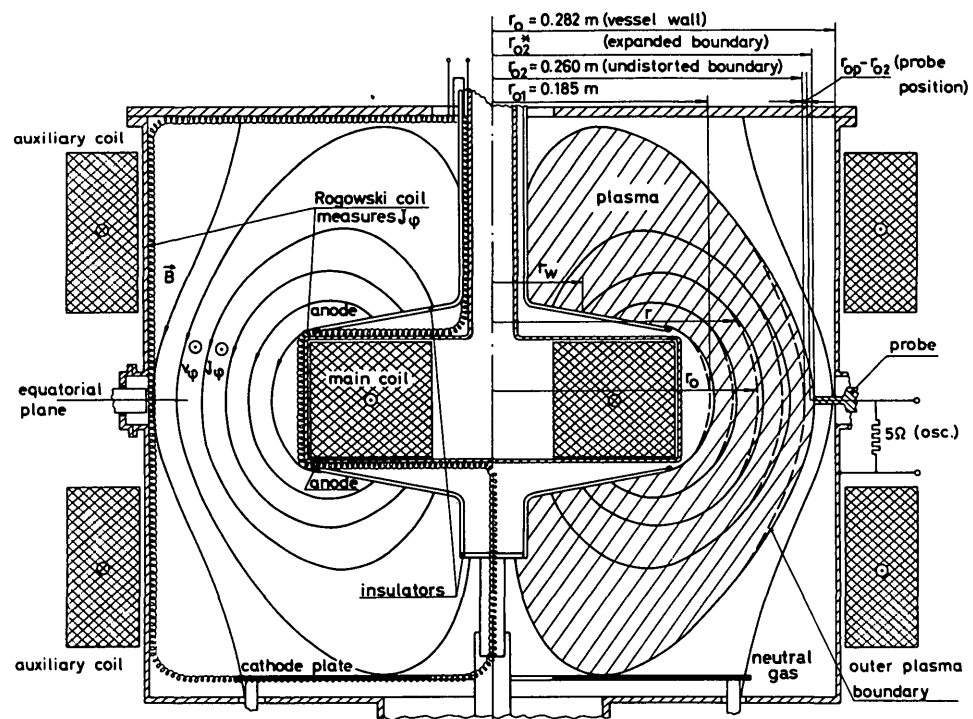


FIG. 8. F I device which is characterized by a curved magnetic field having a large radial ratio.

The shaded area indicates the plasma confinement region in a situation where the magnetic field has been expanded in the radial direction by the centrifugal force. The vacuum magnetic field is given by the dashed lines [160].

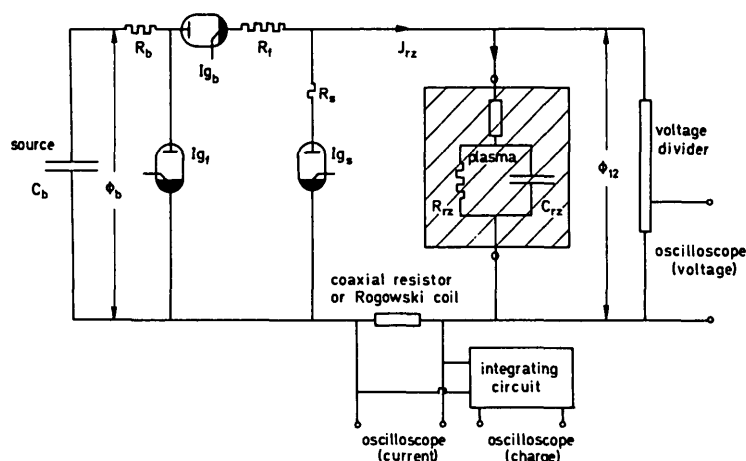


FIG.9. A simple example of the discharge circuits used for rotating plasma production.

The energy source consists of a condenser bank C_b which is connected with the electrodes of the plasma device through the resistances R_b and R_f when the ignitron I_{g_b} is being fired. The plasma can be put into a free-wheeling mode when the energy supply and the current J_{rz} are cut off by firing ignitron I_{g_f} . Ignitron I_{g_b} then stops conducting. Finally, the angular momentum of the plasma can be recovered by applying a short-circuit across the electrodes by means of ignitron I_{g_s} , provided that the resistance is kept small enough.

be used to study azimuthal plasma inhomogeneities and their motions [52, 157].

(g) Small Rogowski coils are suitable for local measurements of the plasma currents [52, 89, 100, 157, 161]. Rogowski coils can also be wrapped around the plasma in a plane through the symmetry axis to measure the total azimuthal current and the integrated effects of the centrifugal force and the thermal pressure, as demonstrated e. g. in Fig. 8 [160, 162].

(h) The state of ionization and the particle balance due to a circulation of matter between the plasma and the neutral gas region can be investigated by means of pressure probes placed in the external neutral gas region [52, 157]. Further, piezoelectric probes may sometimes be placed inside the plasma to measure its pressure fluctuations [106].

(i) Special probe arrangements with valves can be arranged for taking samples from a fully ionized plasma [59].

(j) Neutral particle analysers measure the flux of emitted neutrals due to charge exchange collisions with the rotating plasma. This gives information about the state of ionization and the azimuthal plasma velocity [163, 164].

(k) When the random motion of the ions is large enough their energy and density can be determined from the neutron production in deuterium and tritium plasmas [12, 18, 163].

(l) High-speed photography and registration of the emitted light reveals inhomogeneities, instabilities and the shape and motion of the plasma body [50, 100, 143, 160, 165-168].

(m) Spectroscopic studies are used to determine the impurity levels, the macroscopic velocity field, and the density and temperature of the plasma [12, 13, 38, 52, 100, 157, 169]. It should then be observed that non-uniform conditions of the plasma, as expressed by Eq. (51), give rise to an integrated

effect over the plasma body [169]. Time-resolved spectroscopy by means of monochromators gives information of the instantaneous properties of the plasma [12, 166, 170, 171].

(n) Reflection and transmission of imposed microwaves provides a tool for investigating the behaviour of the plasma, its boundaries and stability, as well as the state of ionization [52, 170, 172-175]. In addition, the emitted microwave radiation can be used to reveal instabilities [152, 176].

(o) The time variations of the plasma density are sometimes recorded by means of laser interferometry [177].

(p) Measurements by means of bolometers, pin-hole cameras or photocells give information about the energy losses due to radiation [161].

3.3. STARTING PERIOD

When a discharge starts in crossed electric and magnetic fields, a number of interaction processes between charged and neutral particles will take place simultaneously. In this section we discuss each of these processes separately.

3.3.1. Ionization process

The build-up of a plasma by a discharge in crossed fields is not always a straightforward procedure. Thus, the main contribution to the ionization is usually expected to arise from the electrons, but these have small Larmor radii and should therefore have difficulties in acquiring energy from an electric field being at right angles to a strong magnetic field.

In early experiments with magnetrons partially ionized plasmas have been created by a sheath mechanism [14-18]. Neutral gas which enters the

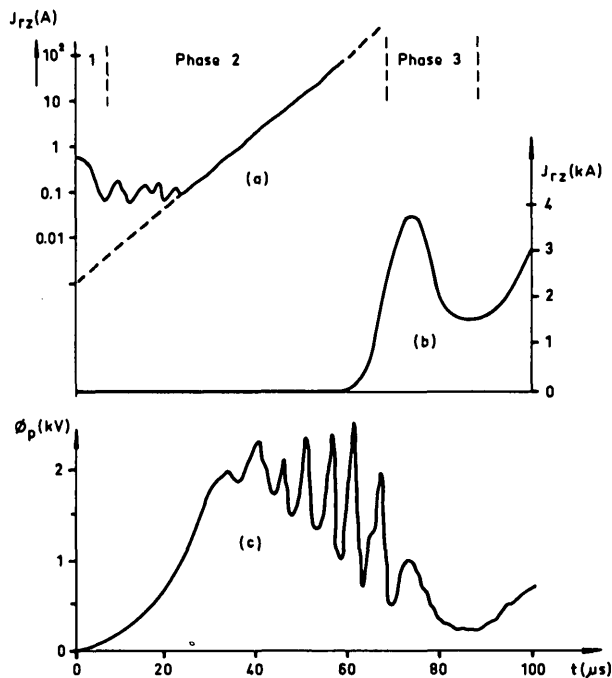


FIG. 10. Measured discharge current J_{rz} and probe voltage ϕ_p in hydrogen as functions of time during starting period of Kruisvuur I device with negative polarity [89].

- Discharge current on (a) logarithmic,
 (b) linear scales,
 (c) probe signal in midplane at the radial distance
 $r = 0.12$ m from axis.

anode sheath is ionized by the high field which exists there, and forms ions which are accelerated out of the sheath to circulate in magnetron orbits.

The discharge build-up at filling gas densities of 10^{-2} mmHg has been studied by Rasmussen et al. [89] in the Kruisvuur I device which is a long homopolar-like arrangement having magnetic mirrors and short central electrodes with pulsed gas inlets at each end. The increasing radial driving current J_{rz} demonstrated in Fig. 10 can here be taken as a measure of an increasing ionization degree. In the range of J_{rz} from 0.1 to 100 A the experiments were found to be consistent with impact ionization by thermal electrons as given by Eq. (40), except in certain cases where also ionization by positive ions becomes important. In this range of currents, which appears to be below the burn-out level (106), the plasma is partially ionized.

Further experimental and theoretical studies of the ionization process and the formative time for breakdown in strong crossed fields have recently been reported by Sherwood and Kunkel [178] for a hydrogen discharge at 0.2-2 mmHg pressure in a homopolar configuration. The experiments were found to be consistent with theory.

The generation of a clean rotating high-temperature plasma in a mirror machine has also been studied by Forsen [197], Forsen and Trivelpiece [198], and by Harder [199].

When the neutral gas pressure before a discharge in the F I and F II devices is chosen in the range below some 10^{-3} mmHg, the discharge does not behave in the quiescent way observed at higher densities. Instead it shows up some irregularities as well as an increasing gassing rate of electrode and wall material [109]. This behaviour may be explained by the theory of Section 2.1., according to which a density limit is predicted by condition (100), of the order of the observed one. Thus, when the density of the filling gas is chosen below its minimum value $n_{\min.}$, one should expect a gassing to start from the walls to provide an extra amount of neutral particles such as to satisfy condition (100).

When the ionization proceeds to a fully ionized state it is not always sure that thermal ionization in an axially symmetric plasma can explain the short build-up times observed in some high-density experiments, being of the order of a few microseconds as measured from the equivalent capacity [36, 38, 39, 109, 160]. This should be the case even if thermal ionization by ions and neutrals is included. We shall later return to this question in connection with Section 3.6.

3.3.2. Spoke formation

The radial currents which are necessary for a rotating plasma to be generated and sustained are large enough for cathode spots, and possibly also anode spots, to be formed at the cold electrode surfaces [50, 102]. Devices with such electrodes should therefore have a natural tendency of forming spoke-shaped azimuthal plasma inhomogeneities. Such conditions are strongly pronounced in all types of devices during the starting period where a large power input is needed for plasma burn-out and acceleration, but they may also prevail during a quasi-stationary holding mode in systems having large plasma losses. The spoke formation may be further reinforced by a pinch effect, and is also affected by the electrode material and plasma injection methods [50]. It is not unlikely that spoke formation becomes necessary for the ionization process itself, especially if non-thermal ionization mechanisms of the types described in Section 2.7.2.(b) are needed for the ionization balance.

Among the plasma mechanisms which affect the spokes, the neutral drag force of Section 2.9.2. is of special interest. With the inner electrode acting as anode, which we shall denote as 'positive polarity' henceforth, the corresponding charge separation yields a radial drift along the spoke towards the axis of the device, and vice versa. Further the centrifugal and pressure forces in magnetic fields with a radially decreasing strength produce an azimuthal charge separation leading to flute-type disturbances and an outward radial drift [132]. Thus, the radial motions produced along the spokes by the neutral drag and the centrifugal force counteract each other at positive polarity and reinforce each other at negative polarity. In addition, deviations of the azimuthal guiding centre

drifts from rigid body rotation will tend to deform the spokes and 'smear' them out across the confinement region.

With these electrode and plasma phenomena in mind, the experiments on rotating plasma spoke formation can be summarized as follows:

(a) In the Homopolar III experiments by Baker et al. [34] and Kunkel et al. [106] it was found that, during the entire period of steady rotation, the discharge current was confined to a set of about 10 to 12 narrow radial spokes rotating with the plasma. An example of this is given by the current-button responses demonstrated in Fig. 11. The average radial ratio of the device was $R_r \equiv r_0/r_w \approx 1.16$ and the polarity negative.

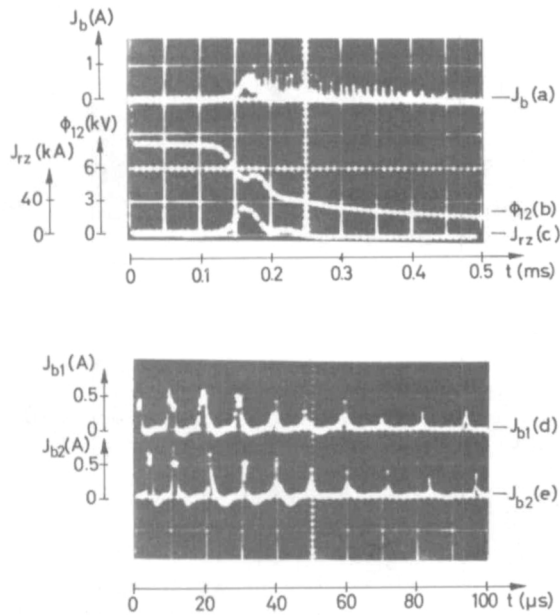


FIG. 11. Experiments with Homopolar III device [106].
(a) Current-button response J_b during the entire course of events.
(b) Electrode voltage ϕ_{12}
(c) Driving current J_{rz}
(d), (e) Current button responses J_{b1} and J_{b2} on a magnified time-scale.

(b) In the Leatherhead Homopolar having the radial ratio 1 and negative polarity, Barber et al. [179] observed one or a few spokes to arise during the initial breakdown of the discharge. They were soon smeared out to form spirals with increasing velocity in the outward radial direction. Later new spokes were formed and the plasma equilibrium appeared to be destroyed by disturbances similar to Rayleigh-Taylor instabilities.

(c) For the Kruisvuur I device, having negative polarity, a radial ratio of about 1 with respect to the end insulators, and an average mirror ratio $R_m = B/B_0 \approx 2.5$ with respect to the ends of the central electrodes, Rasmussen et al. [89] observed a single eccentric $m_\phi = 1$ structure rotating around the axis at a velocity close to E/B . The signal from an electrostatic probe in the midplane of the device showed steep regular fluctuations consistent with the velocity E/B during the later phases of the starting period, as demonstrated by Fig. 10(c).

However, these fluctuations vanished as soon as the radial discharge current reached above some kA, i.e. in the range where burn-out to a fully ionized state should take place.

(d) In the F I device having $R_r = r_0/r_w \approx 3$ and positive polarity, earlier observations of the starting period by means of a segmented outer electrode surface also indicated the existence of a regular single spoke moving at the E/B velocity [102]. Later investigations of this phenomenon are illustrated by Figs 12 and 13 which include recordings made by means of the Rogowski coil in Fig. 8 which measures the azimuthal current J_ϕ and its inhomogeneities [160, 162]. Thus, in Fig. 12 the power input is below the burn-out level (107), the plasma never gets fully-ionized, and there is a single spoke moving at the E/B velocity around the device during the entire period of the discharge up to the instant t_s of an applied short-circuit. In Fig. 13, however, the power input has been chosen above the burn-out level and the spoke formation lasts only up to about 0.2 msec from the start, i.e. as long as the plasma is partially ionized and does not show up any measurable stored angular momentum.

(e) In the plasma source experiments by Harder [199] with positive polarity, there was optical evidence of a single ionizing spoke being responsible for the neutral gas burn-out.

(f) In a study of ionizing waves in a system similar to that of Fig. 2(b), Millar and Watson-Munro [100] observed 1 to 3 radial current filaments which developed at the shock front.

(g) For the Ixion device which has a central plasma cathode, there has to the author's knowledge not been observed any spoke formation [12, 158, 161].

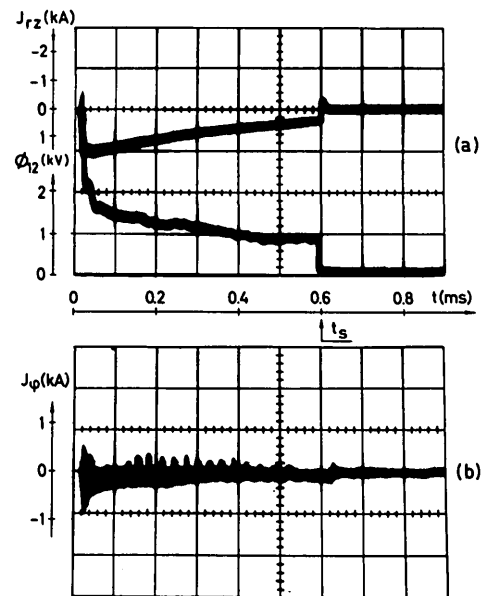


FIG. 12. Voltage ϕ_{12} between electrodes, driving current J_{rz} , and azimuthal current J_ϕ for hydrogen in device F I as functions of time when power input is below burn-out level [160, 162].

A short-circuit is applied between the electrodes at $t = t_s$. Initial bank voltage is $\phi_{b0} = 8$ kV at start of discharge. Total external series resistance $R_b + R_f = 4.11$ ohm (see Fig. 9).

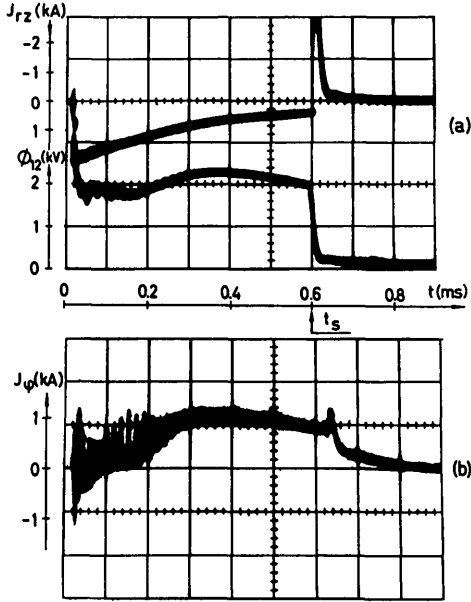


FIG. 13. Same as Fig. 12, but power input has been chosen slightly above burn-out level by increasing initial bank voltage to $\phi_{b0} = 10$ kV.

(h) One of the mechanisms which may contribute to the elimination of the spokes at later stages of the discharge is due to the electromotive field $\vec{v} \times \vec{B}$ of a moving spoke. This field tends to channel the driving currents to regions which have so far not been ionized and set into full rotation.

From the various experiments described here it is therefore seen that spoke formation is not a proof of fully ionized rotating plasmas to be unstable. Spoke formation appears instead to be connected with the starting period in most devices, as well as with lossy systems in general, where radial and mirror ratios close to unity give rise to large quantities of escaping plasma and to a corresponding neutral back-flux. Also electrode phenomena and the polarity of the devices seem to be important to the understanding of this phenomenon.

3.3.3. Burn-out

From Figs 12 and 13 it is seen that a sharp transition takes place from a state with negligible amounts of angular momentum stored in the rotation to a state with a large azimuthal current J_ϕ and recovered charge $Q_e(t_s)$. This occurs as soon as the initial bank voltage ϕ_{b0} is increased from 8 to 10 kV. Further measurements under similar conditions but with a smaller series resistance $R_b + R_f$ in the circuit of Fig. 9 are demonstrated by Fig. 14 [160, 162]. In the latter the equivalent capacity $Q_e/\phi_{12} \approx C_{rz} \propto n_0$ defined by Eq. (75) and the quantity $J_\phi/\phi_{12}^2 = 2n_0 m L_{II}/(r_{02}^2 - r_{01}^2) B_0^3 \propto n_0$ have been plotted as functions of ϕ_{b0} . This clearly indicates that a transition takes place from a lowly ionized state with a small plasma density n to a fully ionized state with a nearly constant plasma density n_0 , as soon as ϕ_{b0} is increased beyond the burn-out voltage $\phi_{boc} \approx 5.2$ kV. The density obtained from

the measurements above ϕ_{boc} is $n_0 \approx 1.6 \times 10^{21} \text{ m}^{-3}$, which is consistent with the initial filling density $n_{n0}/2 \approx 10^{21} \text{ m}^{-3}$ of H_2 molecules within the limits of experimental error. This figure, as well as the horizontal level of the marks and dots in Fig. 14, suggests that the neutral gas gets completely ionized within the shaded region of Fig. 8 as soon as $\phi_{b0} > \phi_{boc}$.

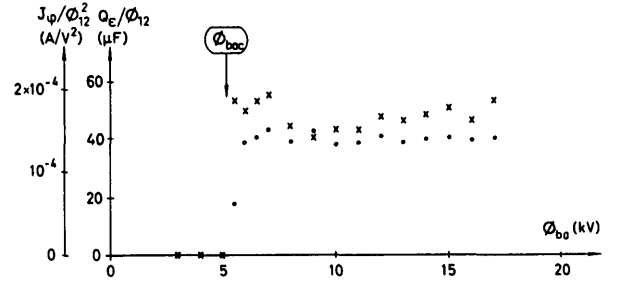


FIG. 14. Equivalent capacity Q_e/ϕ_{12} (dots) and the quantity $J_\phi/\phi_{12}^2 \propto n_0$ (crossed marks) as functions of the initial bank voltage ϕ_{b0} of device F I [160, 162].

The critical voltage ϕ_{boc} corresponds to the burn-out limit.
 $R_b + R_f = 1.48$ ohm.

Measurements at varying series resistances $R_b + R_f$ have given a linear relation between ϕ_{boc} and $R_b + R_f$ with B_0 as parameter, yielding a corresponding critical burn-out current J_{rzc} which agrees with Eq. (106) within a factor of about 1.5 [162].

3.3.4. Neutral gas flow

During the starting and burn-out process a large outflux of neutral gas should arise from the confinement region to the outer wall-near regions, on account of elastic and charge-exchange collisions with the rotating plasma ions. Such a flux of neutrals has in fact also been detected in Homopolar V by means of a particle analyser [163].

Further, Bannenberg et al. [157] have placed a specially constructed pressure gauge in the wall-near regions and have recorded a tenfold increase in neutral gas density, 500 μs after the start of the discharge (Fig. 15). A considerable part of the recorded increase may be due to released impurities, but it is also possible that some pressure rise is caused by the outflow of neutrals from the plasma confinement region during the discharge.

3.4. ACCELERATION PERIOD

We now turn to the acceleration process which follows the starting period in systems which are able to become completely ionized, with the exception of existing narrow wall layers and boundary regions. Consequently, we now disregard the large particle, momentum, and heat losses which arise from neutral gas interaction within the main plasma body during the early ionization, spoke formation, and burn-out processes.

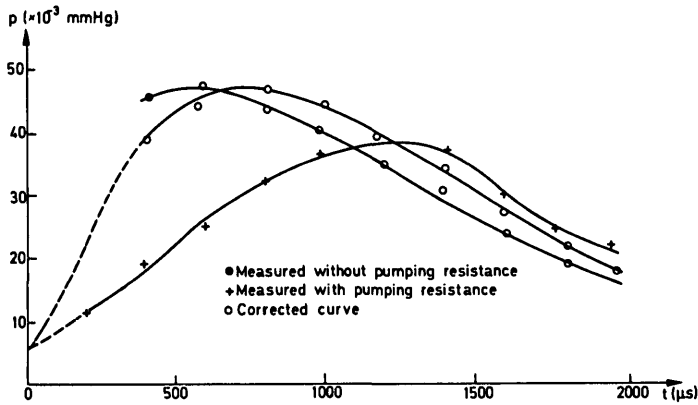


FIG. 15. Time variation of the neutral gas pressure as recorded by an ionization gauge placed at the outer electrode wall, in the plane of a magnetic bottle-neck in Kruisvuur I device [52, 157].

3.4.1. Azimuthal acceleration

Direct indications that the plasma is set into rotation have already been obtained by Anderson et al. [13] and Boyer et al. [12] among others from spectroscopic measurements of the Doppler shift, which also revealed gas impurities released from the walls. Further direct measurements of the rotation have been reported by Gow et al. [17] using a 'flag' of light material in the ion magnetron, and by Bannenberg et al. [157] and Insinger [52] who have studied the momentum transfer to small glass spheres falling through the plasma. In a hollow cathode discharge Boeschoten and Demeter [200] detected a rotation of the plasma by means of a small pendulum and by spectroscopy. The driving force was due to radial electric currents from non-ambipolar diffusion [201]. Similar conclusions were drawn by Vlasov et al. [202] for experiments with a hot-cathode discharge.

According to Section 2.5.2. and Eqs (75) and (79) the stored angular momentum should also be detectable as a reversed current pulse when a short-circuit is applied between the electrodes and the rotating plasma works as an electric generator. Such short-circuit or 'crow-bar' measurements have confirmed the existence of the equivalent capacity and the corresponding stored angular momentum [13, 36-39, 52, 102, 155, 160, 162]. An example is given by recent investigations [169] shown in Fig. 16. After being ionized, preheated, short-circuited and kept in a non-rotating state during a waiting time of about $100 \mu\text{s}$, a fully ionized plasma is accelerated at time t_3 and short-circuited at time t_{s3} . As seen from Fig. 16(b) about 80% of the stored angular momentum and its associated charge can be recovered by the short-circuit. The missing 20% are partly due to particle and viscous losses acting during a period of about $100 \mu\text{s}$ after time t_3 , partly to deviations from reversibility of the velocity profile as defined in connection with Eq. (79). If the waiting time is increased by choosing $t_3 = 0.76 \text{ ms}$ from the start, less than 5% of the charge is recovered at t_{s3} , on account of volume recombination and interaction with neutral gas which penetrates into the plasma during the waiting time, as explained in Section 2.4.

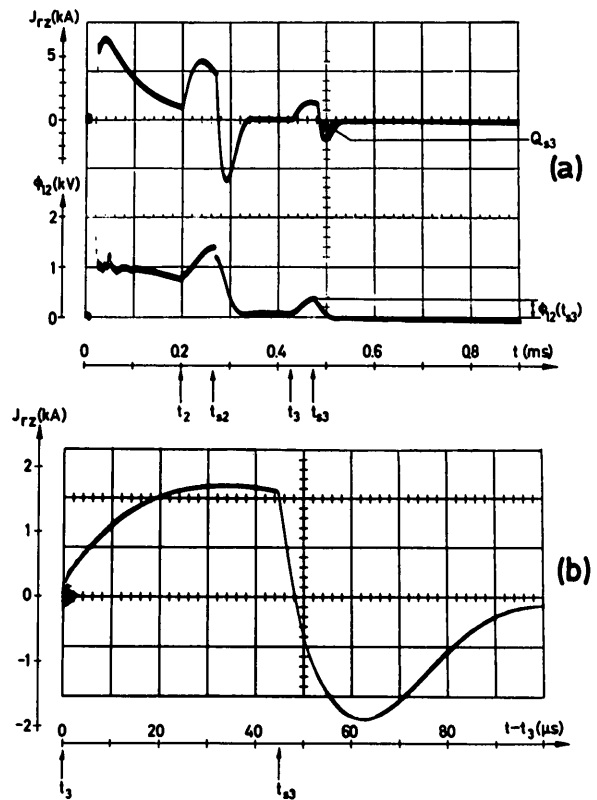


FIG. 16. Driving current J_{rz} and electrode voltage ϕ_{12} in device F II [169].

- (a) Entire course of events.
- (b) The current during the last acceleration period on a magnified time-scale. The figure shows the behaviour of a fully ionized plasma accelerated from rest, without being involved in an ionization or burn-out process.

There are several methods by which the equivalent capacity C_{rz} can be measured:

- (a) In the puffatron devices [163, 177] a fully ionized plasma is created during a few microseconds out of an injected neutral gas blob, somewhat as outlined in Fig. 2(a). Provided that the ionization is rapid enough, one should expect the momentum losses from neutral gas interaction to become small compared to the momentum gain of the plasma during the acceleration. Consequently,

the storage of momentum should reveal itself as a step in the applied bank voltage ϕ_{b0} and in the plasma voltage ϕ_{12} , just as when a charged condenser is connected with a loss-free uncharged one. Measurements of this kind are demonstrated by Fig. 17 and are in good agreement with Eq. (75) from which C_{rz} is expected to become proportional to the filling gas density [177, 180].

(b) In type 'B' systems practically all the injected neutral gas becomes ionized. In the systems of type 'A' shown in Fig. 1, however, the situation becomes more complicated because there is a communication between the plasma and a surrounding neutral gas blanket. In this case C_{rz} will vary more slowly with the magnetic field than $1/\bar{B}_0^2$, as predicted by the theory of Section 2.5.1. in combination with Eq. (75). This is demonstrated by Fig. 18 [92].

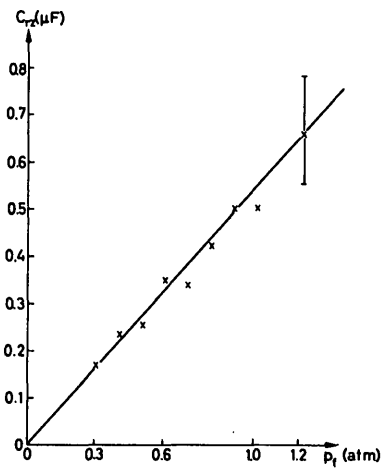


FIG. 17. Equivalent capacity C_{rz} as a function of the filling pressure p_f in the Risø puffatron [177, 180].

The capacity has been determined from the drop in the electrode voltage which is caused by the plasma acceleration $\bar{B}_0 = 1.5 \text{ V} \cdot \text{s} \cdot \text{m}^{-2}$ and $\phi_{b0} = 25 \text{ kV}$.

(c) With a known inductance L_{rz} included in the circuit of the driving current J_{rz} of Fig. 5, the plasma can be put into an oscillatory mode of operation. The measured frequency $2\pi/(L_{rz}C_{rz})^{1/2}$ is then used to determine C_{rz} . This method has been applied to the Ixion yielding the result shown in Fig. 19 [158]. The experimentally obtained values of C_{rz} are proportional to $1/\bar{B}_0^2$, except at high bank voltages ϕ_{b0} , where the deviations from this relationship can be explained by conduction across the end insulators. The difference between the $1/\bar{B}_0^2$ dependence of C_{rz} in Fig. 19 and the more complicated behaviour in Fig. 18(a) can be explained by the fact that little space is left in the Ixion for a surrounding neutral gas blanket, and that the gas is admitted into the chamber by means of a fast-acting valve as shown in Fig. 7.

We therefore see that the theoretically predicted behaviour of the equivalent capacity C_{rz} has been verified by experiments, at least to its first approximation (75). The experiments are so far not accurate enough for a test of the influence on C_{rz} by the potential energy of the induced magnetic

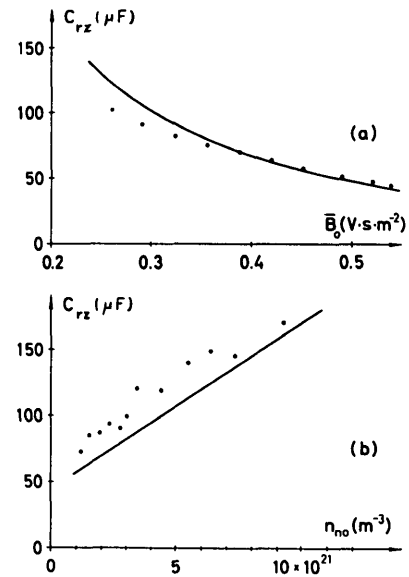


FIG. 18. Magnetic field and density dependence of the equivalent capacity C_{rz} of the rotating plasma in device F I having a neutral gas blanket [92].

(a) Variation of C_{rz} with the magnetic field strength at a fixed filling density $n_{n0} = 1.8 \times 10^{21} \text{ m}^{-3}$ of neutral atoms.

(b) Variation of C_{rz} with the filling density at a fixed magnetic field strength $\bar{B}_0 = 0.41 \text{ V} \cdot \text{s} \cdot \text{m}^{-2}$.

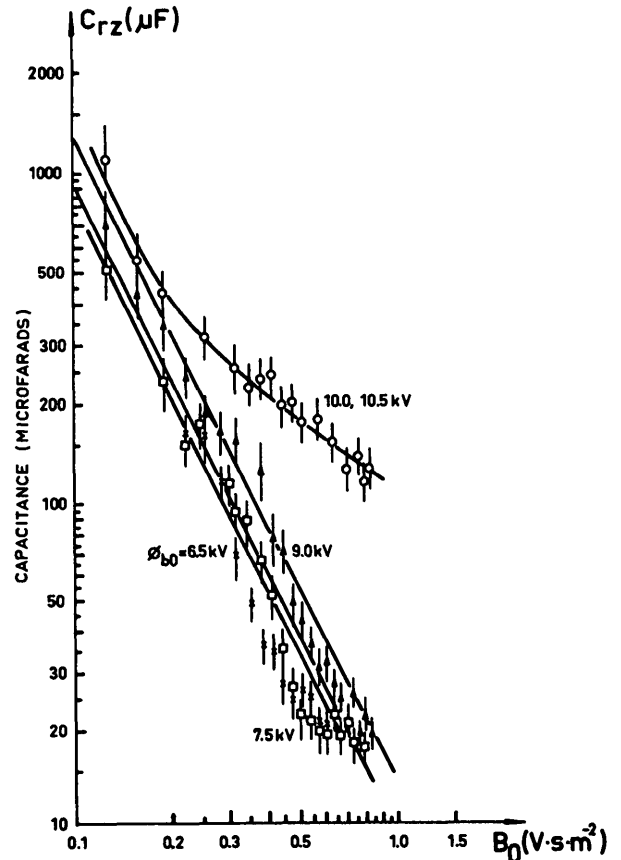


FIG. 19. Ixion equivalent capacity C_{rz} as deduced from period of oscillation [158].

field as described by Eq.(82). On the other hand, there are experiments which confirm that the centrifugal force and its corresponding beta value in the long cylinder approximation

$$\beta_c = 2 \mu_0 n_0 m \bar{G}_0^2 v_0^2 (r_{02} - r_{01}) / (r_{02} + r_{01}) \bar{B}_0^2 \quad (118)$$

becomes appreciable at least in certain devices. Thus, evidence that the plasma is at least partly leaning on the field lines and not only on the walls was already given by Anderson et al. [13]. It was further verified for the Ixion [20, 158], where the magnetic field was found to decrease at the centre of the device in the presence of the plasma. An example of these measurements, which were performed with small magnetic probes, is given in Fig. 20. In this connection it should be observed that there were probably also contributions from heavy plasma impurities in the Ixion device [12].

Finally, measurements with the F I device have shown that large induced magnetic fields and regular radial displacements $r_{02}^* - r_{02}$ of the outer

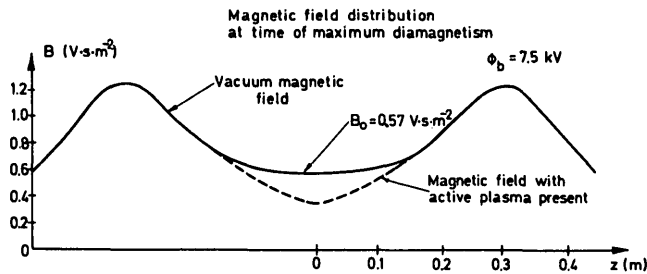


FIG. 20. Axial distribution of the plasma diamagnetic effect in Ixion [20].

Solid curve shows the axial component of the applied vacuum field B as a function of the axial distance z from the midplane, and dashed curve the corresponding field when the plasma is present.

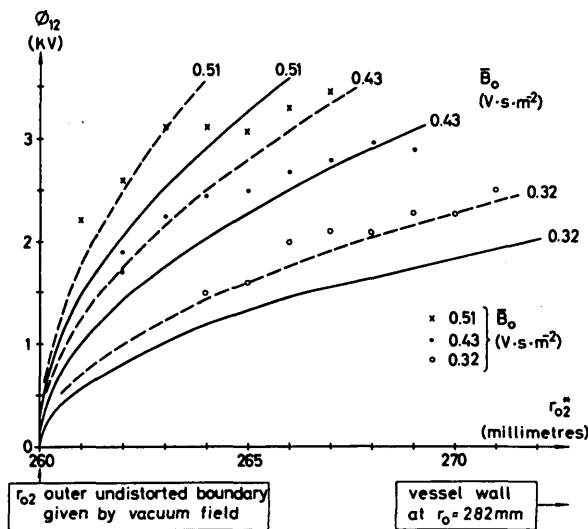


FIG. 21. Plasma voltage Φ_{12} in device F I at which a signal appears from the probe when it is placed at the radial position r_{02}^* [160]. Marks indicate experiments. Full and broken curves represent Eq.(84) with $n_0 = 3 \times 10^{21}$ and $2 \times 10^{21} \text{ m}^{-3}$.

plasma boundary are obtained in agreement with Eq.(84). The measurements were made by means of the probe and the Rogowski coil in Fig. 8 and are demonstrated in Figs 13 and 21 [160].

3.4.2. Axial acceleration

Rotating plasmas can be used as guns for plasma propulsion or for injection into magnetic mirrors by means of a diverging magnetic field. Thus, the Homopolar V device has been modified by Halbach and Baker [181] to work also as a homopolar gun. The magnetic field in the valve region is then chosen 10% higher than in the long drift region to transform 10% of the transverse energy into axial motion.

3.5. QUASI-STATIONARY PERIOD

We now proceed to the quasi-stationary state which enters after the acceleration has been completed and an equilibrium has been established between the connected energy source and the plasma. The capacity C_{rz} of Fig. 5 has then been charged to its final voltage and the current J_{rz} is mainly flowing through the resistance R_{rz} , on account of existing losses.

3.5.1. Holding mode

When the voltage across the electrodes is sustained by the energy source during times which exceed the relaxation times of the plasma balance described in Section 2.5., the plasma can be said to have reached the state of a holding mode. In most fully ionized cases of experimental interest the plasma has then become nearly isotropic and has expanded axially to get into full contact with the end insulators. During this mode the velocity and temperature distributions (51) approach quasi-stationary forms which depend on the particular experimental geometry, and are different from those developed during the acceleration process. The stored angular momentum is considerable even at the relatively small driving currents J_{rz} which are sometimes required during the holding mode. This has been demonstrated by applying a short-circuit and recovering the corresponding electric charge at various times, in some cases even as long as 5 ms after the end of the acceleration period [39]. This is further demonstrated by the upper trace of Fig. 13(a), where a short-circuit occurs at time t_s .

Investigations of the velocity and electric field distributions over the plasma cross-section have given the following results:

(a) From probe measurements of the time difference in correlated ripples in the discharge Bannenberg et al. [157] concluded that the velocity in the Kruisvuur I device was nearly independent of the radius. The radial ratio of this device is close to unity and the end insulators should affect its momentum balance considerably. We return to this problem in Section 3.6.

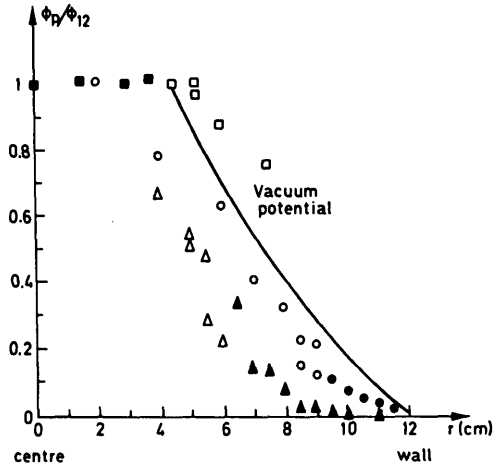


FIG. 22. Ratio between the measured probe voltage ϕ_p and the voltage ϕ_{12} between the electrodes as a function of the radial distance r from the axis of the Ixion [158].

Circular and square marks give voltage at midplane, and triangles at 19 cm from midplane. Open circles represent data where serious probe perturbations are evident.

(b) In the Ixion [158] probe measurements of the radial electric potential distribution were found to be nearly consistent with the vacuum potential field which arises when the magnetic field lines act as equipotential surfaces (Fig. 22). In addition, the electric field was found to drop to a small value near the outer wall. From the obtained voltage profile, the major portion of the rapidly rotating plasma was concluded to be situated in an annular region in the interior of the confinement region.

(c) Results similar to those mentioned under (d) have been obtained from spectroscopic investigations on device F II [169]. Thus, no Doppler shift could be detected for spectral lines of neutral hydrogen, Si I and Cl, whereas an increasing shift was found for the succession of lines SiII, SiIII, CII, CIII, and SiIV up to a maximum velocity of $1.5 \times 10^5 \text{ m} \cdot \text{s}^{-1}$ when special measures were taken to get impurities into the plasma. These results and those mentioned under (b) confirm the theoretical model of Fig. 1 and Section 2.5. for plasmas with comparatively small loss rates. According to this model the velocity v_ϕ and the temperature T should decrease to low values at the electrode surfaces which act as boundaries. The observations also support the model of a thin partially ionized boundary region inside of which there is practically no neutral gas present, and where the isorotation law (12) is valid.

(d) In certain experiments by Angerth et al. [182] and Hopfgarten et al. [164] at low densities and extremely large mirror ratios, in the range $20 < R_m < 100$, there are indications that the parallel electric field $E_\parallel$ is large enough for the isorotation law to be violated. Instead a type of 'partial co-rotation' is established, where the angular momentum seems to be constant along the field lines.

3.5.2. Free wheeling

The most convincing way of proving whether the centrifugal force is being balanced by the magnetic field, and whether the plasma losses are of the order expected from the classical theory of Section 2.5, is to cut off the plasma from its energy source and leave it in a free-wheeling mode. This can be done by firing ignitron I_{gf} in Fig. 9 and cutting off the driving current J_{rz} . The equivalent capacity C_{rz} of Fig. 5 is then discharged only through its leakage resistance R_{rz} . In such a case the time constant $\tau = R_{rz} C_{rz}$ becomes a direct measure of the particle and momentum losses of the plasma.

Free-wheeling operation with the Ixion has earlier given time constants of about $\tau = 50 \mu\text{s}$ [161]. Some recent results obtained with the F I device are demonstrated in Fig. 23 which shows nearly exponential decay rates of the electrode voltage ϕ_{12} and the azimuthal current J_ϕ , and a nearly constant value of J_ϕ/ϕ_{12}^2 which is a measure of the average plasma density [160, 183]. In certain experiments an external load or a short-circuit have also been applied during free-wheeling, and large electric charges have then been recovered in absence of any external power input [102]. The time constant τ determined from the slope of the voltage curve has been plotted in Fig. 24 as a function of the magnetic field strength, both for the Ixion and F I [160, 161]. Finally, free-wheeling operation has been applied to the F I device at power input levels being slightly lower than the experimentally determined burn-out level. Below burn-out there was then observed a drastic decrease in the free-wheeling time, to values of the order of $15 \mu\text{s}$ and less [160].

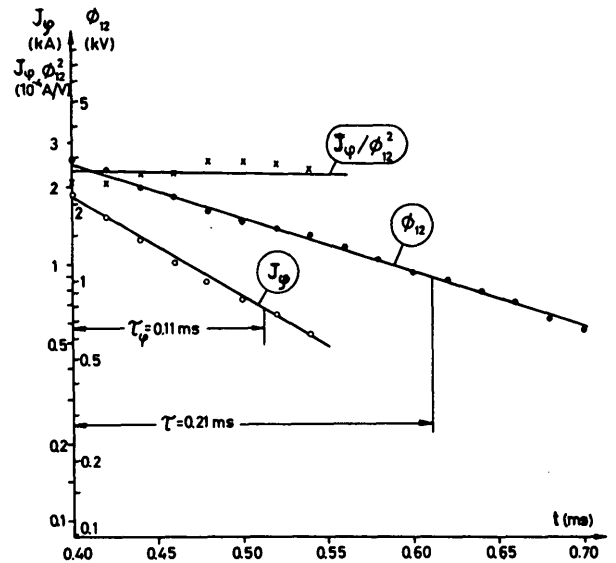


FIG. 23. Electrode voltage ϕ_{12} and azimuthal current J_ϕ of device F I during a free-wheeling mode beginning 0.4 m.s from the start of the discharge [160].

The marks indicate points taken from the oscilloscope traces of one single shot. The experimental deviation from shot to shot is comparable to the size of the marks. The quantity J_ϕ/ϕ_{12}^2 is proportional to the average plasma density, $\bar{E}_0 = 0.55 \text{ V} \cdot \text{s} \cdot \text{m}^{-2}$.

The following statements should therefore apply at least to the free-wheeling experiments, with the Ixion, F I and F II devices:

(a) Without any external energy source there remains an appreciable stored angular momentum in the plasma of F I for times of the order of several hundred microseconds. The equivalent beta values (118) of the centrifugal force are then even of the order of 0.1 under certain conditions. In the Ixion the decay times are shorter on account of the smaller radial ratio, but the general tendencies are the same as in F I.

(b) The fact that large azimuthal currents and recovered charges can be measured during the entire free-wheeling period, rules out the possibility that the exponential decay of the voltage is merely an electrostatic phenomenon in a lowly ionized gas.

(c) The measured free-wheeling times τ agree with the calculated ones, within the limits of experimental error and the uncertainties in the coefficients involved in the theory of Section 2.5. The curves given in Fig. 24 have been obtained from Eqs (72) - (74) in combination with the particle and heat balance equations of Sections 2.5.1 and 2.5.3. with $R = 0$ in Eq. (96). The value $L_{||} = 0.3$ m of the effective longitudinal length corresponds rather well to the geometry of the F I device in Fig. 8. In the Ixion the actual temperature may have been lower than that calculated from Eqs (86) - (96) with $R = 0$, on account of the large observed radiation losses produced by impurities [161]. However, since the time constants given by Eqs (73) and (74) are slow functions of temperature, this should only have a small effect on the calculated free-wheeling times.

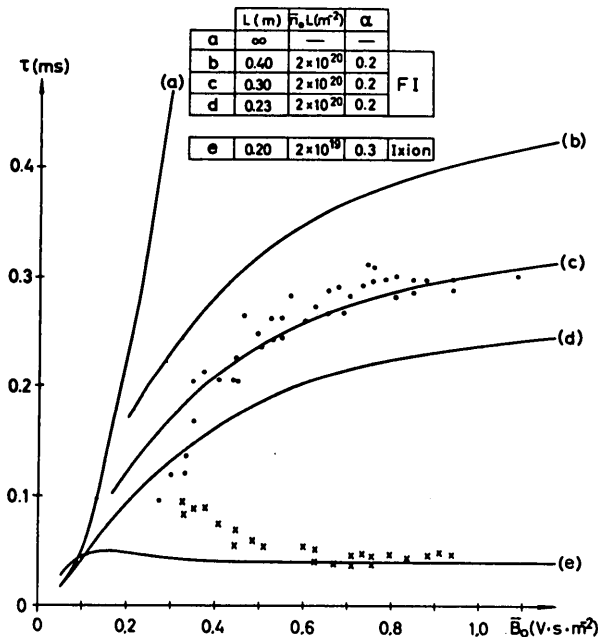


FIG. 24. Free-wheeling time τ at different magnetic field strengths. Crossed marks refer to Ixion [161] and dots to F I [160]. The curves have been calculated from the theory of Section 2.5.

(d) The constancy of J_ϕ/ϕ_{12}^2 in Fig. 23 is consistent with the conclusions of Sections 2.4.1., 2.4.2., and 2.5.1. that the recycling of neutrals is being limited.

(e) The strong decrease of the free-wheeling time in the range below the burn-out level gives additional confirmation of the statement that there are only small traces of neutral gas inside the main plasma body in the type 'A' systems outlined in Fig. 1.

3.5.3. Radiation and recombination losses

In the Ixion [161] bolometer measurements of the energy flux to the walls, and similar measurements of the emitted radiation by means of pinhole cameras and photocells, have indicated that the main part of the instantaneous power input goes into radiation losses. Similar conclusions have been drawn by Tozer [50] for a homopolar device and by Cooper et al. [171] for a device similar to that demonstrated in Fig. 2(b). Since the Ixion was reported to be strongly influenced by impurities [12], probably being introduced by the centre plasma electrode, and since the other two devices just mentioned were subject to a strong neutral gas and end insulator interaction, it is also likely that large radiation losses should arise under such conditions and should affect the plasma heat balance. This does, however, not imply that the plasma should be free from other losses such as those from escaping particles.

In the experiments reported by Tozer [50] and Cooper et al. [171] electron temperatures were observed being as low as about 1600°K. The rate of plasma decay was then found to be consistent with a collisional-radiative recombination process.

3.5.4. Final stage of holding mode

When energy is being gradually extracted by the plasma from a limited energy source, the electrode voltage will slowly drop until it reaches a level where a fully ionized state can no longer be sustained. This behaviour is demonstrated by Fig. 25, where a transition from a fully to a lowly ionized state takes place at the minimum voltage ϕ_{12m} during a short time interval around $t = t_m$ [56, 93]. The experiments have further shown that a measurable angular momentum can be recovered by a short-circuit at times $t < t_m$, but not at times $t > t_m$. The minimum voltage and the corresponding current and power input are found to vary with the magnetic field strength $\bar{B}_0$ and the filling gas density n_{n0} in a way expected from the theory of Section 2.7.5. [93, 184]. This also applies to the case where high-frequency heating has been superimposed on the rotating plasma [93].

3.6. VOLTAGE LIMITATION

We now discuss the effects which limit the applied voltage ϕ_{12} in systems of the types classified in Section 2.1.

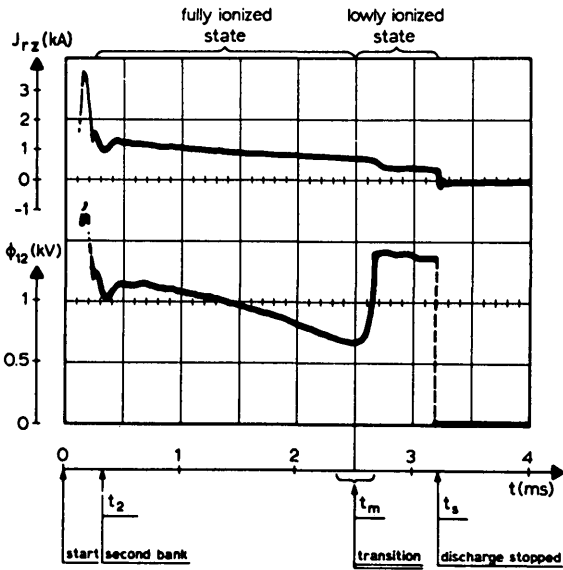


FIG. 25. Transition from a fully to a lowly ionized state at the minimum voltage ϕ_{12m} for device F II A [56, 93].

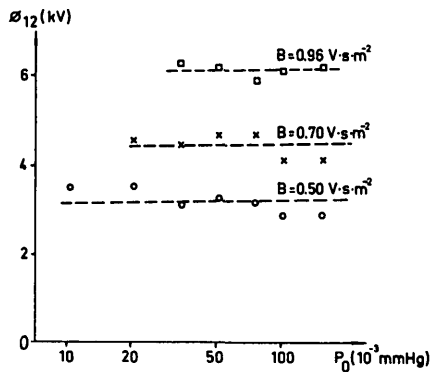


FIG. 26. Burning voltage ϕ_{12} of the rotating plasma in the homopolar of Angerth et al., as function of initial pressure P_0 of hydrogen discharges [38].

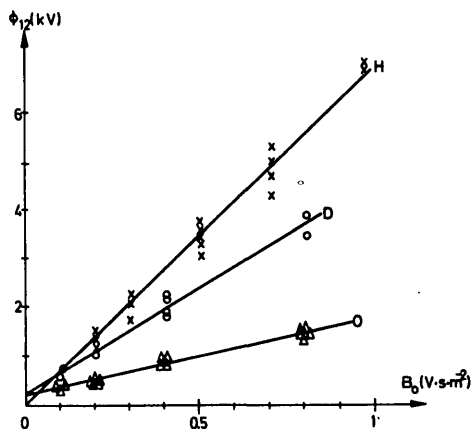


FIG. 27. Voltage limit ϕ_{12c} as a function of field strength B_0 in the homopolar of Angerth et al., as observed during the quasi-stationary period at discharge currents below 2000 A [38].

3.6.1. Type 'A' systems

The current-voltage characteristics of a FIG discharge were studied by Simon [185] who observed that the voltage quickly dropped to a constant level at discharge currents in the range below 100 A.

A more clear evidence of such a voltage level was given by Block and Fahleson [36] and Angerth et al. [38] who studied the electric characteristics of a flat homopolar device. They found that the electrode voltage ϕ_{12} was limited to a critical level ϕ_{12c} within the range of discharge currents J_{rz} from some 50 to 2000 A. An increase in ϕ_{12} with J_{rz} was only observed for currents above 2000 A. The burning voltage ϕ_{12} was observed to be independent of the filling pressure of the neutral gas in the range from 10^{-2} to 0.2 mmHg, as shown in Fig. 26. Further, there was a linear relationship between the critical voltage level and the magnetic field strength as given by Fig. 27 for three different gases. The observed voltage limit was consistent with the critical velocity proposed by Alfvén [5], i.e. the experimental values of ϕ_{12c} agree with Eq. (102) and with Eq. (104) for a flat homopolar, where $G(r_0/r_w) \approx 1$. The increase in ϕ_{12} for currents J_{rz} above some 2000 A can be at least partly understood from the potential drop which arises at large currents across the series impedances in the left circuit branch of Fig. 5. This increase need not necessarily imply that there is an increased velocity v_ϕ and voltage $\phi_{x/rz}$ at high currents.

For other devices also having radial ratios close to unity, such as Homopolar I, II, and III, similar voltage limitation effects have been observed [31, 33, 106].

In the Ixion device the discharge is started by a first bank and is further accelerated by a second 'boost' bank being switched on at time $t_2 = 0.26$ ms from the start. According to Fig. 28 there is a steady increase in the maximum Ixion voltage ϕ_{12max} with the applied voltage ϕ_{b20} of the second bank, but only up to a certain saturation level [161].

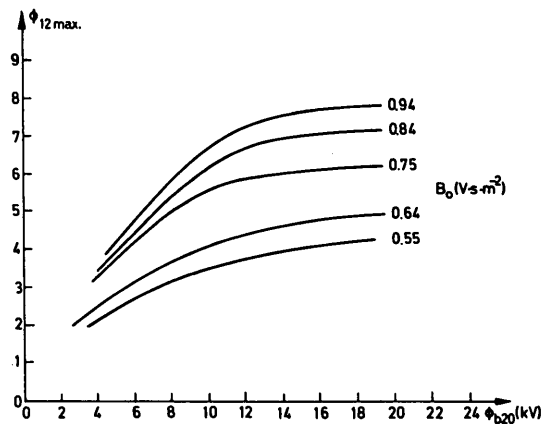


FIG. 28. Saturation in the Ixion peak voltage ϕ_{12max} as a function of increasing initial voltage ϕ_{b20} of second bank, with the magnetic field strength as parameter [161].

From scaling the electrostatic probe data of Fig. 22 and the corresponding velocity distribution $v_{\phi_0} = Gv_0$ in the midplane, the corresponding limiting velocity was estimated to $v_0 \approx 1.3 \times 10^5 \text{ m} \cdot \text{s}^{-1}$. Even if the limits of error in this estimate cannot be neglected, it seems to indicate that the obtained velocity v_0 is higher than the critical velocity v_c of Eq. (101). This is also to be expected from Eqs (103) and (104) at the radial ratio $R_r \approx 1.41$ of the Ixion. A further possible contribution to the observed voltage limitation is due to the fact that parts of the plasma may be scraped off at the outer electrode as stated at the end of Section II.7.b [40]. Further experiments with devices somewhat similar to the Ixion have been reported by Shvets et al. [174] and Bannenberg et al. [157]. The former authors claim to have reached voltages exceeding that corresponding to $v_{\phi_0} = v_c$ in the midplane, but it is not clear whether the plasma was fully ionized, because 3 cm microwave signals were partly transmitted through the plasma and the applied magnetic field appeared to be too weak to contain a fully ionized plasma at velocities comparable to or above the value v_c . The latter authors [157] observed a velocity being independent of the radius of the device and suggested that this fact could very well be reconcilable with the braking mechanism due to the critical velocity v_c .

The voltage limit has also been clearly seen in the device of Fig. 8, having the radial ratio $R_r \approx 3$. The values of ϕ_{12c} were found to be in close agreement with Eq. (104) [155, 160]. To compare the results with those by Block and Fahleson, a disc-shaped insulator plate was introduced close to the midplane to diminish the radial ratio to $R_r = 1$, and the critical voltage level was then found to decrease as expected from Eq. (104) [39]. Further experiments on the voltage limitation were performed with device F II being similar to that described in Fig. 8 but having the main coil suspended by a rod-shaped support [156]. To approach the critical level ϕ_{12c} slowly from below, an inductance was introduced in series with the second condenser bank which was connected at $t_2 = 0.21 \text{ ms}$, as seen from Fig. 29. At low bank voltages ϕ_{b20} the plasma behaved regularly as in Fig. 29(a). At higher voltages, however, there was an abrupt change to an irreproducible state with large losses, as soon as the electrode voltage reached the critical level ϕ_{12c} demonstrated in Figs 29(b) and (c). A strong gassing rate and a heavy damage of the insulator surface also occurred under such conditions. These experiments clearly show that a sharp transition takes place at ϕ_{12c} which hardly can be explained by the ordinary plasma balance equations of Section 2.3. The critical level ϕ_{12c} was found to increase linearly with the magnetic field strength, in close agreement with Eq. (104), with experimental deviations from shot to shot of only a few per cent. Finally, by introducing a turnable tube of ceramics in F II, the radial ratio R_r could be varied in the same way as if the device would have been provided with conical insulator walls having adjustable values of R_r . The critical voltage ϕ_{12c} was then found to increase monotonically by a factor of about 2.3 when

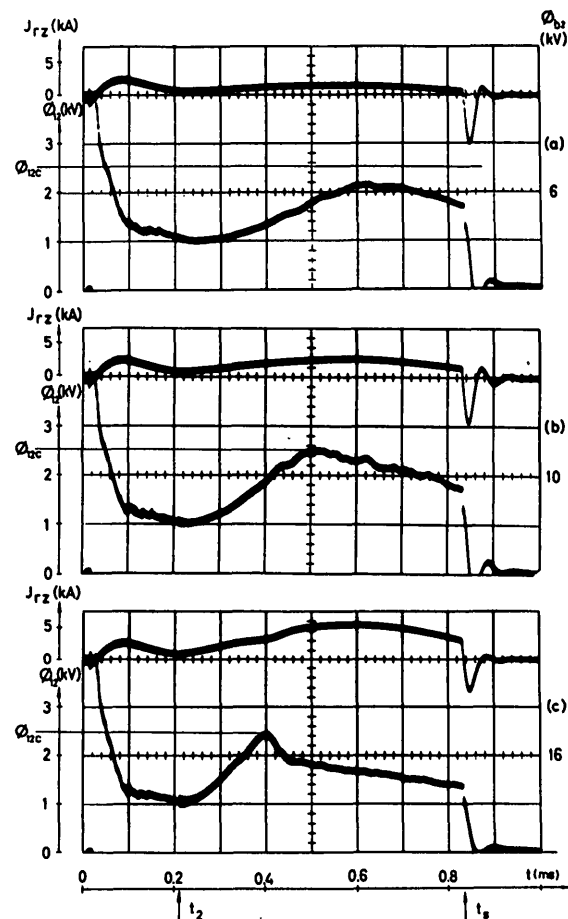


FIG. 29. Limitation of electrode voltage ϕ_{12} in F II [156].

Three shots are shown at different voltages ϕ_{b20} of second bank which is switched on at time t_2 . A short-circuit is applied at time t_3 .

(a) $\phi_{b20} = 6 \text{ kV}$. Electrode voltage does not reach critical level ϕ_{12c} and discharge behaves in a regular way.

(b) $\phi_{b20} = 10 \text{ kV}$. Critical level ϕ_{12c} is reached at 0.5 ms from start. At times $t > 0.5 \text{ ms}$ the plasma behaves in an irregular way.

(c) $\phi_{b20} = 16 \text{ kV}$. The same critical level is reached at $t = 0.4 \text{ ms}$.

the radial ratio R_r increased by a factor of about 2.2. This supports the theory of Eq. (104) and also indicates that the velocity limitation in fully ionized type 'A' systems is localized to the end insulators.

It should finally be mentioned that an F III device has been constructed having the extremely large radial ratio of 9.3 [186]. However, it has so far not been possible to increase the voltage levels with this device according to Eq. (104). The explanation of its behaviour is not clear, but with the extreme geometry of F III, complications are likely to arise from instabilities, insufficient line-tying, and from neutral gas penetrating the narrow confinement region near the end insulators.

3.6.2. Type 'B' systems

To avoid end insulator interaction with the plasma Halbach et al. [163] constructed Homopolar V being a type 'B' system having the basic features outlined in Fig. 2(a). About 30 to 50 μl of deuterium gas were injected into the central part of the device.

A fully ionized plasma was created within a few microseconds which reached velocities v_0 up to about $16 v_c$, i.e. $6 \times 10^5 \text{ m} \cdot \text{s}^{-1}$. After about $19 \mu\text{s}$ the plasma appeared to have expanded axially to get in touch with the end insulators, which resulted in a breakdown of the voltage and in an increased driving current. Neutrons were detected during the main phase of the discharge indicating a randomized ion energy of 1 to 4 keV at an ion density of 10^{20} to 10^{21} m^{-3} .

These experiments give an additional clear confirmation that the velocity limitation of fully ionized plasmas is localized to the end insulators.

3.6.3. Type 'C' systems

In studies of the ionizing front propagation in a type 'C' system similar to that of Fig.2(b) Brennan et al. [170] obtained a linear relationship between the magnetic field strength and the applied electrode voltage at low discharge currents. This suggests a constant azimuthal particle velocity behind the shock front, similar to that observed in type 'A' systems.

3.7. IONIZING WAVES

One of the first experimental studies on ionizing waves were carried out by Wilcox et al. [166] with a device similar to that outlined in Fig.2(b). Magnetic probe measurements indicated that the width of the ionizing front is a few centimetres, in accordance with the theory by Kunkel and Gross [99].

Further extensive studies on ionizing waves with a similar type of device have been described in a series of reports by Watson-Munro and collaborators [100, 170, 172]. Their results which were obtained by means of current buttons at the cylinder wall, mutually orthogonal current loops, photo-multipliers, and a monochromator, can be summarized as follows:

(a) At neutral gas densities in the range from about 5×10^{20} to 5×10^{21} molecules per m^3 the measured velocity U_f of the ionizing front and the generalized impedance ϕ_{rz}/J_{rz} were in good agreement with the theoretical predictions of Section 2.4.3. [99, 100]. An example is given by Fig.30 for experiments with the Supper II device, where relation (50) between the front velocity and the generalized impedance has been experimentally confirmed.

(b) At lower densities the measured velocity was lower than that predicted by theory. A possible explanation of this behaviour was given by dissipative effects such as viscosity [170]. At the end of Section 2.4.3. attention has also been drawn to another mechanism which may reduce the front velocity, namely by neutral gas penetrating far into the fully ionized plasma behind the front. Thus, according to Eq.(41) the penetration length L_n at $T < 5 \times 10^4 \text{ K}$ becomes larger than 0.1 m when the plasma density is less than $5 \times 10^{20} \text{ m}^{-3}$, provided that thermal ionization takes place by electrons only.

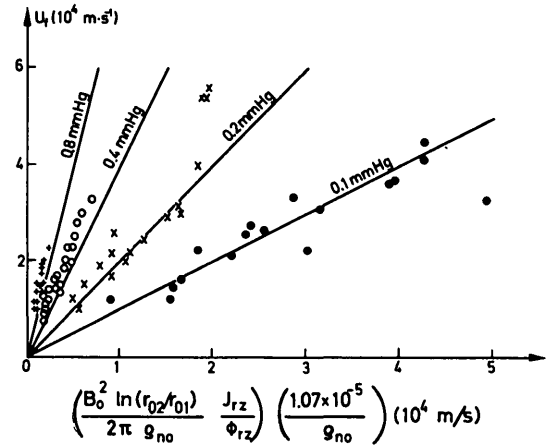


FIG.30. Marks indicate measurements of front velocity U_f and generalized impedance ϕ_{rz}/J_{rz} in Supper II device with B_0 in the range from 0.5 to $1 \text{ V} \cdot \text{s} \cdot \text{m}^{-2}$ and the hydrogen neutral gas density n_{n0} as parameter [100].

Straight lines represent Eq.(50).

(c) At higher densities the measured velocity was higher than predicted by theory, as seen from the plus and circle marks of Fig. 30. This may be connected with the fact that the thermal pressure and axial fluid velocity have been neglected in the deduction of Eq.(50) [100].

(d) The experiments under (a) - (c) were all done at positive polarity. It was also possible to propagate ionizing fronts with negative polarity, except at high magnetic field strengths and low densities. In the latter case electrons from the front were accelerated towards the end of the device and established an axial discharge by a short-circuit across the end wall. Such a discharge was also found to interfere with the attempts to make two oppositely rotating and moving fronts collide at the centre of the device.

3.8. ANODE SHEATHS AND CATHODE SPOTS

In the earlier described ion magnetrons containing lowly ionized plasma an anode sheath develops around the centre electrode being fundamental to the operation of the device [14-18]. However, such sheaths may also develop at the early stage of operation in devices for fully ionized rotating plasmas. In the Ixion negligible current was drawn during a 'waiting' phase about $200 \mu\text{s}$ after application of the voltage, being due to the formation of an anode sheath [12]. The waiting time could be reduced by means of auxiliary electrodes which help the electric field to penetrate the gas. During later stages of operation, however, both the experiments with the Ixion and with Homopolar V [163] and many other devices have shown that the voltage drop ϕ_{orz} in Fig.5 due to electrode sheaths and space charges plays a minor role in the plasma balance. One exception of this is the recent experiment with the Risø puffatron [177], where metal end walls have been used in some cases.

At positive polarity an anode sheath, a few millimetres in thickness, was then observed to sustain almost the entire potential drop across the device at later stages of the discharge.

Cathode spot formation is important to the understanding of the discharge conditions of rotating plasma devices [50, 102]. Special investigations of the influence of the cathode material have been made by Tozer [50] who showed that there are more electrode spots on molybdenum than on copper electrodes.

3.9. ELEMENT AND ISOTOPE SEPARATION EFFECTS

In earlier experiments with rotating plasmas heavy impurity ions were observed to be separated from the discharge under the influence of the centrifugal force [38, 175, 181]. Further studies of such processes in type 'C' devices were reported by Brennan et al. [187] and May et al. [188] who concluded that the separation was due to electron-ion collisions. It is likely that part of the observed separation effect is also due to ion-ion collisions in these experiments, as given by Eq.(111).

Experiments have further been performed by Bonnevier [59] with the F I device to verify the theory described in Section 2.8.1. [60]. Samples were collected during a discharge by means of a probe inserted into the confinement region and being connected with a container through a fast valve. The samples were analysed by a mass spectrometer. In a hydrogen-argon mixture the relative concentration ratio between the heavier and lighter component was found to increase strongly in the direction towards the outer plasma boundary of Fig.8, sometimes even by a factor of about 50. The amount of collected gas also increased steeply when the probe was moved into the plasma from the other boundary, in agreement with the expected velocity profile (51). Further experiments with hydrogen-deuterium mixtures have also given indication of an isotope separation effect caused by the centrifugal force.

According to Eqs (109) - (111) and the discussion at the end of Section 2.8.1., the separation effects observed in mixtures of elements and isotopes with $|A_i - A_j|$ of the same order as $A_i + A_j$ should mainly be caused by the drag force arising from collisions between non-identical ions. In mixtures of heavy isotopes, however, also electron-ion collisions and contributions from viscous forces given by the last term of Eq.(110) may have to be taken into account.

3.10. INSTABILITY PHENOMENA

In this section we shall mainly treat instability phenomena of fully ionized plasmas, and unstable modes involving neutral gas interaction will only be subject to short discussions.

3.10.1. Flute disturbances

For the development of flute disturbances under various experimental conditions, no general statement can be made about the stability of rotating plasmas. The different experiments which have been performed so far are summarized as follows:

(a) When a sufficiently rapid plasma rotation occurs in theta pinches, they are often observed to become unstable [189-192]. Since the pinches are highly compressed in the radial direction and are separated from external walls by a vacuum region, no stabilizing line-tying effects are expected to operate at their plasma boundary. They should therefore become unstable according to theory [80, 135, 145].

(b) In a stellarator, ions which have small velocities parallel to $\vec{B}$ do not greatly benefit from the rotational transform and tend to drift out of the discharge, leaving a net negative charge which sets up a radial electric field and a rotation. This improves the rotational transform and stops further ion loss [140]. However, if a strong radial electric field arises in the plasma or is imposed by certain external means, flute instabilities may arise according to theory. Gorman and Rietjens have imposed such fields on the B-3 stellarator by means of a positively biased Langmuir probe [159]. The confinement was found to be improved by the rotation at velocities below some $7 \times 10^3 \text{ m} \cdot \text{s}^{-1}$. At higher velocities this effect became less efficient, but the confinement time never reached below the value for zero imposed radial electric field. According to the complicated conditions of the stellarator it is difficult to understand these results in detail at the present stage.

(c) In the experiments with the Kruisvuur I device [89, 143, 193] an eccentric $m_\phi = 1$ mode was observed during the partially ionized build-up phase. It is likely to be of the spoke type discussed in Section 3.3.2. and need not necessarily be connected with flute instabilities in a fully ionized plasma. During the main phase of the discharge, however, Barbican and Rasmussen [143] observed high modes with m_ϕ between 5 and 15, the growth rates of which were consistent with Eq.(115) with $\gamma = 0$, including the influence from the Coriolis force. In these experiments the radial ratio R_r associated with the end insulator was close to unity, and the system should then suffer from large longitudinal plasma losses. It is therefore possible that the observed flute-like disturbances during the main phase of the discharge were of the same type as the set of spokes earlier observed in Homopolar III [106].

(d) Decay times of considerable length have been obtained for the plasma voltage and azimuthal current in a free-wheeling mode of the Ixion and of F I, as demonstrated by Figs 23 and 24. During free-wheeling there is no external power input into the plasma. These times are in some cases about two orders of magnitude longer than those obtained from the elementary growth rate $(g_c/\gamma_0)^{1/2}$ of Eq.(114), in the absence of all stabilizing effects.

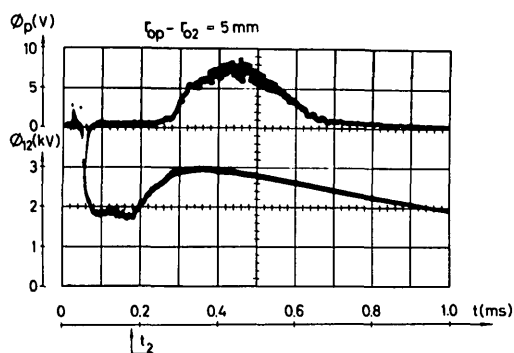


FIG. 31. Signal ϕ_p recorded by the probe of Fig. 9 being loaded by 5 ohm [160].

Measurable signals appear only when the electrode voltage ϕ_{12} is above some 2.5 kV. Distance between probe and unexpanded plasma boundary is $r_{op} - r_{02} = 5$ mm.

In the presence of sufficiently strong line-tying and finite Larmor radius effects, however, the stability condition (116) may be fulfilled. Further experimental indications of the flute stability of F I are given by the probe measurements of Fig. 21. This is also demonstrated by Fig. 31, where the probe of Fig. 8 has been placed at a distance $r_{op} = 5$ mm from the initial position of the plasma boundary at r_{02} , as defined by the undisturbed magnetic vacuum field [160]. There is no probe signal, and a nearly field-free region between the plasma and the outer vessel wall, as long as the voltage ϕ_{12} and the corresponding centrifugal force are small enough for the radially expanded plasma not to touch the probe. This, as well as pictures taken by means of high-speed photography [160] and signals from microwaves reflected from the plasma [173], strongly indicates that the outer plasma boundary in device F I is at least flute stable and sharply defined within a few millimetres. At this stage it cannot be judged for certain whether microinstabilities or Bohm diffusion might have been present in the Ixion or F I because such modes will probably introduce losses being too small to be distinguished from the classical losses discussed in Section 2.5. and represented by the curves of Fig. 24.

(e) Further indications for line-tying to play an important role to the plasma stabilization have been given by Wilcox et al. [166] who studied the flute-like 'swirls' which develop when a rotating plasma is suddenly being stopped by a short-circuit. The swirls were inhibited when a mesh steel screen was mounted just inside the remote end insulator of a system similar to that outlined in Fig. 2(b).

3.10.2. Neutral drag phenomena

At the comparatively low densities of reflex discharges the plasma becomes anisotropic and the stability conditions differ from those of high-density rotating plasmas. At these densities neutral gas is able to penetrate into the central parts of the plasma as discussed in Section 2.4., and a

strong ion-neutral drag and an associated instability [146, 147] may then develop in crossed electric and magnetic fields. The instabilities observed in some experiments are likely to be explained by this mode [194, 195], whereas the unstable behaviour in some other cases is likely to be produced by different mechanisms such as those from micro-instabilities [176].

3.10.3. Velocity shear phenomena

Instabilities similar to the Kelvin-Helmholtz mode have earlier been observed in a strongly sheared magnetohydrodynamic flow in mercury [41, 42]. Similar phenomena should also arise in rotating plasmas and it is likely that they are present in experiments with the so-called Q-machines [149, 195].

3.10.4. Cyclotron resonance phenomena

Rozkov et al. [152] have reported ion-cyclotron oscillations being energized by the external power supply of the rotating plasma. The growth rate of the associated instability was found to be of the order of the ion gyrofrequency, as expected from theory.

4. CONCLUSIONS

The investigations on rotating plasmas can be summarized as follows:

(a) Even if there still exist a number of questions to be settled, satisfactory agreement has now been largely achieved between theory and experiments in the description of rotating plasmas. Thus, earlier uncertainties and apparent contradictions in the interpretation of the experiments, have to a great deal been caused by the manifold of existing experimental conditions and parameters, such as the ionization degree, the relative magnitude of the plasma losses along the magnetic field, the boundary and line-tying conditions, the magnitudes and distributions of density, velocity and temperature, and the presence of impurities. When all these conditions are being properly taken into account in every particular experiment, most of the plasma behaviour can now be understood sufficiently well by theory, including the effects associated with the stored angular momentum, the diamagnetism and the balance of the centrifugal pressure, the macroscopic relaxation times, the overall plasma balance, and stability.

(b) Rotating plasmas suffer from several limiting effects such as the critical velocity phenomenon, the burn-out and minimum power effects, and the saturation of the losses by longitudinal effects in the strong magnetic field limit. These effects are of general physical interest, also in cosmical applications, but they introduce severe restrictions in the use of rotating plasmas for fusion research and other purposes. Especially if the mechanism of the velocity limitation can be removed, however,

the possible use of rotating plasmas as fusion devices may run into a new and more favourable situation.

(c) In spite of these limitations rotating plasmas have a number of important fields of application:

(i) From the point of view of basic plasma research they represent an interesting approach where macroscopic fluid motions are included in the plasma balance. In research on the stability of confined plasmas the centrifugal force provides a simple and well-defined driving force for the study of flute instabilities.

(ii) They provide simple means of creating fully ionized plasmas of high density and purity.

(iii) They can be used as plasma sources and ionization mechanisms for dense plasmas in fusion research, but should also in principle to an improvement of the confinement in open magnetic bottles.

(iv) They are important to a number of cosmic phenomena, probably to a greater extent than what has been realized so far. This is partly due to the fact that rotational motions can hardly be avoided when electric currents are flowing across a magnetic field.

(v) They can be used in a number of interesting technical applications such as in fully and partially ionized plasma centrifuges, in hydromagnetic adjustable condensers for electric power engineering, and in propulsion systems for space research and other purposes.

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